RANK-EIGHT GROTHENDIECK RINGS: CLASSIFICATION THROUGH MULTIPLICITY TWO WITH ONE OPEN CASE

Abstract. We study complex categorification of the complete census of 973 rank-eight fusion rings of multiplicity at most two. Theoretical and exact arithmetic obstructions exclude 895 rings, full multiplication-tensor comparisons identify unitary categories for 77, and one ring, $R8_2_24$, remains open in the general complex setting. The multiplicity-one classification is complete, with exactly 48 Grothendieck rings; at multiplicity two there are at least 29, with at most one additional case. A rank-eight Wedderburn lemma extends the formal-codegree d-number obstruction to noncommutative rings. Further arguments use graph pairs, Morita theory, coherence, character parity and conductor-indicator identities. For the open ring we obtain a pseudo-unitary reduction, forced indicators and an inner-square-root condition, and isolate the categorical transport identities needed by a possible local obstruction. The source-bound decision ledger and exact positive tensor comparisons accompany the manuscript.

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1. Introduction and exact scope

The fusion-ring census [11] supplies all 96 rank-eight based fusion rings of exact multiplicity one and all 877 of exact multiplicity two. A fusion ring need not be the Grothendieck ring of a fusion category. We work over $\mathbb{C}$ and do not assume pivotality, sphericality, braiding, commutativity or unitarity when generally excluding a ring. A positive construction is a unitary category with the exact based-isomorphism type of its full multiplication tensor.

The lower-rank criteria and their finite verifiers are developed in [12, 13, 14, 16, 17, 18]. We cite their theorems and do not reprove them. These new manuscripts are cited by title, with no byline. Our rank-eight observation, Proposition 3.1, extracts ordinary formal codegrees from the Casimir spectrum even though the ring may be noncommutative. Known near-group, zero-spectrum and cyclotomic tests provide other general obstructions [25, 28, 21]. Standard quantum group, pointed, Tambara–Yamagami and representation categories supply the positive side [29, 42, 22, 43].

The accepted study of integral Grothendieck rings [19, 20] supplies a necessary divisibility test and explicit group-theoretical rank-eight models and exclusions. Being an integral or Drinfeld ring does not itself supply a fusion category, so only its explicit categorical constructions enter our positive list.

Theorem 1.1 (Rank-eight classification with one open case)**.** *Among the rank-eight based fusion rings through multiplicity two, the established decisions and the remaining open case are as follows.*

Exact multiplicity	*Fusion rings*	*Excluded generally*	*Unitarily realized*	*Undecided generally*
1	96	48	48	0
2	877	847	29	1
Total	973	895	77	1

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Every excluded ring is ruled out over $\mathbb{C}$ *with no pivotality assumption. Each positive model is matched to the entire tensor under a unit-preserving permutation, and is therefore a Grothendieck ring. Thus there are exactly* 48 *Grothendieck rings of exact multiplicity one and at least* 29 *of exact multiplicity two. The only undecided ring is R*8_2_24*; it has no unitary categorification by Proposition 18.1. The number of general complex Grothendieck rings of exact multiplicity two is therefore either* 29 *or* 30*.*

The open case `R8_2_24` is studied in Section 13. The ring `R8_1_3`, of global FP dimension 20, is realized below by a twisted group-theoretical category. Its dimension vector agrees with that of the commutative ring $\mathrm{Rep}(\mathrm{Dic}_5)$, although its multiplication is noncommutative. Thus full tensor comparison remains essential even after a suitable dimension profile is found.

2. Trusted input and stable records

A rank-r fusion ring has a distinguished basis $B = (b_0 = 1, b_1, \dots, b_{r-1})$, structure constants $N_{ij}^k \in \mathbb{Z}_{\geq 0}$, and a duality involution $i \mapsto i^*$ satisfying

$$b_i b_j = \sum_k N_{ij}^k b_k, \qquad N_{ij}^0 = \delta_{j,i^*}, \qquad N_{ij}^k = N_{i^*k}^j = N_{kj^*}^i.$$

Its exact multiplicity is $\max_{i,j,k} N_{ij}^k$. A based isomorphism permutes the basis, fixes b_0, and preserves every structure constant; it automatically preserves duality. We do not identify tensors by character table, spectrum or FP dimension vector alone.

The source is the public *Fusion Ring Census* [11] at Git commit `b86eb6dc6eeeafe5fc5f6906d7b074e0322e9726`. It contains `results/rank8/tables.txt.gz` and `results/rank8/counts.csv`, with audit file `verification/rank8/audit.json`. The SHA-256 of the uncompressed tensor file is `97b86d2de8ca0352c5c6e14686cd1d43dbd52180dc53b2ff1ccf7a61788c9600`. The census verifies all duality types, associativity, nonnegative coefficients and uniqueness; its terminated-strata evidence justifies completeness through multiplicity two. We trust that source for counting and do not claim an independent exhaustive enumeration here.

Within each exact multiplicity m, sort full flattened $8 \times 8 \times 8$ tensors lexicographically and call the jth one $R8_m_j$. This is a deterministic local index, not an invariant of its original basis labelling. The files `data/decisions.json`, `data/realized_models.json` and `data/rank8_tables.txt.gz` give, respectively, one decision per input, the exact positive model tensors and the source multiplication tables. A SHA-256 of each JSON tensor prevents accidental source-record substitution. The comparison code minimizes the full tensor under all relabellings compatible with inexpensive intrinsic signatures; this is an isomorphism comparison, not a filter by numerical fingerprint.

3. Necessary criteria and their exact application

3.1. **Short tensor and subring tests.** If X is self-dual and $X^2 = 1 + kX$ with $k \geq 2$, its span with 1 is a fusion subring that cannot arise over $\mathbb{C}$ by [12, 23]. When exactly seven basis elements are invertible and one additional object X satisfies

$$gX = Xg = X, \qquad X^2 = \sum_{g \in G} g + \kappa X,$$

the rank-eight near-group case with $\kappa = 1$ or 2 is commutative and satisfies $0 < \kappa < |G| - 1$. Schopieray's general Proposition 3.2 [25] excludes it. The machine certificate checks the complete group and all displayed equations, not just the value of κ.

The zero-spectrum test is the finite nine-index condition of [28, Theorem 2.14] in the form reproduced in [17, Section 3]. It is a necessary condition for an associative tensor lift, with no pivotality assumption. Its C++ scanner searches the nine indices on the full tensor and the Python

checker independently recomputes every coefficient, positivity condition and vanishing contraction. The native arrays are sized for rank eight, and the checker has no rank-six restriction.

Every duality-closed, multiplication-closed based subset of a category gives a fusion subcategory. We compare every such proper subring of ranks two through six against the *complete positive list* in the corresponding lower-rank theorem. The lower-rank classifications are unrestricted for ranks two, three and four, and complete through multiplicity 20 for rank five and 2 for rank six [12, 13, 14, 16, 17]; every rank-eight subring in this manuscript also has multiplicity at most two. A rank-seven subring is not excluded merely because it is absent from an incomplete rank-seven list.

The rank-seven analysis [18] provides four specific, fully checked negative multiplicity-two tensors that occur as rank-eight closed subrings. We transfer their exact signed-sphericalization witnesses by complete seven-dimensional based isomorphism and recheck all four proof trees. This separate stage gives exclusions only for rings actually containing one of those four proved negative subrings; it makes no claim that the rank-seven positive catalogue is exhaustive.

The involution-module theorem of [17, Section 5] says that an invertible involution fixing basis elements forces a finite anti-invariant module representation. Where its stated hypothesis is met, the script searches *all* bounded integer matrices and verifies multiplication, transpose, parity and entry bounds. The FP dimension quotient of a hypothetical fusion subcategory must likewise be an algebraic integer [22, Proposition 8.15]; a rational coefficient in its exact minimal polynomial excludes categorification.

3.2. **A rank-eight noncommutative codegree lemma.** For any fusion ring, write

$$\Omega = \sum_i b_i b_{i^*}, \qquad C_R = \sum_i N_i N_i^{\mathsf{T}}.$$

For an irreducible representation E of $R \otimes \mathbb{C}$ of degree e_E, the central element Ω acts as the scalar $e_E f_E$, where f_E is its ordinary formal codegree. The regular representation contains E with multiplicity e_E, so that scalar has multiplicity e_E^2 in the characteristic polynomial of C_R [24, 17].

Proposition 3.1 (Noncommutative rank-eight codegrees)**.** *Let R be a noncommutative fusion ring of rank eight. Its semisimple complex algebra has Wedderburn decomposition $\mathbb{C}^4 \oplus M_2(\mathbb{C})$. The formal codegree of the unique matrix-block representation is a rational integer. Every irreducible factor of the characteristic polynomial of C_R of degree at least two consists entirely of formal codegrees of one-dimensional representations. Consequently, if any such factor fails Ostrik's d-number divisibility condition, R has no complex fusion categorification.*

Proof. The FP dimension character is one-dimensional, so at least one scalar summand exists. Noncommutativity requires a matrix summand of dimension at least two. Since the sum of the squares of the degrees of simple modules is eight, there can be only one matrix summand and it must have degree two; the four remaining dimensions are one. The rational based algebra is split by some number field. Galois automorphisms permute its irreducible representations preserving their degrees, so the unique two-dimensional representation is Galois-fixed. Its formal codegree is an algebraic integer by [24] and rational by Galois invariance, hence belongs to $\mathbb{Z}$. Its Casimir eigenvalue is therefore rational. Every nonlinear irreducible factor of the integral Casimir polynomial comes from the four one-dimensional representations; on each, the eigenvalue is exactly the formal codegree. Finally, formal codegrees of complex fusion categories are d-numbers [24]. For an irreducible polynomial $q(t) = t^d + a_1 t^{d-1} + \cdots + a_d$, this requires $|a_d|^j \mid a_j^d$ for every $j = 1, \ldots, d$. Failure of one exact integer divisibility gives the assertion. □

The commutative branch uses the ordinary d-number test on every Casimir factor. The script never treats raw Casimir eigenvalues as formal codegrees of noncommutative representations without Proposition 3.1. We also ran the rational-indivisibility screen of [26, Theorem 4.4], noncommuting field-automorphism witnesses from [17], and finite-field cyclotomic witnesses from [15]. These three screens add no exclusions in the present bound; their zero rows and exact code remain in the stage

data. Schopieray's greedy formal-codegree obstruction $2, 3, 7$ [27] cannot occur before multiplicity five, since its forced rank-four subring has a coefficient five, so it is not an active step here.

3.3. **Integral divisibility and the complete signed quotient test.** The integral Drinfeld condition [19, Section 2.3] says that, if all simple FP dimensions of a categorifiable ring are integers, every ordinary formal codegree is a positive integer dividing its global FP dimension. For a commutative tensor, the verifier finds its unique positive FP character by an exact integral eigenvector of the strictly positive matrix $\sum_i N_i$. It then compares every exact linear factor of $\det(tI - C_R)$ with $D = \sum_i \mathrm{FPdim}(b_i)^2$. A nonintegral factor, or an integer codegree that does not divide D, is an unconditional complex no-go. This stage includes concrete multiplicity-two failures with $D = 156$ and 110, respectively, at $R8_2_288$ and $R8_2_392$.

The archived rank-eight appendix of [20] prints full tensors and explicit categorification decisions. Its items C.3.2(5) and C.3.2(7) are excluded by the paper's no-induction-matrix Lemmas 4.3 and 4.2. The appendix-to-census verifier checks all 8^3 entries under a unit-preserving permutation before transferring either published exclusion. We rely on those cited lemmas for the induction obstruction; their original Normaliz computation is not repeated here. The same transfer finds group-theoretical models in items C.3.2(3), C.3.2(6), and C.3.2(11). Item C.3.2(4) matches $R8_1_3$ but supplies no existence or exclusion claim; the explicit twisted group-theoretical construction below now realizes that ring.

Finally, the oriented signed-sphericalization theorem of [17, 18] applies even when the alleged category has no pivotal structure. The canonical sphericalization has two lifts of each simple; quotienting its Grothendieck ring by the negative central invertible yields an ordered signed multiplication tensor. We enumerate every cyclic and duality orbit of signed coefficients within the exact parity bounds, require all ordered associativity equations, then check every possible spherical character by the inherited norm, duality, and dimension-order criteria. A finite search that times out or has an unrefuted branch supplies no exclusion. The attached campaign records each attempted multiplicity-one tensor, incomplete result, and timeout separately; only certificates whose complete signed branches independently regenerate are admitted to the ledger.

3.4. **Grading obstructions with rank-one components.** The following short arguments give five further exclusions without any pivotality assumption. In a faithful grading $\mathcal{C} = \bigoplus_{g\in G} \mathcal{C}_g$, every component is an invertible bimodule category over $\mathcal{D} = \mathcal{C}_e$, and $g \mapsto [\mathcal{C}_g]$ is a homomorphism to $\mathrm{BrPic}(\mathcal{D})$ [7, Theorem 1.3 and Section 6].

Lemma 3.2. *If a pointed category* Vec_A^ω *has a module category with one simple object, then* $[\omega] = 0$.

Proof. Identify the module category with Vec. Each invertible object acts by the identity functor, up to isomorphism. The scalar module constraints form a nonzero 2-cochain whose coboundary is ω, by the module pentagon. Hence the associator class is trivial. □

Proposition 3.3. *The rings* $R8_1_26$, $R8_2_116$, *and* $R8_2_189$ *have no complex fusion categorification.*

Proof. In the original zero-based census labels, $R8_1_26$ has a faithful C_4 grading with components

$$\{0, 3, 4\}, \quad \{1\}, \quad \{5, 6, 7\}, \quad \{2\}.$$

Its neutral group is $A = C_3$. Lemma 3.2 identifies its neutral category with Vec_{C_3}. By [7, Corollary 1.2], its Brauer–Picard group is the orthogonal group of $A \oplus A^*$ with hyperbolic quadratic form. Over $\mathbb{F}_3$ this consists of $I, -I, J, -J$, where J exchanges the two coordinates. Every square is the identity, so $[\mathcal{C}_2] = [\mathcal{C}_1]^2 = 1$. On the other hand, the degree-two invertible object $s = 5$ conjugates $a = 3$ to $a^{-1} = 4$. Its component is the regular left module with right action twisted by inversion. It cannot be equivalent to the regular bimodule: such an equivalence would send s to some $b \in A$ and the relation $as = sa^{-1}$ would imply $ab = ba^{-1}$, which is false in C_3.

For $R8_2_116$ and $R8_2_189$, the universal grading has five components, of ranks $4, 1, 1, 1, 1$, so its group is C_5. Their neutral groups are respectively C_4 on $\{0, 1, 2, 7\}$ and C_2^2 on $\{0, 5, 6, 7\}$. Again their associator classes are trivial. In the first case the Brauer–Picard group embeds in $\mathrm{GL}_2(\mathbb{Z}/4)$, of order 96; in the second it is $O_4^+(\mathbb{F}_2)$, of order 72. Neither group has an element of order five. Hence the component homomorphism is trivial, forcing every component to be equivalent to the rank-four regular bimodule, contrary to the rank-one components. The verifier enumerates the small orthogonal groups and checks every grading support identity exactly. □

Proposition 3.4. *The rings $R8_2_78$ and $R8_2_168$ have no complex fusion categorification.*

Proof. Each has a faithful grading with a rank-one component and neutral fusion rules $\mathrm{Rep}(S_3)$: $g^2 = 1$, $gx = xg = x$, and $x^2 = 1 + g + x$. A rank-one module category gives a fiber functor on the neutral category $\mathcal{D}$, so $\mathcal{D} \simeq \mathrm{Rep}(H)$ for a semisimple Hopf algebra of dimension six. The dimension-six classification [10] and the rank-three fusion rules force $H \simeq \mathbb{C}[S_3]$. The dual category of its fiber-functor module is $\mathrm{Rep}(H^*) \simeq \mathrm{Vec}_{S_3}$, of rank six [22, Example 7.12.26]. But invertibility of the homogeneous $\mathcal{D}$-bimodule would identify that dual with $\mathcal{D}$, of rank three, a contradiction. □

3.5. **Solvability, equivariantization and module ranks.** We give three further general exclusions using only small groups and exact fusion rules. By [3, Theorem 1.6], a fusion category of FP dimension $p^a q^b$ is solvable. The characterization in its Definition 1.2 and Proposition 4.4 expresses it as successive cyclic-prime extensions and equivariantizations. If its adjoint subcategory is the whole category, its last nontrivial step cannot be an extension. Thus it is an actual C_p-equivariantization, with a central $\mathrm{Rep}(C_p)$ whose underlying invertibles embed into the category. We use the orbit/stabilizer description of equivariant simples [6, Section 2]; cyclic stabilizers have only one-dimensional irreducible projective representations.

Proposition 3.5. *The ring $R8_1_29$ has no complex fusion categorification.*

Proof. Its invertibles form C_6, its other two simples X, Y are self-dual of dimension three, and its adjoint ring is the whole ring. Its dimension is 24, so a putative category $\mathcal{C}$ is $\mathcal{D}^{C_p}$ with $p = 2$ or 3.

For $p = 3$, the order-three character subgroup fixes X, Y and has two free orbits on the six invertibles. The forgetful images of X, Y split into three distinct invertibles each; those of the invertibles give two more. Thus $\mathcal{D} = \mathrm{Vec}_A^\omega$, $|A| = 8$, with two free C_3-orbits. The only groups of order eight admitting an order-three automorphism are C_2^3 and Q_8. On C_2^3 the action has a fixed line and an irreducible plane. The three nonzero elements in that plane form a free orbit whose product has coefficient two into the same free orbit. The output stabilizer is trivial, so associator and tensorator choices cannot change this coefficient. On Q_8, inversion exchanges the orbits $\{i, j, k\}$ and $\{-i, -j, -k\}$; their two induced dimension-three objects are distinct duals. These conclusions contradict multiplicity one and self-duality, respectively.

For $p = 2$, the character involution exchanges X, Y. De-equivariantization has the fusion rules

$$G(\mathcal{D}) = C_3 = \langle a \rangle, \qquad a\rho = \rho a = \rho, \qquad \rho^2 = 1 + a + a^2 + 2\rho.$$

It has dimension twelve and trivial universal grading. The same last-step argument gives $\mathcal{D} = (\mathrm{Vec}_V^\omega)^{C_3}$, where $V = C_2^2$ and C_3 permutes the three nonzero elements transitively. No assumption on ω is made. A monoidal automorphism of $\mathrm{Id}_\mathcal{D}$ has scalar one on ρ, because ρ occurs in ρ^2, and then on all invertibles. Consequently the map from central invertibles to underlying invertibles is injective, and the central $\mathrm{Rep}(C_3)$ is unique.

The C_2 action on $\mathcal{D}$ fixes every simple: exchanging a, a^2 would produce a dimension-two simple upstairs. Every such tensor autoequivalence preserves the unique central $\mathrm{Rep}(C_3)$. Its restriction is tensor-isomorphic to the identity, since $H^2(C_3, \mathbb{C}^\times) = 0$. Equivariantization/de-equivariantization [4, Theorem 4.18(i) and Proposition 4.19] therefore lifts it to a C_3-equivariant tensor autoequivalence of Vec_V^ω. Its object permutation centralizes C_3 in $\mathrm{Aut}(V) = S_3$, so lies in C_3. Composing with the inverse action functor removes that permutation without changing the descended equivalence class.

Identity-on-objects tensor autoequivalences of Vec_V^ω are classified by $H^2(V,\mathbb{C}^\times) = C_2$: the two associators cancel in the tensorator equation. Their equivariant structures have ambiguity $H^1(C_3,\widehat{V}) = 0$, since multiplication by three is invertible on $\widehat{V}$. Both possible classes have explicit representatives. Besides the identity, take

$$J(u,v) = (-1)^{u_1v_1+u_1v_2+u_2v_2}, \qquad u,v \in \mathbb{F}_2^2.$$

This nontrivial normalized 2-cocycle is C_3-invariant and satisfies $J^2 = 1$. Thus it commutes even with nontrivial scalar tensorators of the existing C_3 action and descends with identity comparisons. There are no further C_2 coherence choices, because $\mathrm{Aut}_\otimes(\mathrm{Id}_\mathcal{D}) = 1$.

The identity action leaves the coefficient two in ρ^2. For the J action, all degree-zero channels have $J(u,u) = -1$ for $u \neq 0$; for each nonzero output degree, the two ordered channels have opposite signs. Hence each lift $\rho_\pm$ has square containing the three negative-character invertibles and one copy of each $\rho_\pm$. The unit occurs in $\rho_+\rho_-$, not in either square. The two lifts are distinct duals, again contradicting the self-duality in $R8_1_29$. This exhausts both prime branches for arbitrary pointed associators. □

Lemma 3.6 (Module-rank divisibility)**.** *For a fusion category $\mathcal{B}$, let $g_\mathcal{B}$ be the greatest common divisor of the ranks of its indecomposable finite semisimple module categories. If $\mathcal{B} \subseteq \mathcal{D}$ is a fusion subcategory, then $g_\mathcal{B} \mid \mathrm{rank}(\mathcal{D})$.*

Proof. Decompose $\mathcal{D}$, with its left tensor action by $\mathcal{B}$, into indecomposable module categories. Their ranks add to $\mathrm{rank}(\mathcal{D})$. This statement uses ranks in the specified module presentation; it makes no assertion that module ranks are Morita invariant. □

Proposition 3.7. *The rings $R8_1_45$ and $R8_1_47$ have no complex fusion categorification.*

Proof. Each ring has simple dimensions $(1,1,2,3,3,4,4,4)$ and admits labels $1,h,E,P,Q,W_0,W_1,W_2$ with

$$h^2 = 1, \quad hE = E, \quad hP = Q, \quad hW_i = W_i, \quad P^* = Q,$$
$$E^2 = 1 + h + E, \qquad EP = EQ = P + Q, \qquad PQ = 1 + E + P + Q.$$

The adjoint ring is the whole ring. Dimension 72 and the invertible group C_2 therefore force a putative categorification to be $\mathcal{C} = \mathcal{D}^{C_2}$ with $\mathrm{FPdim}(\mathcal{D}) = 36$. The forgetful images are

$$F(1) = F(h) = 1, \quad F(E) = a + a^2, \quad F(P) = F(Q) = Y, \quad F(W_i) = U_i + V_i.$$

Here a has order three, Y is simple of dimension three, and the six U_i, V_i are distinct simples of dimension two. Thus $\mathrm{rank}(\mathcal{D}) = 10$. The full fusion subcategory $\mathcal{B}$ on $1, a, a^2, Y$ satisfies

$$aY = Ya = Y, \qquad Y^* = Y, \qquad Y^2 = 1 + a + a^2 + 2Y.$$

The C_2 action exchanges a, a^2 and fixes Y; its equivariantization inside $\mathcal{C}$ has the displayed nonselfdual P, Q.

As in the preceding proof, solvability and trivial universal grading imply $\mathcal{B} = (\mathrm{Vec}_V^\omega)^{C_3}$, $V = C_2^2$, with the transitive action. The restriction map

$$H^3(V,\mathbb{C}^\times) \longrightarrow \prod_{\substack{L \leq V \\ |L|=2}} H^3(L,\mathbb{C}^\times)$$

is an isomorphism $C_2^3 \to C_2^3$. In integral-cohomology generators u^2, v^2, uv, the three restrictions of (r,s,t) are $(r,s,r+s+t)$. Since C_3 permutes the factors, there are exactly two invariant classes: zero and the class nontrivial on every subgroup of order two.

Suppose first $[\omega] = 0$. The lift of the object C_3 action is unique in $\mathrm{Aut}_\otimes(\mathrm{Vec}_V) = C_2 \times S_3$, and its coherence is unique because $H^2(C_3,\widehat{V}) = 0$. Therefore $\mathcal{B} \simeq \mathrm{Rep}(A_4)$. Its central $\mathrm{Rep}(C_3)$ is unique, since $\mathrm{Aut}_\otimes(\mathrm{Id}_\mathcal{B}) = 1$, and is preserved by the C_2 action. Sequential equivariantization and de-equivariantization [3, Section 2.6 and Proposition 2.10][4, Proposition 4.19] identify $\mathcal{B}^{C_2}$ with

$(\mathrm{Vec}_V)^{S_3}$. The action on nonzero elements is the usual faithful S_3 permutation action; inversion on the character C_3 forces the group $C_3 \rtimes C_2 = S_3$.

There are exactly two lifts of this object action to $C_2 \times S_3$, given by the trivial or parity homomorphism into $H^2(V, \mathbb{C}^\times) = C_2$. Their coherence ambiguity vanishes: $H^j(S_3, \widehat{V}) = 0$ for $j > 0$, by the normal C_3 subgroup, whose positive cohomology and fixed space on $\widehat{V}$ both vanish. Both lifts yield self-dual dimension-three orbit objects. For completeness choose a stabilizing transposition τ swapping coordinates and its fixed nonzero element $v = (1, 1)$. For the trivial lift, the equivariance scalars $u = \pm 1$ act trivially on the unit channel of $v \otimes v$. For the other lift choose tensorator $\mu(a, b) = (-1)^{a_1 b_2}$. Its square is trivialized by $\eta_a = (-1)^{a_1 a_2}$, so $\eta_v = \mu(v, v) = -1$. The equivariance scalar satisfies $u^2 = \eta_v$ and acts on the unit channel by $u^2/\mu(v, v) = 1$. A τ-invariant character change of η is one on v, so cannot change the conclusion. Frobenius reciprocity for the orbit induction gives a tensor unit in each of the two squares. This contradicts $P^* = Q$.

It remains that $[\omega]$ is nontrivial on every order-two subgroup. The crossed product of Vec_V^ω by C_3 is pointed, and realizes $\mathcal{B}$ as the group-theoretical category

$$\mathcal{B} \simeq \mathcal{C}(A_4, \Omega, K, \psi), \qquad K = C_3, \qquad [\Omega]|_V = [\omega], \quad [\Omega]|_K = 0.$$

By the classification of pointed module categories and its double-coset rank formula [5, Example 2.1, Theorem 3.1 and Proposition 3.1], indecomposable $\mathcal{B}$-module categories correspond to subgroups $H \leq A_4$ with $[\Omega]|_H = 0$ and permissible cochains. If $|H|$ were even, Cauchy's theorem would give a subgroup of order two, contradicting the restriction of Ω. Thus H is trivial or has order three. For $H = 1$, the four double cosets in $K\backslash A_4/H$ have trivial intersections, giving module rank four. For $H = C_3$, conjugate it to K. There are two double cosets, with intersections C_3 and 1; projective cocycles on C_3 are coboundaries, so their contributions are three and one. Every indecomposable $\mathcal{B}$-module category therefore has rank four, regardless of the permissible cochains. Lemma 3.6 would force $4 \mid \mathrm{rank}(\mathcal{D}) = 10$, a contradiction. □

The portable verifiers check the source-bound tensors, the displayed fusion identities, the complete small cocycle identities, and all relevant A_4 subgroups and double cosets. These finite checks certify the hypotheses; the categorical exhaustion is supplied by the proofs above. Neither argument assumes a braiding or a pivotal structure.

3.6. **Rank-one modules and semisimple Hopf algebras.**

Proposition 3.8. *The rings R8_2_41, R8_2_86, R8_2_162, R8_2_140 and R8_2_784 have no complex fusion categorification.*

Proof. For the first three rings, a faithful C_2 grading has a rank-one nonneutral component and a dimension-sixteen neutral category $\mathcal{D}$ of type $(1^4, 2^3)$. A rank-one module supplies a fiber functor, so $\mathcal{D} = \mathrm{Rep}(H)$ for a semisimple Hopf algebra of dimension sixteen. Invertibility of the component further requires $\mathrm{Rep}(H^*) \simeq \mathcal{D}$; in particular both groups of grouplikes have order four. Kashina's classification [45, Theorem 1.2 and Table 1] leaves only $H_{C:\sigma_1}$, whose representation ring is $\mathrm{Rep}(D_{16})$: its invertibles form C_2^2, and all three dimension-two simples are self-dual. The rings *R*8_2_41, *R*8_2_86 instead have cyclic invertibles, while *R*8_2_162 has a nonselfdual dimension-two pair.

For *R*8_2_140, census indices $\{0, 1, 2, 3, 7\}$ give the neutral Tambara–Yamagami category on C_4, of dimension eight. Each other simple separately gives a rank-one left module: an invertible acts as one and the dimension-two object acts as two. Thus the neutral category has a fiber functor and is $\mathrm{Rep}(H)$ with $G(H^*) = C_4$. Kashina's prime-power theorem [45, Theorem 1.1] says that a semisimple Hopf algebra of prime-power dimension with cyclic grouplikes is a group algebra. It would give $H^* \simeq \mathbb{C}[C_8]$, which has eight grouplikes, a contradiction. This argument does not require the rank-one modules to be grading components.

For *R*8_2_784, the neutral component has dimension twenty-two, rank seven and type $(1^2, 2^5)$; the other component has a unique simple of dimension $\sqrt{22}$. Its fiber functor reconstructs a Hopf

algebra of dimension $2 \cdot 11$. The Hopf pq classification [10] and the nonpointed neutral rules force $\mathcal{D} = \mathrm{Rep}(D_{22})$. Its fiber-functor dual is $\mathrm{Vec}_{D_{22}}$, of rank twenty-two. The invertible grading component would identify this dual with $\mathcal{D}$, of rank seven. This is impossible. □

3.7. **Preserved Tannakian subcategories.**

Proposition 3.9. *The rings* R8_2_697*,* R8_2_808 *and* R8_2_181 *have no complex fusion categorification.*

Proof. For R8_2_697, dimension sixty-eight, type $(1^4, 4^4)$ and trivial universal grading, with all four invertibles fixing each noninvertible, force a C_2 equivariantization with descent $\mathcal{D}$ of dimension thirty-four and type $(1^2, 2^8)$. Its adjoint is the whole category: a proper adjoint has dimension $1, 2$ or 17; a prime-dimensional adjoint is pointed, but these possibilities cannot contain the required dimension-four squares with only two invertibles. Solvability and prime descent now give $\mathcal{D} = (\mathrm{Vec}^{\omega}_{C_{17}})^{C_2}$, with inversion. Since $U(\mathcal{D}) = 1$, central lifts of its invertibles are unique. The outer involution therefore preserves its central $\mathrm{Rep}(C_2)$, and the combined equivariantization group is C_2^2, as its four linear characters are the census invertibles. Every $C_2^2 \to \mathrm{Aut}(C_{17})$ has image of order at most two. A nonzero orbit has size two and cyclic stabilizer; it therefore supplies dimension-two equivariant simples. The target has only dimension-four noninvertibles.

For R8_2_808, dimension $186 = 2 \cdot 3 \cdot 31$ and the integral pqr theorem make any categorification group-theoretical [3, Theorem 9.2]. Groups of order 186 are solvable, so the category is solvable. The adjoint, type $(1^2, 2, 6^5)$, and the character involution fixing all noninvertibles force C_2 descent of type $(1^3, 3^{10})$ and dimension ninety-three. The same prime-dimensional-adjoint argument shows that its universal grading is trivial. Thus it is $(\mathrm{Vec}^{\omega}_{C_{31}})^{C_3}$ with nontrivial order-three action. Uniqueness of the central C_3 preserves it under the outer involution. The combined group is S_3, because its embedded representation subcategory has type $(1^2, 2)$. Every homomorphism $S_3 \to \mathrm{Aut}(C_{31})$ kills its normal C_3, contradicting that action.

For the commutative ring R8_2_181, the neutral category has dimension twenty-one, type $(1^3, 3^2)$ and trivial universal grading. Solvability makes it $(\mathrm{Vec}^{\omega}_{C_7})^{C_3}$, with multiplication by two or four on C_7. The action on $H^3(C_7, \mathbb{C}^\times) = C_7$ is multiplication by four or two, so its invariant associator class is zero. Coprime action cohomology then identifies the neutral category with $\mathrm{Rep}(C_7 \rtimes C_3)$. Its central C_3 has unique lifts. The three odd simples have free, identical left and right C_3 actions, so the odd bimodule fixes these central lifts pointwise. Relative-center equivariantization [48] and $H^2(C_2, \mathbb{C}^\times) = 0$ give six central invertible lifts upstairs. The unique order-three subgroup has trivial self-braiding, inherited from the neutral $\mathrm{Rep}(C_3)$, and is an embedded Tannakian subcategory. Its de-equivariantization has the Tambara–Yamagami rules on C_7. The induced C_3 symmetry must preserve its nondegenerate symmetric bicharacter $\chi(a, b) = \exp(2\pi i k a b/7)$, $k \neq 0$. Multiplication by two or four cannot do so: neither square is one modulo seven. This proves the contradiction without a pivotal or a fermionic-transparency assumption. □

3.8. **A dimension-twenty-eight adjoint criterion.**

Lemma 3.10. *A nonpointed integral fusion category of dimension twenty-eight has universal grading* C_2 *and adjoint fusion ring* $\mathrm{Rep}(D_{14})$*.*

Proof. Jordan–Larson's theorem [46, Theorem 1.1] applies with $p = 7, q = 2$ and $7 \nmid q + 1$. Write the category as $\mathcal{C}(G, \omega, H, \psi)$, where $|G| = 28$. Sylow theory gives $G = C_7 \rtimes P$, with $P = C_4$ or C_2^2; the action is trivial or inversion. The possible subgroup orders are $1, 2, 4, 7, 14, 28$. For orders one and seven, or a central subgroup of order two, the cyclic stabilizers give a pointed category. For a noncentral subgroup H of order two, its normal closure is $C_7 \rtimes H = D_{14}$, of index two; support gives a faithful C_2 grading. For $|H| = 14$, H is normal and support again gives such a grading. An abelian $H = C_{14}$ would give a pointed category; otherwise $H = D_{14}$, whose Schur multiplier is

trivial. For $H = G$ one obtains $\mathrm{Rep}(G)$; a nonabelian group of order twenty-eight has center C_2 and quotient D_{14}. In each of these nonpointed cases the neutral fusion ring is $\mathrm{Rep}(D_{14})$.

It remains to justify this ring identification for $|H| = 4$, and also the cocycle reduction used for a reflection complement above. Conjugate H to P. The positive-degree cohomology of C_7 is seven-primary, so the coprime Lyndon–Hochschild–Serre sequence has no mixed positive-degree terms. Since $\omega|_P = d\psi$, the associator class is represented by a P-invariant class on C_7. Choose a normalized μ_7 representative, average over P, and raise to the inverse of $|P|$ modulo seven. This gives a strictly invariant representative α. On $C_7 \rtimes P$ use

$$\omega_\alpha((n_1, p_1), (n_2, p_2), (n_3, p_3)) = \alpha(n_1, p_1 n_2, p_1 p_2 n_3).$$

It is identically one on P. The remaining 2-cocycle ψ is removed by its inflation along $G \to P$. Thus the category is $(\mathrm{Vec}^{\alpha}_{C_7})^P$ with strict permutation action. A trivial action is pointed. Otherwise its kernel C_2 acts strictly trivially and its characters grade the category, with neutral $(\mathrm{Vec}^{\alpha}_{C_7})^{C_2}$. This has the $\mathrm{Rep}(D_{14})$ fusion ring; the same argument with a reflection complement supplies the previous order-two case.

Finally this neutral ring is adjoint to itself. For its three dimension-two objects, indexed by nonzero elements modulo inversion, $X_i^2 = 1 + \varepsilon + X_{2i}$, and doubling cycles the three indices. The adjoint of the full category is contained in the displayed neutral component and contains its adjoint, so it equals that component. Consequently the universal grading is exactly C_2. □

Proposition 3.11. *There is no fusion category with dimension fifty-six, type* $(1^4, 2, 4^3)$ *and trivial universal grading. In particular* R8_2_694 *is generally excluded.*

Proof. Solvability and trivial universal grading force a last equivariantization. A C_7 equivariantization would supply seven invertibles, so the last group is C_2. Its descent $\mathcal{D}$ has dimension twenty-eight and is integral by the orbit/stabilizer dimension formula. It cannot be pointed, since a C_2 equivariantization of a pointed category has all dimensions at most two. Lemma 3.10 therefore applies. The outer involution preserves $\mathcal{D}_{\mathrm{ad}}$ and permutes its three dimension-two simples, fixing at least one. Its cyclic stabilizer has trivial projective multiplier, so its two characters produce two distinct dimension-two equivariant simples. The proposed category has only one. This is a contradiction. □

3.9. **A two-object orbit obstruction in dimension ninety-six.**

Proposition 3.12. *A fusion category of type* $(1^4, 2, 4, 6^2)$ *has a nontrivial universal grading. In particular* R8_2_404*,* R8_2_418*,* R8_2_604*,* R8_2_692 *have no complex fusion categorification.*

Proof. Suppose such a category has trivial universal grading. Its dimension is ninety-six, and its invertible group has order four. Solvability forces a last C_2 equivariantization $\mathcal{C} = \mathcal{D}^{C_2}$. Whichever nontrivial character occurs, it fixes the unique dimension-two and dimension-four simples. Thus the outer involution exchanges the two dimension-two simples of $\mathcal{D}$. Depending on whether it fixes or exchanges the dimension-six pair upstairs, $\mathcal{D}$ has type $(1^4, 2^2, 3^4)$ or $(1^4, 2^2, 6)$ and dimension forty-eight.

Its canonical adjoint is preserved by the outer involution. A proper adjoint's dimension therefore has the form $a + 4b + 9c + 36e$, where $a \in \{1, 2, 4\}$, $b \in \{0, 2\}$, c is even, and e is zero or one according to the displayed type. The proper divisors of forty-eight among these possibilities are only $1, 2, 4, 12$. A pointed adjoint of dimension at most four cannot contain the square of a dimension-three or dimension-six simple, since each invertible occurs there with coefficient at most one. An adjoint of dimension twelve consists of all four invertibles and both dimension-two simples. Every other grading component would then have dimension twelve using only dimension-three or dimension-six simples; a sum of their squared dimensions is divisible by nine, so this is impossible. Hence $\mathcal{D}_{\mathrm{ad}} = \mathcal{D}$.

A second prime descent gives $\mathcal{D} = \mathcal{E}^{C_2}$, $\mathrm{FPdim}(\mathcal{E}) = 24$. The complete cyclic-orbit dimension partitions are $(1^6, 3^2)$ and $(1^2, 2, 3^2)$. In the first case the dimension-two pair of $\mathcal{D}$ comes from two free orbits in the invertible group A of order six. The inner involution fixes exactly two elements of

A. For $A = C_6$ it is inversion, and for $A = S_3$ it is conjugation by a transposition. In either group the coefficients of the two induced objects in their own squares are respectively zero and one, by ordinary orbit addition. The output stabilizers are trivial, so associator or tensorator phases cannot change these coefficients. The outer involution cannot exchange this pair as required.

In the second case the unique dimension-two simple x of $\mathcal{E}$ is fixed by the inner involution, with $x^2 = 1 + g + x$. Its two equivariant lifts x_+, x_- are exchanged by the central character ε. Therefore their squares are equal, because $\varepsilon^2 = 1$ and ε is central. Each square has precisely one constituent from the dimension-two pair, by forgetting to the single x channel. The outer involution exchanges x_+, x_- but preserves their common square. Its unique dimension-two constituent would have to be fixed, which is impossible. Both dimension partitions and every initial character choice have therefore been excluded. The supplementary check binds the full four source tensors and exhausts these small dimension and orbit cases. □

3.10. **An order-four action on rank-one components.**

Proposition 3.13. *The ring R8_2_68 has no complex fusion categorification.*

Proof. Its exact type is $(1^4, 4^2, 6^2)$, its invertibles form C_4, its adjoint is the whole ring, and its order-two invertible fixes all four noninvertibles. Solvability in dimension 108 therefore forces $\mathcal{C} = \mathcal{D}^{C_2}$ with $\mathcal{D}$ integral of dimension 54 and type $(1^2, 2^4, 3^4)$. The outer involution exchanges both noninvertible dimension classes in pairs and fixes the invertibles. The possible proper dimensions of its invariant adjoint are $1, 2, 9, 18, 27$. Pointed dimensions one and two cannot contain the square of a dimension-three object; dimensions nine and twenty-seven would contain dimension-two objects, contradicting strong Frobenius divisibility [3]. Thus either $\mathcal{D}_{\mathrm{ad}} = \mathcal{D}$ or its adjoint $\mathcal{E}$ has type $(1^2, 2^4)$ and dimension eighteen.

In the first case prime descent gives $\mathcal{D} = \mathcal{F}^{C_2}$, with $\mathcal{F}$ of type $(1^9, 3^2)$. In the second case $U(\mathcal{E}) = 1$: a grading of order two would have a dimension-nine neutral containing a dimension-two simple, again contradicting strong Frobenius; order-three components of dimension six would each require both invertibles; higher grading orders cannot accommodate a dimension-two simple. Hence $\mathcal{E} = (\mathrm{Vec}_A^{\omega})^{C_2}$, $|A| = 9$, with inversion on A. Its unique nontrivial central invertible g is the canonical Tannakian lift. The relative-center C_3 action of the grading of $\mathcal{D}$ fixes this lift. Since $H^2(C_3, \mathbb{C}^{\times}) = 0$, its central equivariant lifts, together with the unit lifts, form an abelian group of order six. The unique order-two lift restricts to g and inherits its self-braiding $+1$; it therefore gives a central $\mathrm{Rep}(C_2)$ in $\mathcal{D}$. This explicitly verifies bosonicity, rather than inferring it from centrality alone. De-equivariantization again gives $\mathcal{F}$ of type $(1^9, 3^2)$: a dimension-three object fixed by g would descend to dimension $3/2$, which is not an algebraic integer.

In both cases the order-two central lift in $\mathcal{D}$ is unique: its kernel is trivial when $U(\mathcal{D}) = 1$, and has order three when $U(\mathcal{D}) = C_3$. The outer involution preserves this Tannakian subcategory. Combining the two equivariantizations gives $\mathcal{C} = \mathcal{F}^{C_4}$, because the intermediate symmetric subcategory has dimension four and its four characters form the census C_4. Let β generate this coherent action. Its square acts by inversion on $A = \mathrm{Inv}(\mathcal{F})$ and fixes the two dimension-three objects.

The adjoint of $\mathcal{F}$ is Vec_A^{ω}, with a C_3 grading of ranks $9, 1, 1$. Rank-one components force $[\omega] = 0$ by their module pentagon. The action of β exchanges these two components; otherwise the original $\mathcal{C}$ would inherit a nontrivial C_3 grading. For $A = C_9$ no automorphism squares to inversion. For $A = C_3^2$, its matrix B satisfies $B^2 = -I$, so its minimal polynomial is the irreducible $t^2 + 1$ over $\mathbb{F}_3$ and $\det(B) = 1$. Its tensorator class $\eta \in H^2(A, \mathbb{C}^{\times}) = C_3$ is fixed by B; order-four coherence gives $4\eta = 0$, hence $\eta = 0$. Such a tensor autoequivalence fixes every rank-one left-module class, since those classes are $H^2(A, \mathbb{C}^{\times})$. But the two nonneutral components have different left-module classes: an equivalence, tensored with the inverse of one invertible bimodule, would identify the rank-nine regular module with a rank-one component. Thus β cannot exchange them, a contradiction. The supplement checks every source coefficient, adjoint dimension partition and matrix calculation; the central lift and action tower use [48, 4, 3]. □

3.11. **Integral extensions from pointed Morita data.**

Lemma 3.14. *Suppose $\mathcal{B}$ is Morita equivalent to Vec_G^ω and every normal abelian subgroup of G has square order. Then every faithful finite-group graded extension of $\mathcal{B}$ is integral.*

Proof. An invertible Vec_G^ω bimodule is, as a left module, a transitive pointed module associated to a subgroup $N \leq G$ and a cochain trivializing $\omega|_N$. Its dual is pointed [7, Proposition 4.2]. In the twisted double-coset description [47, Theorems 4.1 and 5.1], a simple over NaN has FP dimension

$$[N : N \cap aNa^{-1}] \dim(\pi),$$

where π is an irreducible projective representation of the intersection. Pointedness forces N to be normal. The identity coset subcategory is ordinary $\mathrm{Rep}(N)$, even with the general associator and subgroup cochain [47, Theorem 4.1]; thus N is abelian as well. The left module has $[G : N]$ simples with equal normalized FP dimension. Since a graded component has total squared dimension $|G|$, each dimension is $\sqrt{|N|}$, an integer. Every component of an extension of Vec_G^ω is an invertible bimodule, so that extension is integral.

Morita equivalence transports the complete graded-extension data, including multiplication and associativity, and the resulting complete extensions are Morita equivalent [7, Corollary 3.13 and Theorem 1.3]. Integrality is preserved under this equivalence in the present setting. Indeed, the center of an integral category is integral because forgetting preserves FP dimensions. A Morita equivalent category $\mathcal{C}$ has the same center and integer total dimension. For a simple $W \in \mathcal{C}$, induction has dimension $\mathrm{FPdim}(\mathcal{C})\,\mathrm{FPdim}(W)$, an integer. Hence $\mathrm{FPdim}(W)$ is rational and an algebraic integer, and therefore is an integer. □

Proposition 3.15. *The rings R8_2_53 and R8_2_93 have no categorification over $\mathbb{C}$.*

Proof. Each source tensor has FP type $1^4, 4^2, (3\sqrt{2})^2$ and a faithful C_2 grading with integral neutral subring D of dimension 36. Its invertibles form $C_4 = \langle g \rangle$. They fix both noninvertibles X, Y, which are self-dual, and

$$X^2 = 1 + g + g^2 + g^3 + X + 2Y, \qquad Y^2 = 1 + g + g^2 + g^3 + 2X + Y.$$

In particular $D_{\mathrm{ad}} = D$. Any categorification $\mathcal{D}$ is solvable by the two-prime theorem [3, Theorem 1.6]. Its last prime step must be an equivariantization since its universal grading is trivial. The prime cannot be three because $\mathrm{Inv}(\mathcal{D}) = C_4$ contains no C_3. Thus $\mathcal{D} = \mathcal{E}^{C_2}$, with $\mathrm{FPdim}(\mathcal{E}) = 18$. The sign object is g^2 and fixes X, Y. The equivariantization orbit formula [6] makes $\mathcal{E}$ of type $1^2, 2^4$.

Let η be its nontrivial invertible. In every ZZ^* with $\mathrm{FPdim}(Z) = 2$, omitting η would leave dimension three after the unit, which cannot be a sum of two-dimensional simples. Therefore $ZZ^* = 1 + \eta + T$ with $\mathrm{FPdim}(T) = 2$. The adjoint subcategory contains both units and a two-dimensional simple. If proper, its integer dimension divides 18; the only possibilities for a subcategory of type $1^2, 2^n$ are 2 and 6. Dimension two omits T. At dimension six, the other universal-grading components contain only two-dimensional simples and cannot have total squared dimension six. Hence $\mathcal{E}_{\mathrm{ad}} = \mathcal{E}$. Solvability again forces $\mathcal{E} = \mathcal{P}^{C_2}$, where $\mathrm{FPdim}(\mathcal{P}) = 9$ and $\mathcal{P}$ is pointed [21, 22].

The canonical central $\mathrm{Rep}(C_2) \subset Z(\mathcal{E})$ has underlying object η. Its half-braiding is unique: two half-braidings on an invertible differ by a tensor character, and $\mathrm{Aut}_\otimes(\mathrm{Id}_\mathcal{E}) = \mathrm{Hom}(U(\mathcal{E}), \mathbb{C}^\times) = 1$. The outer C_2 action therefore preserves this central subcategory. Its equivariantization embeds as a symmetric subcategory of $Z(\mathcal{E}^{C_2})$: equivariant half-braidings act on every equivariant object, and the morphism conditions give full faithfulness. Forgetting equivariance followed by the fiber functor on $\mathrm{Rep}(C_2)$ gives a fiber functor, so this subcategory is $\mathrm{Rep}(H)$ for a group of order four. Its four underlying invertibles exhaust the C_4 in $\mathcal{D}$, forcing $H = C_4$. Central de-equivariantization [4, 22] now gives

$$\mathcal{D} \simeq (\mathrm{Vec}_A^\alpha)^{C_4}, \qquad |A| = 9.$$

The dimension-four noninvertibles force C_4 to act freely on all nonidentity elements of A. Thus $A = C_3^2$, since $\mathrm{Aut}(C_9)$ has order six. Its generator acts by an irreducible matrix J over $\mathbb{F}_3$ with $J^2 = -I$.

The pointed crossed product is Morita dual to this equivariantization. Consequently $\mathcal{D}$ is Morita equivalent to Vec_G^ω, where $G = C_3^2 \rtimes_J C_4$; the group extension splits by coprimality. The only normal abelian subgroups of G are 1 and C_3^2. Indeed, an element with nonzero C_4 exponent has commutators with C_3^2 filling $(J^k - I)C_3^2 = C_3^2$. A normal subgroup containing such an element contains C_3^2 and cannot be abelian. Within C_3^2, irreducibility of J leaves only 1 and C_3^2. Their orders are squares, so Lemma 3.14 makes every extension of $\mathcal{D}$ integral. The alleged census category would be a C_2 extension with two simples of dimension $3\sqrt{2}$, a contradiction. The exact verifier checks both full source tensors, the grading and every finite-group premise. □

Proposition 3.16. *The rings R8_2_191 and R8_2_299 have no categorification over* $\mathbb{C}$.

Proof. Their source tensors have FP type $1^4, 2^2, (\sqrt{6})^2$ and a faithful C_2 grading with integral neutral subring D of dimension 12. Its invertibles form V_4 and fix two simples X, Y, with

$$X^* = Y, \qquad X^2 = 2Y, \qquad Y^2 = 2X, \qquad XY = YX = \sum_{g \in V_4} g.$$

Thus a hypothetical neutral category $\mathcal{D}$ is faithfully C_3 graded over its pointed subcategory $\mathcal{B}$ on V_4, with rank-one nonneutral components. A rank-one module trivializes the pointed associator, so $\mathcal{B} \simeq \mathrm{Vec}_{V_4}$.

The ambient odd component $\mathcal{M}$ has two simples permuted transitively by V_4, with a stabilizer $H = C_2$; its stabilizer in source labels is $(0, 5)$ for R8_2_191 and $(0, 7)$ for R8_2_299. Its restriction $\mathcal{N}$ to $\mathcal{B}$ is therefore an indecomposable rank-two module. Since $H^2(C_2, \mathbb{C}^\times) = 0$, its cochain is trivial. A decomposition $V_4 = H \times K$ identifies its dual with

$$\mathcal{B}_\mathcal{N}^* \simeq \mathrm{Rep}(H) \boxtimes \mathrm{Vec}_K \simeq \mathrm{Vec}_{V_4},$$

including the trivial associator. The actual multiplication action of $\mathcal{D}$ on $\mathcal{M}$ is a coherent extension of this $\mathcal{B}$ module.

Apply [59, Theorem 1.3] to the *induced* module $\mathcal{L} = \mathcal{D} \boxtimes_\mathcal{B} \mathcal{N}$. It makes the dual $\mathcal{D}_\mathcal{L}^*$ a semidirect C_3 product with neutral component $\mathcal{B}_\mathcal{N}^*$, and hence pointed. The theorem concerns this induced-module dual; it does not assert that the dual of the original two-object $\mathcal{M}$ is pointed.

The original C_3 Brauer–Picard action on $\mathcal{B}$ is nontrivial: a rank-one component is not the regular rank-four module. Its class therefore has order three. Morita transport conjugates the action and preserves this order. The tensor-automorphism classes of Vec_{V_4} have kernel $H^2(V_4, \mathbb{C}^\times) = C_2$ over $\mathrm{Aut}(V_4) = S_3$ [22]. An order-three class must therefore act nontrivially on V_4. The pointed Morita group is $V_4 \rtimes C_3 = A_4$, so $\mathcal{D}$ is Morita equivalent to $\mathrm{Vec}_{A_4}^\omega$ for some ω. The only normal abelian subgroups of A_4 are 1 and V_4, of square orders 1 and 4. Lemma 3.14 makes every extension of this compatible neutral category integral, contradicting the odd FP dimensions $\sqrt{6}$. The source checker verifies both gradings, the complete neutral tensors, the odd module action and all ten subgroups of A_4. The neutral ring by itself is not claimed to have only pointed Morita models; the coherent odd module is essential. □

Proposition 3.17. *The rings R8_2_158, R8_2_186, R8_2_579 and R8_2_682 have no categorification over* $\mathbb{C}$.

Proof. All four source tensors have FP type $1^4, 2, 4, (2\sqrt{3})^2$ and a faithful C_2 grading. Their common integral neutral subring D, of dimension 24, has V_4 invertibles fixing Y, Z and

$$Y^* = Y, \quad Z^* = Z, \qquad Y^2 = \sum_{g \in V_4} g, \quad YZ = ZY = 2Z, \quad Z^2 = \sum_{g \in V_4} g + 2Y + 2Z.$$

Thus $D_{\mathrm{ad}} = D$. Solvability [3, Theorem 1.6] forces a neutral categorification to be $\mathcal{D} = \mathcal{E}^{C_2}$ with $\mathrm{FPdim}(\mathcal{E}) = 12$; its invertibles exclude a final C_3 equivariantization. The sign ϵ fixes Y, Z. The orbit formula makes $\mathcal{E}$ of type $1^4, 2^2$, with an outer involution swapping the two noninvertibles and swapping a pair of invertibles.

The finite rank-six check exhausts both groups $A = C_4, V_4$, their trivial and transitive actions on the two nonunits, both dualities, and every oriented nonunit product allowed by FP dimension and reciprocity. There are 100 candidate tables and six associative ones, of which exactly two admit the required outer involution. Their rings are $R_{3,A}$:

$$X^* = X', \qquad X^2 = 2X', \qquad (X')^2 = 2X, \qquad XX' = X'X = \sum_{g \in A} g,$$

with all units fixing X, X'. The transitive C_4 action in this exhaustion is the homomorphism $(0, 1, 0, 1)$ in cyclic labels. Jordan–Larson [46, Theorem 1.4] excludes $A = C_4$, since its cyclic prime-power factor occurs with odd multiplicity. Thus $\mathcal{E}$ is C_3 graded over $\mathcal{B} \simeq \mathrm{Vec}_{V_4}$; its rank-one components trivialize the pointed associator.

Let $\mathcal{M}$ be the two-object ambient odd module over $\mathcal{D}$. The central regular algebra $1 + \epsilon$ in $\mathcal{D}$ gives an actual de-equivariantized module $\mathcal{M}_{1+\epsilon}$ over $\mathcal{E}$. This construction uses the central algebra inside $\mathcal{D}$; it requires no half-braiding on ϵ in the ambient category. Its rank is one if ϵ swaps the source simples, and four if it fixes them: each fixed simple has two algebra-module structures, using $H^2(C_2, \mathbb{C}^\times) = 0$.

The six indecomposable Vec_{V_4} modules have ranks $4, 1, 1, 2, 2, 2$, by their stabilizers and cochains. The C_3 action sends the regular rank-four class to the X component, then to the X' component. These are distinct rank-one classes, so they form a three-cycle of total rank six. The three rank-two classes are either fixed or form another orbit of total rank six. Galindo's Clifford theorem [59, Proposition 4.1 and Theorem 1.1] now excludes rank one for $\mathcal{M}_{1+\epsilon}$, and makes its rank-four alternative two fixed rank-two submodules. The group C_3 cannot permute two actual summands; each is a coherent $\mathcal{E}$-module extension of a rank-two $\mathcal{B}$ module.

For one such module, the induced-module dual of [59, Theorem 1.3] is a semidirect C_3 product with neutral Vec_{V_4}. Its action class has order three, as in Proposition 3.16. Since the tensor-automorphism classes are $C_2 \times S_3$, this class is a pure order-three group action. The remaining action coherence lies in $H^2(C_3, \widehat{V_4}) = 0$ by coprimality [58, Proposition 3.2]. Hence the action is equivalent to the strict group action and the dual is the untwisted Vec_{A_4}. Its graded tensor section is part of Galindo's semidirect-product conclusion; no uncontrolled C_3 associator twist is discarded.

The equivariantization $\mathcal{D}$ is Morita equivalent to $\mathcal{E} \rtimes C_2$. Morita transport gives a C_2 extension $\mathcal{T}$ of Vec_{A_4} with the same center as $\mathcal{D}$. The center of Vec_{A_4} has exactly three Lagrangian subcategories, all in the orbit of the canonical $\mathrm{Rep}(A_4)$ [58, Proposition 7.6 and Section 8.3]. Consequently each has untwisted Vec_{A_4} as pointed Morita dual. The involution fixes at least one of these three. For its invariant pointed module $\mathcal{N}$, the dual of $\mathcal{T} \boxtimes_{\mathrm{Vec}_{A_4}} \mathcal{N}$ is a C_2 crossed product with pointed neutral Vec_{A_4} [59, Proposition 5.3]; every component is pointed. This step requires module-class invariance, rather than a coherent extension of $\mathcal{N}$. It gives a pointed Morita group

$$1 \longrightarrow A_4 \longrightarrow G \longrightarrow C_2 \longrightarrow 1.$$

Because A_4 is centerless with $\mathrm{Out}(A_4) = C_2$, G is S_4 or $A_4 \times C_2$. The latter has six identity-supported central characters for every associator. But $\mathrm{Inv}(Z(\mathcal{D})) \to \mathrm{Inv}(\mathcal{D})$ is injective: its kernel is $\mathrm{Hom}(U(\mathcal{D}), \mathbb{C}^\times) = 1$. The four source invertibles therefore exclude $A_4 \times C_2$. Thus $\mathcal{D}$ is Morita equivalent to $\mathrm{Vec}^{\omega}_{S_4}$.

The only normal abelian subgroups of S_4 are 1 and its V_4. Lemma 3.14 makes every extension of $\mathcal{D}$ integral, contradicting the two odd dimensions $2\sqrt{3}$. The exact checker validates all four source tensors, the small-ring exhaustion, module-rank alternatives and all thirty subgroups of S_4. □

4. Further general obstructions

The following arguments exclude nine further source tensors without a pivotality assumption. The proofs use a conductor bound, a Frobenius–Perron index, cyclic-module divisibility, and changes of Morita representative.

Proposition 4.1. *The rings R8_2_35, R8_2_38, R8_2_346 and R8_2_406 have no categorification over* $\mathbb{C}$.

Proof. Each tensor has four invertibles and four simples of FP dimension $d = 3 + \sqrt{10}$. By the published [25, Lemma 3.1], existence of any categorification implies existence of a pseudo-unitary Galois conjugate, whose canonical spherical structure has global dimension

$$D = 4(1 + d^2) = 80 + 24\sqrt{10}, \qquad DD' = 640.$$

Its simple-dimension and global-dimension fields both equal $\mathbb{Q}(\sqrt{10})$. The nontrivial quadratic character has conductor 40, so [64, Theorem 5.1] requires

$$\left|\frac{1}{D} - \frac{1}{D'}\right|^2 \leq \frac{1}{40}.$$

The left side is $9/160 > 1/40$, a contradiction. □

Proposition 4.2. *The ring R8_2_209 has no categorification over* $\mathbb{C}$.

Proof. Put $d = (5 + \sqrt{29})/2$. The source dimension vector is $(1, 1, 1, d, d-1, d+1, d, d)$, and its first three basis elements form a pointed subcategory of dimension 3. Hence

$$\mathrm{FPdim}(\mathcal{C}) = 5(1 + d^2) = \frac{145 + 25\sqrt{29}}{2}.$$

By [21, Proposition 8.15], its quotient by 3 must be an algebraic integer. Its field trace is $145/3$, which is not an integer. □

Lemma 4.3 (Cyclic-module divisibility)**.** *Let $H \cong C_p$, with p prime, be a group of invertibles in a fusion category over* $\mathbb{C}$. *Suppose that simple objects X, Y, Z satisfy $hX \cong X$, $Yh \cong Y$, and $hZ \cong Z \cong Zh$ for every $h \in H$. If* $\mathrm{FPdim}(Z)/p$ *is not an algebraic integer, then $p \mid N_{XY}^{Z}$.*

Proof. The rank-one module category generated by X trivializes the pointed associator on H. Thus $A = \bigoplus_{h \in H} h$ admits its regular connected separable algebra structure. Since $H^2(C_p, \mathbb{C}^\times) = 0$, the stabilizer cocycles can be trivialized to give X a left A-module structure and Y a right A-module structure. Hence $X \otimes Y$ is an A-bimodule.

The A-bimodules supported on the fixed simple Z correspond to projective representations of C_p^2: the projective-representation dimension equals the underlying multiplicity of Z. A scalar two-cocycle on C_p^2 has either zero or nondegenerate commutator pairing. Its twisted group algebra is accordingly $\mathbb{C}^{p^2}$ or $M_p(\mathbb{C})$. The first possibility would give a simple bimodule of FP dimension $\mathrm{FPdim}(Z)/\mathrm{FPdim}(A) = \mathrm{FPdim}(Z)/p$, contrary to algebraic integrality. In the second possibility every simple bimodule supported on Z has underlying object pZ. Decomposing the Z-isotypic bimodule summand of $X \otimes Y$ proves the assertion. This is the regular cyclic-algebra method of [65, Lemmas 6.1–6.2], with the stabilizer cocycle retained rather than assumed trivial. □

For later use, a free left C_p orbit whose right orbit agrees with it has cyclic left-right stabilizer of order p. Its block in A-*bimod*$(\mathcal{C})$ therefore consists of p simples of the original FP dimension. A label fixed on both sides instead contributes either p^2 simples of dimension $\mathrm{FPdim}(X)/p$ or one simple of dimension $\mathrm{FPdim}(X)$, by the preceding proof.

Proposition 4.4. *The ring R8_2_260 has no categorification over* $\mathbb{C}$.

Proof. Put $d = 3+2\sqrt{3}$. Its dimension vector is $(1,1,1,d,d,d,d,3)$, and the first three basis elements form C_3. They fix label 3 on both sides. Since $\mathrm{Norm}(d/3) = -1/3$, Lemma 4.3 requires $3 \mid N_{33}^3$. The source tensor has $N_{33}^3 = 2$. □

Proposition 4.5. *The rings R8_2_380 and R8_2_673 have no categorification over* $\mathbb{C}$.

Proof. Write $x = 4+\sqrt{6}$ and $y = 3+2\sqrt{6}$. Each ring has a subcategory $\mathcal{E}$ with the representation ring of the dihedral group D_{10} of order ten. It occupies labels $(0,3,4,5)$ in *R8_2_380* and $(0,1,6,7)$ in *R8_2_673*. Its nontrivial invertible g fixes the two dimension-two labels, so the regular algebra $A = 1+g$ exists. By [65, Lemma 6.2 and the proof of Theorem 6.3], A-*bimod*$(\mathcal{E})$ is pointed with group D_{10}.

The outside labels form two free g-orbits, of dimensions x and y, with the same left and right actions. They are $(1,2),(6,7)$ in the first tensor and $(2,5),(3,4)$ in the second. The cyclic stabilizer calculation gives two x-dimensional and two y-dimensional simples in the dual category $\mathcal{B} = A$-*bimod*$(\mathcal{C})$; all its other simples are the ten invertibles of D_{10}.

The rotation subgroup $H = C_5$ fixes every noninvertible on both sides, since each dimension fiber has only two elements. Moreover

$$\mathrm{Norm}(x/5) = \frac{2}{5}, \qquad \mathrm{Norm}(y/5) = -\frac{3}{5}.$$

For an x-dimensional simple X, Lemma 4.3 therefore makes every noninvertible multiplicity in XX^* divisible by 5. Write $x^2 = a+bx+cy$ for the resulting dimension equation, where a,b,c are nonnegative integers. The $\sqrt{6}$ coefficient gives $b+2c = 8$, contradicting $5 \mid b,c$. □

Proposition 4.6. *The ring R8_2_745 has no categorification over* $\mathbb{C}$.

Proof. Put $d = 1+\sqrt{2}$ and let g be label 1. The dimension vector is $(1,1,d,d,2,2,2d,2d)$. The subsets $\mathcal{D} = (0,1,2,3)$ and $\mathcal{E} = (0,1,4,5)$ are fusion subcategories, the latter with the representation ring of the dihedral group D_{10} of order 10. The element g exchanges labels $2,3$ and fixes labels $4,5,6,7$ on both sides. Thus $A = 1+g$ exists.

In $\mathcal{B} = A$-*bimod*$(\mathcal{C})$, the subcategory $\mathcal{E}' = A$-*bimod*$(\mathcal{E})$ is pointed with group D_{10}, by [65, Lemma 6.2 and the proof of Theorem 6.3]. The subcategory $\mathcal{D}' = A$-*bimod*$(\mathcal{D})$ has two invertibles and two simples of dimension d, by the cyclic-stabilizer calculation above. Each of labels $6,7$ contributes either four simples of dimension d or one of dimension $2d$. These are all the simples of $\mathcal{B}$, so it has precisely ten invertibles and 2, 6 or 10 simples of dimension d.

For a d-dimensional simple X, the dimension of XX^* is $a+bd$ with $a,b \in \mathbb{Z}_{\geq 0}$, where a is its invertible stabilizer order. The identity $d^2 = 1+2d$ forces $a = 1$. Thus the D_{10} action on these simples is free, and their number must be 10. It follows that $\mathcal{B}$ is generalized near-group, with no $2d$-dimensional simples remaining.

By [66, Proposition 3.6(1)], the adjoint subcategory $\mathcal{B}_{\mathrm{ad}}$ is nonpointed and is contained in every nonpointed fusion subcategory, hence in $\mathcal{D}'$. There is no proper nonpointed fusion subcategory of $\mathcal{D}'$: one noninvertible would give the forbidden rank-two rule $X^2 = 1+2X$ [23], while two noninvertibles force the other invertible by reciprocity. Therefore $\mathcal{B}_{\mathrm{ad}} = \mathcal{D}'$. Its order-two invertible group must be normal in D_{10}, since conjugation preserves the adjoint subcategory. No such normal subgroup exists. □

The source-bound verifiers in `evidence/theoretical_20261016/rank8` check all dimension characters and subcategory tensors used above, including the indicated left and right actions. The categorical implications are the preceding proofs and their explicitly cited dependencies.

5. An anti-invariant module obstruction

We give eight further exclusions that require no pivotal structure.

Lemma 5.1. *Let g be an invertible simple of order two in a fusion category, and let $x_1, \dots, x_f$ be all simples fixed by right tensoring by g, where $f > 0$. There is an integral f-dimensional representation T of the fusion ring such that, with $B_X = (N_{Xx_a}^{x_b})_{a,b}$,*

$$|(T_X)_{ab}| \le (B_X)_{ab}, \qquad (T_X)_{ab} \equiv (B_X)_{ab} \pmod 2, \qquad T_{X^*} = T_X^t.$$

In particular T_X is normal if $XX^* = X^*X$.

Proof. A fixed simple supplies a rank-one module category over the pointed category generated by g, trivializing its associator. Hence the regular connected separable algebra $A = \mathbf{1} + g$ exists; compare [65, Lemmas 6.1–6.2]. Each fixed simple has two simple right A-module structures m_a^+, m_a^-, whereas each free right orbit gives one simple module. Twisting the right action by the algebra automorphism that acts as -1 on g exchanges m_a^+ with m_a^- and fixes the free-orbit modules. It commutes with left tensoring by every object. Thus the anti-invariant subgroup has basis $v_a = [m_a^+] - [m_a^-]$. The two lift multiplicities have sum $(B_X)_{ab}$ and difference $(T_X)_{ab}$, proving the bound and parity. Rigidity gives the transpose relation, since the orthogonal difference vectors all have squared norm two. All fusion relations are preserved. □

Proposition 5.2. *The rings R8_2_318, R8_2_319, R8_2_320, R8_2_368, R8_2_370, R8_2_735, R8_2_758 and R8_2_760 have no categorification over $\mathbb{C}$.*

Proof. Set $q = z^2 - 6z + 1$, $r = z^2 + 2z - 1$, and $h = z^2 - 6z - 1$, and put

$$A = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}.$$

The following source identities use zero-based labels. Each indicated X commutes with X^*; the last column is a polynomial vanishing on X. The stated fixed labels are all the right-fixed simples of g.

suffix	g	fixed labels	X	B_X	polynomial
318	4	$(1,2,3,7)$	5	A	$(z^2-1)qr$
319	4	$(1,2,3,7)$	5	A	$(z^2-1)q$
320	6	$(1,2,5,7)$	3	A	$(z^2-1)(z^2+1)q$
368	3	$(4,5,6,7)$	1	C	$(z^2-1)(z^2+1)h$
370	3	$(4,5,6,7)$	1	B	$(z^2-1)(z^2+1)q$
735	1	$(4,5,6,7)$	2	B	$(z^2-1)qr$
758	1	$(4,5,6,7)$	2	C	$(z^2-1)h$
760	1	$(4,5,6,7)$	2	B	$(z^2-1)q$

Let $T = T_X$ from Lemma 5.1. Normality gives $\|T\|_F^2 = \sum_{j=1}^4 |\lambda_j|^2$. Integrality forces conjugate quadratic eigenvalues to occur with equal multiplicity. For bounds A, B, parity gives $\|T\|_F^2 \in \{12, 16\}$. The roots $3 \pm 2\sqrt{2}$ of q contribute 34 and so are absent. For suffixes $319, 320, 370, 760$, all remaining roots have modulus one, forcing $\|T\|_F^2 = 4$, a contradiction.

For $318, 735$, the additional roots $-1 \pm \sqrt{2}$ contribute 6. The possible squared norms are therefore $4, 8, 12$, according to the number of copies of this pair. Only two copies are possible, giving norm squared 12 and trace -4. But norm squared 12 forces the last diagonal entry to vanish, while the other diagonal entries are two signs and a zero. Their trace lies in $[-2, 2]$, a contradiction.

For bound C, the squared norm is in $\{12, 16, 20\}$. The roots $3 \pm \sqrt{10}$ of h contribute 38 and are absent. All remaining roots for $368, 758$ have modulus one, again giving the impossible squared norm 4. These arguments give unrestricted exclusions for $318, 735$, in addition to their pivotal obstructions. □

The portable verifier `evidence/theoretical_20261016/rank8/verify_anti_module_eight.py` checks the displayed source identities and spectral arithmetic directly; the proof does not rely on an enumeration of candidate matrices.

5.1. **Ordered exclusion ledger.** An excluded tensor is removed after the first verified certificate in the following order. For each row, "left" still counts rings subsequently recognized as positive, so this is an exclusion table rather than a table of outstanding questions.

Criterion	New $m=1$	New $m=2$	Left $m=1$	Left $m=2$
Near-group stabilizer	1	1	95	876
Rank-two closed subring	0	10	95	866
Zero-spectrum tensor condition	7	80	88	786
Proved rank 2–6 subring	3	121	85	665
Four proved rank-seven subrings	0	12	85	653
Commutative codegree d-number	23	462	62	191
Rank-eight noncommutative codegree	1	17	61	174
Integral formal-codegree divisibility	1	6	60	168
Published no-induction-matrix lemma	2	0	58	168
Involution-module obstruction	0	2	58	166
Field-automorphism obstruction	0	0	58	166
Fusion-subcategory dimension quotient	0	1	58	165
Primitive formal-codegree test	0	0	58	165
Cyclotomic eigenvalue test	0	0	58	165
Complete oriented signed quotient	3	71	55	94
Graded Brauer–Picard obstruction	1	2	54	92
Rank-one fiber-functor obstruction	0	2	54	90
Solvability and equivariantization	3	0	51	90
Prime descent and Hopf obstructions	0	9	51	81
Dimension-96 two-object orbit	0	4	51	77
Dimension-108 order-four action	0	1	51	76
Galois-invariant pointed associator	0	3	51	73
Normal abelian square-order kernels	0	8	51	65
Dimension-block conductor bound	0	4	51	61
Quadratic subcategory index	0	1	51	60
Cyclic-module divisibility (direct or Morita)	0	3	51	57
Morita adjoint normality	0	1	51	56
Conductor-32 signed Jacobi obstruction	0	1	51	55
Anti-invariant involution module	0	8	51	47
Algebraic $2D2$ graph-pair rigidity	0	4	51	43
Odd-involution Morita	0	1	51	42
Twisted half-braiding support	0	2	51	40
Quantum-group inverse Morita	0	1	51	39
Pivotal parity and inverse Morita	1	0	50	39
Fibonacci character parity	0	2	50	37
Conductor-seventeen indicator balance	0	5	50	32
Weighted Galois third indicator	0	1	50	31
Conductor-forty center trace	0	1	50	30
Cyclic induction phase multiplicity	1	0	49	30
Full center STS and fourth moment	1	0	48	30

The 78 tensors left by the exclusion ledger comprise 77 positive models and the open case $R8_2_24$. Every negative stage is a theorem applied to tensor data or an exact based-tensor transfer of a published and proved no-go. Proposition 3.1 and the integral-divisibility stage both have certified first applications already at multiplicity one; the latter also resolves two multiplicity-two cases not

excluded by the preceding criteria. These *first appearances* refer only to the two multiplicities of the complete rank-eight census. Nothing in this calculation establishes a multiplicity from which no new criterion is needed.

6. Exact positive models

Each standard family in this section admits a unitary complex realization. A pointed group ring of order eight is $K_0(\mathrm{Vec}_G)$ for the group reconstructed from its tensor. The five groups are C_8, $C_4 \times C_2$, C_2^3, D_8 and Q_8. The near-group ring $C_7 + 0$ is realized by a Tambara–Yamagami category [42]. If A is a rank-four unitary model from the complete catalogue of [14] and B is either Vec_{C_2} or Fibonacci, the Deligne product $A \boxtimes B$ is a unitary rank-eight model with multiplication tensor $N_A \otimes N_B$. Its multiplicity is at most two precisely for the displayed matches.

The three group-theoretical models $\mathcal{C}(F_5, 1, D_5, 1)$, $\mathcal{C}(S_4, 1, A_4, 1)$, and $\mathcal{C}(\mathrm{PSL}(2,7), 1, C_7 \rtimes C_3, 1)$ are untwisted group-theoretical fusion categories, hence admit unitary realizations. Their exact appendix tensors are matched against the census in all 512 positions, with a stored basis permutation for every match [20, Appendix C.3.2(3), (6), (11)]. Some admit a second subgroup construction, also recorded in the model catalogue.

Proposition 6.1 (Representation and quantum-group templates)**.** *All three character-ring templates* $\mathrm{Rep}(D_{20})$, $\mathrm{Rep}(D_{26})$, *and* $\mathrm{Rep}(\mathrm{Dic}_5)$ *have rank eight and multiplicity one. The standard unitary* $SU(2)_7$ *category, and the even subcategories of* $SU(2)_{14}$ *and* $SU(2)_{15}$*, also have rank eight and multiplicity one. Their generated multiplication tensors occur in the supplied census and are matched by full based-ring isomorphisms.*

Proof. For $SU(2)_k$, the simples are the labels $0, \dots, k$ and

$$N_{ab}^c = 1 \quad \Longleftrightarrow \quad |a-b| \le c \le \min(a+b, 2k-a-b), \qquad a+b-c \equiv 0 \pmod 2;$$

all other coefficients are zero. The even labels close under multiplication and yield eight objects for $k = 14, 15$, while $k = 7$ already has eight labels [29]. The explicit tensor generator implements this formula directly.

For the dihedral group of order $2n$, take $n = 10$ or 13. Its two-dimensional characters ρ_j obey $\rho_i\rho_j = \rho_{i+j} + \rho_{|i-j|}$, with $\rho_0 = 1 + \beta$ and $\rho_{n-j} = \rho_j$. For $n = 10$, $\rho_5 = \alpha + \alpha\beta$, the four linear characters form C_2^2, $\beta\rho_j = \rho_j$ and $\alpha\rho_j = \rho_{5-j}$. For $n = 13$ there are two linear characters, with $\beta\rho_j = \rho_j$. Thus their ranks are respectively $4+4$ and $2+6$. For Dic_5 of order twenty the four linear characters χ_i form C_4, with

$$\rho_0 = \chi_0 + \chi_2, \quad \rho_5 = \chi_1 + \chi_3, \quad \rho_{10-j} = \rho_j, \quad \chi_i\rho_j = \begin{cases} \rho_j & i \text{ even}, \\ \rho_{5-j} & i \text{ odd}; \end{cases}$$

the same product $\rho_i\rho_j = \rho_{i+j} + \rho_{|i-j|}$ completes the ring. These are ordinary finite-group character identities [43]. The supplied generator expands each product and compares all 8^3 ordered coefficients after unit-fixing relabelling. A match is a ring isomorphism and the corresponding representation category supplies the construction. □

Multiplicity	Ring ID	One exact unitary model
1	`R8_1_1`	`Rep(Dic5)`
1	`R8_1_3`	$\mathcal{C}(C_5 \rtimes C_4, \omega_2, C_2, 1)$
1	`R8_1_4`	`Vec(G), |G|=8`
1	`R8_1_5`	`Vec(C4) x Fib`
1	`R8_1_6`	Bosonic quotient of $PSU(2)_6 \boxtimes \mathcal{C}(C_4, q)$
1	`R8_1_7`	Central square-root extension of pseudo-$PSU(2)_6$
1	`R8_1_10`	`Vec(G), |G|=8`
1	`R8_1_11`	Central-twisted flip extension of pseudo-$PSU(2)_6$

Multiplicity	Ring ID	One exact unitary model
1	R8_1_12	Isotropic orbifold of $NG(C_{16}, 16)$
1	R8_1_13	Vec(G), |G|=8
1	R8_1_14	Central-twisted flip extension of $PSU(2)_6$
1	R8_1_15	$\mathrm{HI}(C_4)$
1	R8_1_16	Vec(G), |G|=8
1	R8_1_19	$(\mathrm{Fib} \boxtimes \mathrm{Fib}) \rtimes C_2$
1	R8_1_20	TY(C7)
1	R8_1_22	Diagonal parity in $\mathrm{TY}(C_3) \boxtimes \mathrm{Vec}_{C_4}$
1	R8_1_23	$\mathrm{Vec}_{Q_8}^{C_3}$
1	R8_1_24	C(F5,1,D5,1)
1	R8_1_25	$PSU(2)_6 \rtimes C_2$ with the flip action
1	R8_1_27	TY(C3) x Vec(C2)
1	R8_1_30	$\mathrm{TY}(C_6)^{C_2}$ with odd coherence -1
1	R8_1_32	pseudo-PSU(2)_6 x Vec(C2)
1	R8_1_34	$\mathrm{TY}(C_9)^{C_2}$ with odd coherence -1
1	R8_1_35	$\mathrm{TY}(C_3^2)^{C_2}$ with odd coherence -1
1	R8_1_36	TY(C3) crossed C2
1	R8_1_39	TY(C3) x Fib
1	R8_1_42	Pseudo-$PSU(2)_6 \rtimes C_2$ with the flip action
1	R8_1_43	pseudo-PSU(2)_6 x Fib
1	R8_1_53	Isotropic orbifold of $NG(C_4^2, 16)$
1	R8_1_59	Rep(D10) x Vec(C2)
1	R8_1_61	One-half SO(12)_2
1	R8_1_65	Vec(G), |G|=8
1	R8_1_66	Vec(C2xC2) x Fib
1	R8_1_67	PSU(2)_6 x Vec(C2)
1	R8_1_69	$\mathrm{HI}(C_2^2)$
1	R8_1_70	Fib x Vec(C2) x Fib
1	R8_1_71	PSU(2)_6 x Fib
1	R8_1_72	PSU(2)_7 x Vec(C2)
1	R8_1_73	PSU(2)_14
1	R8_1_78	SO(9)_2
1	R8_1_79	Z(Vec(S3))
1	R8_1_83	Integral part of the $m = 0$ triality gauging
1	R8_1_85	Neutral part of S3-gauged SO(8)_1
1	R8_1_91	Rep(D_{26})
1	R8_1_92	PSU(2)_15
1	R8_1_93	Rep(D10) x Fib
1	R8_1_94	PSU(2)_7 x Fib
1	R8_1_95	Fib x Fib x Fib
2	R8_2_37	Nonsplit C_2 extension of the unitary $2D2$ category
2	R8_2_39	Asaeda–Haagerup AH_4
2	R8_2_143	$\mathrm{Vec}_{C_{17}}^{C_4}$
2	R8_2_154	$\mathrm{TY}(C_2^2)^{S_3}$ with parity coherence
2	R8_2_174	Rep(A4) x Vec(C2)
2	R8_2_179	$\mathrm{Vec}_{C_2^4}^{C_5}$
2	R8_2_202	C(S4,1,A4,1)
2	R8_2_212	Rep(A4) x Fib
2	R8_2_237	$\mathrm{Rep}(A\Gamma\mathrm{L}(1,8))$
2	R8_2_239	C_3-algebra Morita dual of an S_3 extension of Vec_{C_7}
2	R8_2_241	$\mathcal{C}(\mathrm{PSL}(2,7), 1, C_7 \rtimes C_3, 1)$
2	R8_2_244	$\mathrm{Vec}_{C_4^2}^{C_3}$

Multiplicity	Ring ID	One exact unitary model
2	R8_2_289	$\mathrm{Vec}_{C_2^4}^{C_3}$
2	R8_2_290	Charge-conjugation defects of $PSU(3)_4$
2	R8_2_313	Diagonal bosonic quotient of the two pseudo-$PSU(2)_6$ categories
2	R8_2_477	even 2D2 x Vec(C2)
2	R8_2_484	Asaeda–Haagerup AH_5
2	R8_2_491	even 2D2 x Fib
2	R8_2_493	Haagerup H1 x Vec(C2)
2	R8_2_535	Dual of the $1+\rho$ Q-system in $HI(C_6)$
2	R8_2_557	Diagonal bosonic quotient of $PSU(2)_6 \boxtimes PSU(2)_6$
2	R8_2_575	Davydov–Runkel extension of $\mathrm{Rep}(H_{C:\sigma_1})$
2	R8_2_611	$\mathrm{TY}(C_2^2)^{S_3}$ with strict action
2	R8_2_666	Pointed centralizer in $Z(NG(C_4,4))$
2	R8_2_820	$NG(C_5,5)$ center factor, auxiliary C_3^2
2	R8_2_839	Haagerup H1 x Fib
2	R8_2_855	Even D_{14} category
2	R8_2_868	Dual Haagerup–Izumi C_7 category
2	R8_2_869	$NG(C_5,5)$ center factor, auxiliary C_9

Five further multiplicity-one models are imported from exact AnyonWiki tensor matches: R8_1_36 is the crossed product of $\mathrm{TY}(C_3)$ by the order-two inversion action; R8_1_61 is the rank-eight fusion subcategory called one-half $SO(12)_2$; R8_1_78 is $SO(9)_2$; R8_1_79 is $Z(\mathrm{Vec}_{S_3}) \simeq \mathrm{Rep}(D(S_3))$; and R8_1_85 is the neutral component of the universal $\mathbb{Z}_2$ grading after gauging the full S_3 symmetry of $SO(8)_1$ [1, 2]. For the crossed product, the standard bicharacter $\chi(a,b) = \exp(2\pi iab/3)$ and positive $\tau = 1/\sqrt{3}$ are invariant under simultaneous inversion. The induced tensor action squares to the identity, and the crossed product is unitary. Its independently constructed tensor is matched in all 512 positions. The other four are standard unitary categories or fusion subcategories thereof; their complete sourced tensors and unit-preserving permutations are included in the positive catalogue. Existence of these models proves general complex categorifiability. No negative conclusion is imported merely from an AnyonWiki flag; R8_1_26 is excluded by the theoretical proof in Proposition 3.3.

6.1. **Twenty-two further constructions from extensions and subfactors.** The following constructions are verified by generating their multiplication tensors directly and comparing all 512 coefficients. No associator search is needed.

For $G = C_4$ and $G = C_2^2$, write $X_g = g\rho$ and use

$$gh = g + h, \qquad gX_h = X_{g+h}, \qquad X_g h = X_{g-h}, \qquad X_g X_h = (g-h) + m\sum_{a\in G} X_a.$$

At $m = 1$, Izumi's unitary generalized Haagerup constructions [8, Section 9] give R8_1_15 and R8_1_69, respectively. At $m = 2$, these are the Asaeda–Haagerup rings AH_4 and AH_5, giving R8_2_39 and R8_2_484. The C_4 construction is [9, Theorem 4.9]; existence of the C_2^2 category follows from its Lemma 3.15, Theorem 3.17 and Section 4. These are unitary subfactor categories.

The ring R8_1_19 is $(\mathrm{Fib} \boxtimes \mathrm{Fib}) \rtimes C_2$, using the canonical action exchanging the two factors. The ring R8_1_22 is the fusion subcategory of $\mathrm{TY}(C_3) \boxtimes \mathrm{Vec}_{C_4}$ whose simple objects have matching parity: (a, z^{2j}), (m, z) and (m, z^3). Both operations preserve unitarity.

For R8_1_30, take the standard unitary $\mathrm{TY}(C_6)$ with bicharacter $\chi(a,b) = \exp(2\pi iab/6)$ and positive parameter $\tau = 1/\sqrt{6}$. Inversion fixes m, preserves every standard associator coefficient, and gives a tensor functor T with identity tensorator. Equip the C_2 action with coherence $T^2 \Rightarrow \mathrm{Id}$ equal to $+1$ on invertibles and -1 on m. This is a T-invariant monoidal natural automorphism, since the TY grading is $\mathbb{Z}/2$. In the equivariantization, the two structures on m have eigenvalues $\pm i$. The tensor decomposition is obtained by splitting the permutation action on each fusion space into its

two characters; in particular two odd eigenvalues contribute the additional minus sign. The result is the stated full tensor, with four invertibles, two objects of dimension two, and two of dimension $\sqrt{6}$.

Let $F_{20} = C_5 \rtimes C_4$ with

$$(a,b)(c,d) = (a + 2^b c,\ b + d), \qquad \omega(x,y,z) = (-1)^{b(x)\lfloor (b(y)+b(z))/4 \rfloor},$$

where the two coordinates are reduced modulo five and four, respectively, and representatives $0 \le b < 4$ are used in the cocycle. For $H = \langle (0,2) \rangle$, the restriction $\omega|_H$ is trivial. The unitary group-theoretical category $\mathcal{C}(F_{20}, \omega, H, 1)$ realizes *R*8_1_3. The supplied code constructs the eight simple homogeneous $\mathbb{C}[H]$-bimodules and their balanced tensor products. All phases are fourth roots of unity and are stored as integer exponents modulo four; the cocycle identity, bimodule coherence, balanced quotient, decomposition multiplicities, and full census tensor match are checked exactly.

The ring *R*8_1_6 has a bosonic quotient construction. Let $\mathcal{D} = PSU(2)_6$, with simples $1, f, \rho, f\rho$, where f is a fermion and $\rho^2 = 1 + \rho + f\rho$. Let $\mathcal{P} = \mathcal{C}(C_4, q)$ be the unitary pointed modular category with $q(z^a) = \exp(\pi i a^2/4)$. Then $b = f \boxtimes z^2$ in $\mathcal{B} = \mathcal{D} \boxtimes \mathcal{P}$ is an order-two boson: its self-braiding is $+1$. It generates a Tannakian $\mathrm{Rep}(C_2)$, whose regular algebra is $A = 1 \oplus b$. The central inclusion supplied by the braiding gives a fusion category $\mathcal{B}_A$ of all right A-modules, with tensor free-module functor F [22, Theorems 8.4.9 and 8.23.3]. The unitary crossed-product construction preserves its positive $*$-structure [41]. The boson is not transparent in $\mathcal{B}$; thus this uses all modules, rather than only local modules, and asserts no descended braiding.

Tensoring by b acts freely on the sixteen simples by $(V_j, z^a) \mapsto (V_{6-j}, z^{a+2})$, with $j \in \{0,2,4,6\}$. By adjunction every free module $F(V_j \boxtimes z^a)$ is simple, and two are isomorphic exactly when their labels belong to the same orbit. Dominance of F gives precisely eight simples. With $g = F(1 \boxtimes z)$ and $r = F(\rho \boxtimes 1)$, their rules are

$$g^4 = 1, \qquad gr = rg, \qquad r^2 = 1 + r + g^2 r.$$

The ordered model basis $(1, g, g^2, g^3, r, gr, g^2 r, g^3 r)$ maps to census labels $(0,1,7,2,5,3,6,4)$. The verifier reconstructs all coefficients from the truncated $SU(2)_6$ rules and the eight free orbits.

For *R*8_1_7, use either of the two unitary pseudo-$PSU(2)_6$ categories $\mathcal{D}$ of [30, Theorem 4.1]. Their basis $1, g, x, y$ obeys $g^2 = 1$, $gx = y$, $x^2 = y^2 = g + x + y$ and $xy = 1 + x + y$. The two source-bound AnyonWiki associator tables, labelled $(4,1,2,4,1,0,1)$ and $(4,1,2,4,2,0,1)$, are included in the supplement. Over $\mathbb{Q}(\zeta)$, $\zeta^4 + 1 = 0$, the verifier checks all 224 coefficients, 136 invertible associator blocks, unit and rigidity conditions and 2688 pentagons for each table.

Write $a(i) = gi$ and let $F_\ell^{ijk}[e,f]$ take left intermediate e to right intermediate f. In the two tables respectively, the scalars

$$(\lambda_1, \lambda_g, \lambda_x, \lambda_y) = (1, \zeta^2, \zeta, \zeta^3), \qquad (1, -\zeta^2, -\zeta^3, -\zeta)$$

define a half-braiding of g. The exact check is the following identity for each of the 24 fusion channels $N_{ij}^k = 1$:

$$\lambda_i \lambda_j F^{igj}_{a(k)}[a(i), a(j)] = \lambda_k F^{gij}_{a(k)}[a(i), k] F^{ijg}_{a(k)}[k, a(j)].$$

Thus $\widetilde{g} = (g, c)$ is an invertible object of $Z(\mathcal{D})$. Its self-braiding satisfies $\lambda_g^2 = -1$, so a fermionic condensation argument would be inappropriate. The equivalence between the unitary and ordinary centers permits a unitary representative of this lift [33, corrected Remark 3.1].

Instead take the trivial homomorphism $C_2 \to \mathrm{BrPic}(\mathcal{D})$ and modify its product equivalences by the normalized central 2-cocycle $t(1,1) = \widetilde{g}$, with other normalized values the unit. The only nontrivial cocycle identity compares $\widetilde{g}$ with itself, regardless of its order in $\mathrm{Inv}\, Z(\mathcal{D})$. The remaining extension obstruction belongs to $H^4(C_2, \mathbb{C}^\times) = 0$ [7, Theorem 1.3]; this is also the trivial-extension zesting of [32, Section 6.2].

This extension is unitary. Choose regular unitary $\mathcal{D}$-bimodule components and unitary multiplication equivalences from the unitary lift. Preliminary associativity intertwiners can be chosen unitary by polar decomposition, so their normalized pentagon defect is $U(1)$-valued. A normalized

3-cochain on C_2 is determined by u at $(1,1,1)$, and its coboundary at $(1,1,1,1)$ is u^2. A unit-modulus square root therefore removes the defect while keeping every coherence map unitary. The resulting regular-component extension is a unitary fusion category; this argument uses no general assertion that arbitrary extensions of unitary categories are unitarizable.

Its simples are (X,a), $a \in C_2$, with

$$(X,a)(Y,b) = (XYg^{ab}, a+b).$$

For $z=(1,1)$, this gives $z^2 = g$, $zx = xz$ and $x^2 = z^2 + x + z^2 x$. The model basis $(1,g,x,y,z,zg,zx,zy)$ maps to census labels $(0,7,3,4,1,2,6,5)$, and all 512 coefficients agree. The exact associator and central-lift replay is invoked by the finite-update verifier.

The same two rank-four bases have nontrivial flip extensions. For $\mathcal{D} = PSU(2)_6 = \mathrm{Ad}(A_7)$, a tensor autoequivalence τ exchanges ρ and $f\rho$ [34, Lemma 4.10]. Its square is tensor-isomorphic to the identity. The fusion rules imply $\mathrm{Aut}_\otimes(\mathrm{Id}_\mathcal{D}) = 1$, so this square isomorphism is unique after normalization and obeys the C_2 action coherence. Reutter's unitarization and automatic-unitarity theorems give a unitary representative of the entire action [35, Theorems 1–2].

For the second pseudo-$PSU(2)_6$ source table, the permutation $\tau = (x\ y)$ also gives a coherent tensor action, now checked explicitly. Its tensorator J^k_{ij} is 1 when an input is the unit. On the seventeen other admissible channels, ordered lexicographically, write $J = (\sqrt{2})^a \zeta^b$; the exponent vectors are

$$a = (-1,-1,0,-1,-1,-1,-1,-1,0,-1,0,-1,0,-1,0,0,0),$$
$$b = (29,15,10,17,1,5,5,11,0,1,16,15,4,5,0,0,0).$$

All 224 tensor-functor identities hold over $\mathbb{Q}(\zeta)$. The natural isomorphism $\eta : \tau^2 \Rightarrow \mathrm{Id}$ has scalars

$$\eta = \left(1, \frac{1+\zeta^2}{2}, -\frac{1+\zeta^2}{2}, -\frac{1+\zeta^2}{2}\right)$$

in the order $(1,g,x,y)$. The verifier checks all 24 equations $\eta_i \eta_j = \eta_k J^k_{ij} J^{\tau(k)}_{\tau(i),\tau(j)}$ and $\eta_{\tau(i)} = \eta_i$. This is therefore a coherent action, not just a symmetry of the ring. Transport to unitary functors preserves its coherence; the transported monoidal natural isomorphism is automatically unitary by [35, Theorems 1–2].

For either rank-four base, its nontrivial invertible has a unique central lift: the kernel of $\mathrm{Inv}\, Z(\mathcal{D}) \to \mathrm{Inv}\, \mathcal{D}$ is $\mathrm{Aut}_\otimes(\mathrm{Id}_\mathcal{D}) = 1$, and a lift was already constructed above. The flip fixes this lift. Thus one may use either the crossed product or its central 2-cocycle modification. The remaining C_2 obstruction is removed by the unitary phase correction described above. With v the nontrivial invertible, their products are

$$(X,a)(Y,b) = (X\tau^a(Y) v^{\epsilon ab}, a+b), \qquad \epsilon \in \{0,1\}.$$

For $PSU(2)_6$, these give $R8_1_25$ at $\epsilon = 0$ and $R8_1_14$ at $\epsilon = 1$; for pseudo-$PSU(2)_6$, they give $R8_1_42$ and $R8_1_11$, respectively. The exact tensorator and coherent-square replay is `software/verify_pseudo_psu_swap.py`.

The ring $R8_1_23$ is $\mathrm{Vec}^{C_3}_{Q_8}$ for the strict automorphism $i \mapsto j \mapsto k \mapsto i$ of the quaternion group. The fixed objects $1, -1$ each have three equivariant structures, giving six invertibles. The two free orbits $\{i,j,k\}$ and $\{-i,-j,-k\}$ give two simples of dimension three. For a free output orbit, the fusion coefficient is the number of pairs of input group elements with product equal to its chosen representative. For a fixed output, the product fibre is a sum of free C_3-orbits, each contributing its regular character, together with any fixed pairs carrying the product character. These rules independently generate every coefficient. In the order of the three characters over 1, the three over -1, and the two free orbits, the census permutation is $(0,5,6,7,1,2,3,4)$.

The inversion construction for $R8_1_30$ also applies to $A = C_9$ and $A = C_3^2$, producing $R8_1_34$ and $R8_1_35$. Use respectively $\chi(a,b) = \exp(2\pi i ab/9)$ and $\chi(a,b) = \exp(2\pi i(a_1 b_1 + a_2 b_2)/3)$, with

$\tau = 1/3$. Inversion preserves these nondegenerate bicharacters, hence the standard unitary TY associators. The parity coherence is again -1 on m and $+1$ on group elements. There are two equivariant simples above 0, four free pairs of nonzero group elements, and two structures with eigenvalues $\pm i$ above m. Their dimensions are $1, 1, 2, 2, 2, 2, 3, 3$; the same exact character projection generates the recorded tensors.

Finally take $A = \mathbb{F}_2^2$, $\chi(a, b) = (-1)^{a_1 b_2 + a_2 b_1}$ and $\tau = 1/2$. The whole group $S_3 = \mathrm{GL}_2(\mathbb{F}_2)$ preserves χ, so label permutations give a strict unitary action on the standard TY presentation. For $t = 0, 1$, modify its action coherence on m by

$$\alpha_t(g, h) = (-1)^{tp(g)p(h)},$$

where $p : S_3 \to \mathbb{F}_2$ is parity, leaving the group simples unchanged. This is monoidal by the TY grading, and the action pentagon is the elementary normalized 2-cocycle identity for $p(g)p(h)$. Both equivariantizations are therefore unitary. Their orbits $\{0\}$, $A\backslash\{0\}$ and $\{m\}$ have stabilizers S_3, C_2, S_3. Denote their simples by A_ρ, X_ϵ and M_ρ, where ρ ranges over $1, \mathrm{sign}, \mathrm{std}$ and $\epsilon = \pm$. Above m, the equivariance is $i^{tp(g)}\rho(g)$, so the product of two such factors contributes sign^t.

The fusion coefficients are ordinary character inner products. Multiplication by A_ρ is tensoring with ρ, restricted to C_2 on an X-object. The other rules are

$$\begin{aligned} X_\epsilon X_\delta &= A_{\mathrm{Ind}_{C_2}^{S_3}(\epsilon\delta)} + X_+ + X_-, \\ X_\epsilon M_\rho &= M_{(\mathrm{Ind}_{C_2}^{S_3}\epsilon)\rho}, \\ M_\rho M_\sigma &= A_{\rho\sigma\,\mathrm{sign}^t} + X_{\mathrm{Res}_{C_2}^{S_3}(\rho\sigma\,\mathrm{sign}^t)}, \end{aligned}$$

where subscripts extend additively to reducible representations. For XX, the zero fibre is the induced character and the nonzero fibre is a regular C_2 permutation representation. These formulas give $R8_2_611$ at $t = 0$ and $R8_2_154$ at $t = 1$, both of global dimension 48. The model dimensions in the order $(A_1, A_{\mathrm{sign}}, A_{\mathrm{std}}, X_+, X_-, M_1, M_{\mathrm{sign}}, M_{\mathrm{std}})$ are $(1, 1, 2, 3, 3, 2, 2, 4)$. All character arithmetic and tensor comparisons are exact.

One more quadratic model is $R8_1_53$. Yu–Zhang construct a unitary near-group category $NG(G, 16)$ for $G = C_4^2$ [36, Proposition 3.1]. Its bicharacter and quadratic refinement may be taken as

$$\chi(g, h) = i^{g_1 h_1 - g_2 h_2}, \qquad a(g) = \zeta_8^{3(g_1^2 - g_2^2)}.$$

For $H = 2G$, exact finite-group arithmetic gives $H = H^\perp$ and $a(h) = \chi(h, g_a)$ with $g_a = (1, 1) \notin H$. Izumi's general orbifold theorem applies to $H \subseteq H^\perp$ even when $a|_H$ is nontrivial [37, Theorem 12.5]; its crossed-product-factor construction remains unitary. With $A = G/H = C_2^2$, the resulting rules are

$$\alpha_g \alpha_h = \alpha_{g+h}, \qquad \alpha_g \sigma = \sigma \alpha_g, \qquad \sigma^2 = \alpha_{g_a} + \sum_{g \in A} \alpha_g \sigma.$$

There are four invertibles and four noninvertibles of dimension $(8 + 4\sqrt{5})/4 = 2 + \sqrt{5}$, and every noninvertible has a distinct dual. This uses the non-self-dual theorem, rather than its self-dual Lagrangian corollary. In the model order $(00, 01, 10, 11, \sigma, 01\sigma, 10\sigma, 11\sigma)$, the census permutation is $(0, 5, 6, 7, 1, 3, 4, 2)$. The portable verifier checks the subgroup hypotheses, the duality shift and all fusion coefficients; the cited theorem supplies categorical existence.

A cyclic version gives $R8_1_12$. Budinski constructs $NG(C_{16}, 16)$ [38, Theorem 3.1.1 and Table 3.1], with data

$$\chi(x, y) = \exp(2\pi i\, 3xy/16), \qquad a(x) = \exp(-\pi i\, 3x^2/16).$$

Although the table prints decimal arguments, the thesis's Section 3.3.2 gives an exact algebraic certification, with the defining-polynomial scope explicitly including $n = 16$. We use that exact existence statement as a citation dependency; no numerical reconstruction is counted here as a

category. For $H = 4C_{16}$, one has $H = H^{\perp}$ and $a(h) = \chi(h, 2)$. The same unitary orbifold theorem [37, Theorem 12.5] therefore gives

$$g^4 = 1, \qquad \sigma g = g^{-1}\sigma, \qquad \sigma^2 = g^2 + (1 + g + g^2 + g^3)\sigma.$$

In the order $(1, g, g^2, g^3, \sigma, g\sigma, g^2\sigma, g^3\sigma)$, the census permutation is $(0, 1, 7, 2, 3, 6, 4, 5)$. The four noninvertibles are paired by duality, distinguishing this ring from the standard C_4 Haagerup–Izumi ring. The finite verifier checks all subgroup hypotheses and the full tensor exactly.

The ring $R8_1_83$ has a triality-gauging model. The S_3 gauging of the three-fermion category exists by [2, Section 6.2]; the unobstructed gauging data form the usual $H^3(S_3, U(1))$ torsor, including the twist denoted $m = 0$ in [39, Section VI.D and Table IX]. This supplies an actual category. Starting from a pointed category, gauging produces a weakly group-theoretical fusion category, hence one admitting a unitary realization [40, Theorem 5.20]. Its eight integral simples, with dimensions $(1, 1, 2, 3, 3, 4, 4, 4)$, form a fusion subcategory of dimension 72.

The integer fusion formulas for this subcategory are independently checked against the full twelve-object S-matrix of [39, Eq. (6.53), $m = 0$]. Its entries lie in $\mathbb{Q}(\sqrt{2})$; using the unnormalized matrix, the verifier checks $S^2 = 144I$, all 1728 nonnegative integral Verlinde coefficients, and fusion closure of the eight integral simples. It then compares the entire tensor with the pinned census under the stored permutation. Category existence comes from the gauging construction, not merely from the feasible modular datum.

The table provides one construction per positive tensor. Some have two isomorphic presentations; `data/realized_models.json` records every matched constructor, exact model tensor, precise model source and source-to-model isomorphism. Labels such as "pseudo-$PSU(2)_6$" denote the conventional fusion-ring name, not a weaker existence assertion: the cited rank-four table supplies a unitary category [14]. The table exhausts the multiplicity-one positive list and records the 29 established multiplicity-two models. General categorifiability of $R8_2_24$ remains open.

6.2. **Pointed equivariantizations and an affine group.** Four further models are strict unitary equivariantizations of untwisted pointed categories. Their group automorphisms are

$$\begin{aligned}
&R8_2_244: \quad A = C_4^2, \quad (x, y) \mapsto (-y, x - y), \quad G = C_3;\\
&R8_2_289: \quad A = C_2^4, \quad (x, y, z, w) \mapsto (y, x + y, w, z + w), \quad G = C_3;\\
&R8_2_143: \quad A = C_{17}, \quad x \mapsto 4x, \quad G = C_4;\\
&R8_2_179: \quad A = C_2^4, \quad (x, y, z, w) \mapsto (w, x + w, y + w, z + w), \quad G = C_5.
\end{aligned}$$

In each case the displayed generator has the stated order and acts freely off zero. The zero orbit supplies the ordinary characters of G, and each nonzero orbit has trivial stabilizer. Tensor products are calculated by orbit addition and projection to stabilizer characters, giving the exact full source tensors. Strict permutation functors with identity tensorators supply coherent unitary actions.

The ring $R8_2_237$ is $K_0(\mathrm{Rep}(A\Gamma L(1, 8)))$. This group of order 168 has elements (v, a, t) acting on $\mathbb{F}_8$ as $x \mapsto v + ax^{2^t}$. Its simple dimensions are $(1^3, 3^2, 7^3)$. The three dimension-seven characters have the exact expression

$$\chi_s(v, a, t) = \zeta_3^{st} \sum_{\substack{r \in \mathbb{F}_8^{\times} \\ ar^{2^t - 1} = 1}} (-1)^{\mathrm{Tr}_{\mathbb{F}_8/\mathbb{F}_2}(v/r)}.$$

Together with the characters inflated from $\mathbb{F}_8^{\times} \rtimes C_3$, they give an exact character table in $\mathbb{Q}(\zeta_{21})$. Orthogonality and all tensor multiplicities are checked independently by finite group sums.

6.3. **Bosonic quotients and the D_{14} model.** The ring $R8_2_557$ is the central bosonic quotient of $PSU(2)_6 \boxtimes PSU(2)_6$ by its diagonal invertible. Each factor invertible is a fermion, so their product has self-braiding $+1$ and trivial pointed associator. It generates a genuine central $\mathrm{Rep}(C_2)$. All sixteen simples have free diagonal orbits; the eight free-module orbits give the stated ring.

For $R8_2_313$, use instead the two unitary pseudo-$PSU(2)_6$ categories of [30], with the exact sourced associators discussed above. Their central invertibles have self-braidings $+i$ and $-i$, and each has associator $F_{ggg} = -1$. The diagonal product therefore has both trivial self-braiding and trivial associator; its central $\mathrm{Rep}(C_2)$ gives the same eight-orbit quotient construction. For each rank-four factor the universal grading is trivial, so a central lift of an invertible is unique; its central square is the unit. This verifies the order-two condition as well as the associator and self-braiding conditions required for the categorical quotient.

The ring $R8_2_855$ is the even D_{14} category, or the local module category of the $SU(2)_{24}$ algebra $1 \oplus V_{24}$. It is unitary modular [49, 50]. Its basis is $f_0, f_2, f_4, f_6, f_8, f_{10}, P, Q$, with P, Q self-dual. Fold the bulk $SU(2)_{24}$ products by $f_{2t} = f_{2(12-t)}$ and $f_{12} = P + Q$. The remaining rules are

$$Pf_{2j} = \sum_{k=6-j}^{5} f_{2k} + \begin{cases} P & j \text{ even}, \\ Q & j \text{ odd}, \end{cases} \qquad Qf_{2j} \text{ interchanges } P, Q,$$

$$P^2 = P + f_0 + f_4 + f_8, \quad Q^2 = Q + f_0 + f_4 + f_8, \quad PQ = f_2 + f_6 + f_{10}.$$

All these constructions preserve the unitary structure.

6.4. **A Morita construction over C_7.** The ring $R8_2_239$ is a Morita dual of an S_3 extension of Vec_{C_7}. In $\mathrm{BrPic}(\mathrm{Vec}_{C_7}) = O(C_7 \oplus \widehat{C_7})$ take $r = \mathrm{diag}(2, 4)$ and let s exchange the two coordinates. They generate S_3. Its degree-three obstruction vanishes by coprime cohomology with seven-primary coefficients; the next obstruction lies in $H^4(S_3, \mathbb{C}^\times) = 0$. Thus the actual extension exists [7, Theorem 1.3]. Its C_3 restriction is pointed, on $F_{21} = C_7 \rtimes C_3$. The associator class there is inflated from C_3: the order-three action kills the $H^3(C_7, \mathbb{C}^\times)$ invariant part. Surjectivity of $H^3(S_3, \mathbb{C}^\times) \to H^3(C_3, \mathbb{C}^\times)$ permits an outer twist that makes this restriction untwisted.

Take the separable algebra $B = \bigoplus_{h \in C_3} h$ in this restriction. The category of B-bimodules is an actual Morita dual, whose neutral category is $\mathrm{Rep}(F_{21})$. Write its simples as α_i ($i \in \mathbb{Z}/3$), X, Y, and odd m_j ($j \in \mathbb{Z}/3$). The nontrivial rules, with all m_j self-dual, are

$$X^2 = X + 2Y, \quad Y^2 = 2X + Y, \quad XY = \sum_i \alpha_i + X + Y,$$

$$\alpha_i m_j = m_{i+j}, \quad m_j \alpha_i = m_{j-i}, \qquad Xm_j = Ym_j = m_j X = m_j Y = \sum_k m_k,$$

$$m_j m_k = \alpha_{j-k} + X + Y.$$

They follow from the S_3 bimodule action and reciprocity, and match all 512 source coefficients. The extension and its Morita dual are weakly group-theoretical, hence unitarizable [40, Theorem 5.20].

6.5. **Charge-conjugation defects of $PSU(3)_4$.** The ring $R8_2_290$ is a unitary C_2 crossed extension of the rank-five modular category $\mathcal{B} = PSU(3)_4$. Charge conjugation is an actual braided symmetry [51, Corollary 2.7]. The invertible group of $\mathcal{B}$ is trivial, so the degree-three gauging obstruction vanishes, and $H^4(C_2, U(1)) = 0$ removes the remaining obstruction. The unitary gauging theorem [2, Sections 3.2–3.4] therefore supplies an actual unitary extension. Its odd rank is three, the number of charge-conjugation-fixed neutral simples [52, Proposition 1.1].

For completeness, its tensor is forced by a small exact module calculation. In the neutral basis $(0,0), (3,0), (1,1), (0,3), (2,2)$ the nonselfdual generator X has regular characteristic polynomial $(t^2 + t + 2)(t^3 - 2t^2 - t + 1)$, computed by quantum Weyl alternants in $\mathbb{Q}(\zeta_{21})$. Its odd action A has the cubic factor as characteristic polynomial, since its Perron root has degree three. The commuting neutral action and the dual-transpose relation make A normal. Its three roots are real, so A is symmetric. Nonnegative integer entries, $\mathrm{Tr}(A) = 2$ and $\mathrm{Tr}(A^2) = 6$ give, up to relabelling,

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

The five neutral actions are I, A, A^2-I, A, A^2-A. Distinct odd dimensions force each odd object to be self-dual; reciprocity then determines every odd product. The independent constructor matches all 512 census coefficients.

6.6. **A self-dual Hopf extension in dimension thirty-two.** For $R8_2_575$ the neutral Hopf algebra is $H = H_{C:\sigma_1}$ of dimension sixteen from [45]. A symmetric nondegenerate Hopf pairing supplies the extra data of the Davydov–Runkel construction [53, Theorem 2.5]. Here is enough explicit information to fix the construction and its normalization. Write $h(p,q,j)$ for its basis, with p modulo four and q, j modulo two. Put $\zeta = \exp(2\pi i/16)$ and $i = \zeta^4$. The pairing is

$$P(h(p,q,j), h(r,s,k)) = \delta_{s,j}\delta_{k,q}\, i^{-3rp}\zeta^{jq}/4.$$

The supplementary Hopf presentation and checker verify nondegeneracy, symmetry and every pairing identity exactly in $\mathbb{Q}(\zeta)$. Davydov–Runkel use $H^\vee = (H^*)^{\mathrm{op,cop}}$; consequently their self-duality map is $\Gamma = S \circ P^{-1}$, rather than the unmodified inverse pairing. The right cointegral $\lambda(h(p,q,j)) = \delta_{j,0}/2$ satisfies $\Gamma(\lambda) = 4\Lambda$ and $\lambda(\Gamma(\lambda)) = 1$, where Λ is the normalized integral. The distinguished grouplike is one, $S^2 = 1$, and Γ is symmetric, so all conditions of the cited theorem hold. Its rigidity criterion [53, Proposition 3.25] gives an actual fusion category $\mathrm{Rep}(H) \oplus \mathrm{Vec}$. Its neutral ring is $\mathrm{Rep}(D_{16})$; for the odd object M of dimension four,

$$XM = MX = \mathrm{FPdim}(X)M, \qquad M^2 = \sum_{X \in \mathrm{Irr}(H)} \mathrm{FPdim}(X)X.$$

The dimension-32 solvability theorem and [40, Theorem 5.20] make this fusion category unitarizable. The complete model tensor, Hopf identities, and source comparison are separate exact checks.

6.7. **The dual Haagerup–Izumi C_7 category.** The ring $R8_2_868$ is the dual even part of the unitary Haagerup–Izumi C_7 subfactor. Its existence and center are supplied by [54, Theorem 5(d)]. Put $d = (7+\sqrt{53})/2$, so $d^2 = 1 + 7d$. Izumi's dual Q-system analysis [8, Proposition 7.4 and Theorem 7.5] gives rank eight, with dimensions $1, d, (d-1)^3, (d+1)^3$.

We verify the canonical-algebra hypothesis needed to use the full fusion rules in [54, Theorem 7]; matching modular data alone would not suffice. In its published double, the twist-one simples are 1, b of dimension d^2, and six A_i of dimension $1+d^2$. The nonzero C_{53} sector has no twist one, and the other twist-one labels are precisely those with zero second coordinate. For the actual dual category $\mathcal{D}$, its canonical Lagrangian algebra $L = I_{\mathcal{D}}(1)$ is connected, has trivial twist and dimension $7(1+d^2)$. Therefore

$$L = 1 + m_b b + \sum_{i=1}^{6} m_i A_i.$$

Writing $T = \sum_i m_i$, rational and irrational coefficients in the dimension identity give $m_b + T = 7$ and $m_b + 2T = 13$, hence $m_b = 1, T = 6$. Adjunction gives $\dim \mathrm{End}(L) = \mathrm{rank}(\mathcal{D}) = 8$, so $\sum_i m_i^2 = 6$. Thus every $m_i = 1$. The actual canonical algebra has exactly the underlying object required by the cited theorem; no assumption about a guessed algebra multiplication is made.

For a direct exact constructor, use basis $1, a, b_1, b_2, b_3, c_1, c_2, c_3$. Put $B = \sum_i b_i$, $C = \sum_i c_i$, $b_0 = c_0 = 0$, and $b_{-i} = b_i, c_{-i} = c_i$ for indices modulo seven. The rules are

$$\begin{aligned}
a^2 &= 1 + a + B + C, & ab_i &= a + B - b_i + C,\\
ac_i &= a + B + c_i + C, & b_i c_j &= a + B + C,\\
b_i b_j &= \delta_{ij}1 + (1-\delta_{ij})a + B - b_{i+j} - b_{i-j} + C,\\
c_i c_j &= \delta_{ij}1 + (1+\delta_{ij})a + B + c_{i+j} + c_{i-j} + C.
\end{aligned}$$

They match every source coefficient; all 4096 associativity equations and all dimension products in $\mathbb{Q}(d)$ are checked. Both even parts of the cited finite-depth C^* subfactor are unitary.

Model	Model-to-census permutation
$R8_2_143$	$(0,1,7,2,3,5,6,4)$
$R8_2_179$	$(0,1,4,3,2,5,6,7)$
$R8_2_237$	$(0,1,2,5,6,7,4,3)$
$R8_2_239$	$(0,1,2,3,4,5,7,6)$
$R8_2_244$	$(0,1,2,3,7,4,5,6)$
$R8_2_289$	$(0,1,2,3,4,5,6,7)$
$R8_2_290$	$(0,1,6,2,7,3,4,5)$
$R8_2_313$	$(0,7,1,2,3,4,6,5)$
$R8_2_557$	$(0,1,2,7,3,6,5,4)$
$R8_2_575$	$(0,5,1,2,3,7,4,6)$
$R8_2_855$	$(0,1,7,4,5,6,2,3)$
$R8_2_868$	$(0,3,1,2,4,5,7,6)$

The model bases are fixed by the independent constructors in the supplement. Each permutation is checked on the entire ordered tensor; it is not inferred from a character table or dimension vector.

7. A nonsplit extension of the 2D2 category

Proposition 7.1. *The ring $R8_2_37$ admits a unitary categorification.*

Proof. We use the factor realization of a graded extension of the known Haagerup–Izumi category for C_4 in [67, Section 3 and Theorem 4.1]. It supplies automorphisms α_j, β of a factor M, an endomorphism ρ, and unitaries $u, v \in M$ such that

$$\beta\alpha_j = \alpha_j\beta, \qquad \beta\rho = \mathrm{Ad}(u)\alpha_1\rho\beta, \qquad \beta^2 = \mathrm{Ad}(v)\alpha_{1+z}, \quad z \in \{0,2\}.$$

The α_j form an honest C_4 action. The character identities in [67, Lemma 3.2, (3.10) and (3.18)] give $\alpha_2(u) = -u$ and $\alpha_2(v) = -v$.

Form $P = M \rtimes_{\alpha_2} C_2$, with implementing involution λ. Grossman's construction [56, Section 4.1] extends the neutral generators by

$$\widetilde{\alpha}_j(\lambda) = (-1)^j\lambda, \qquad \widetilde{\rho}(\lambda) = \lambda.$$

They generate the unitary $2D2$ category with simple sectors $(1, g, r, gr)$, where $g = [\widetilde{\alpha}_1]$, $r = [\widetilde{\rho}]$, and

$$g^2 = 1, \qquad gr = rg, \qquad r^2 = 1 + 2r + 2gr.$$

This construction depends only on the algebra $1 + \alpha_2$ in the neutral category, so it applies to the neutral realization just chosen.

Extend β to $b \in \mathrm{Aut}(P)$ by $b(\lambda) = \lambda$. The factor identities extend to

$$b\widetilde{\rho} = \mathrm{Ad}(u)\widetilde{\alpha}_1\widetilde{\rho}b, \qquad b^2 = \mathrm{Ad}(v)\widetilde{\alpha}_{1+z}.$$

Indeed, they hold on M by construction. On λ the two right-hand sides are respectively $u(-\lambda)u^* = \lambda$ and $v(-\lambda)v^* = \lambda$. The anticommutation of u, v with λ is essential here.

Put $a = [b]$. These identities imply

$$a^2 = g, \qquad a^4 = 1, \qquad ara^{-1} = gr = a^2r, \qquad ra = a^{-1}r.$$

The full tensor category generated by b and $\widetilde{\rho}$ has exactly the eight simple sectors

$$(1, a, a^2, a^3, r, ar, a^2r, a^3r).$$

The four invertibles are distinct because $a^2 = g \neq 1$. If $a^ir \cong a^jr$, Frobenius reciprocity puts a^{j-i} in $rr^* = 1 + 2r + 2gr$, so $i = j$. The noninvertibles have dimension $2 + \sqrt{5}$. The displayed relations show that these eight sectors are closed under products and duals, and their associativity and unitarity are inherited from $\mathrm{End}_0(P)$.

For $i, j \in \mathbb{Z}/4$, their multiplication is

$$a^i a^j = a^{i+j}, \qquad a^i(a^j r) = a^{i+j} r,$$
$$(a^i r)a^j = a^{i-j} r, \quad (a^i r)(a^j r) = a^{i-j} + 2a^{i-j} r + 2a^{i-j+2} r.$$

The displayed basis maps to zero-based census labels $(0, 1, 7, 2, 5, 3, 6, 4)$. The exact verifier compares all 512 structure constants under this bijection. Thus it identifies an already constructed category with R8_2_37; no coherent order-two autoequivalence is assumed. □

7.1. **Odd isotypic multiplicities and involution algebras.**

Lemma 7.2. *Let X be a self-dual simple in a pivotal fusion category over $\mathbb{C}$. If a self-dual simple S occurs with odd multiplicity in $X^{\otimes n}$, then*

$$\nu_2(S) = \nu_2(X)^n.$$

More generally, in a palindromic tensor word of self-dual simples, every self-dual constituent of odd multiplicity has indicator equal to the product of the indicators of the factors.

Proof. Use pivotal coherence to suppress associators and double-dual identifications. Choose a self-duality $f : X \to X^*$ whose pivotal transpose is $t(f) = \epsilon f$, where $\epsilon = \nu_2(X) \in \{1, -1\}$. The reversed dual word has the same identical factors, so

$$F = f^{\otimes n} : X^{\otimes n} \longrightarrow (X^{\otimes n})^*, \qquad t(F) = \epsilon^n F.$$

This construction needs no braiding: reversal changes neither the object word nor the sequence of component morphisms. On the S-isotypic block the nondegenerate form is $f_S \otimes B$, with $t(f_S) = \nu_2(S) f_S$ and B an invertible bilinear-form matrix on the complex multiplicity space. Hence

$$B^{\mathsf{T}} = \frac{\epsilon^n}{\nu_2(S)} B.$$

An invertible alternating matrix has even size, which proves the assertion for an odd multiplicity. Semisimplicity ensures that the form on a self-dual isotypic block is nondegenerate. The same calculation works for a palindrome, using the same chosen self-duality at each pair of matching positions. □

Proposition 7.3. *The ring R8_1_49 has no categorification over $\mathbb{C}$.*

Proof. Its invertibles are $G = \{0, 3, 4, 5\} \cong C_2^2$, and its four noninvertibles have dimension $d = 2 + \sqrt{5}$, with $d^2 = 1 + 4d$. The published two-dimension theorem [25, Lemma 3.1] therefore replaces any hypothetical categorification by a pseudo-unitary Galois conjugate with the same fusion rules. Equip this conjugate with its canonical positive spherical structure. This supplies the pivotal structure without assuming one on the initial category.

Set $u = 3$, $g = 5$ and $\rho = 6$. The source tensor gives

$$\rho^* = \rho, \qquad \rho^2 = 1 + \sum_{h \in G} h\rho, \qquad \rho^3 = (1, 4, 4, 1, 1, 1, 5, 4)$$

in source-label order. The unit and every invertible occur once in the cube. Lemma 7.2 first gives $\nu_2(\rho) = 1$ and then $\nu_2(h) = 1$ for every $h \in G$. In the positive spherical structure of a pointed C_2 category, the indicator of its nontrivial object is the sign of its associator class. Thus the restriction to $H = \{1, u\}$ is trivial, and its regular algebra $A = 1 \oplus u$ actually exists and is separable. No central lift of u is required.

Consider $\mathcal{B} = {}_A\mathcal{C}_A$. The source relation $u\rho u = g\rho$ gives two pointed H-by-H orbits of size two, $\{0, 3\}$ and $\{4, 5\}$, and the free noninvertible orbit $\{1, 2, 6, 7\}$. Each pointed orbit has stabilizer C_2 and contributes two simple bimodules of dimension one, since every complex projective representation of C_2 can be linearized. The free orbit contributes the single simple $A\rho A$ of bimodule dimension

$4d/2 = 2d$. Thus $\mathcal{B}$ has four invertibles, forming $G' \cong C_4$ or C_2^2, and one noninvertible Z. Reciprocity and dimensions force

$$Z^2 = \sum_{h \in G'} h + 8Z.$$

For $G' = C_2^2$ this contradicts [25, Theorem 1.1]; for $G' = C_4$ it contradicts the general cyclic-four criterion [16, Theorem 6.3] at $k = 2$. The source-bound verifier `verify_r49_fs_morita.py` checks the cube, all double orbits, and the same six-branch terminal certificate used below. Hence the original ring cannot be categorified. □

7.2. **An inverse-Morita obstruction for the split involution.**

Proposition 7.4. *The ring R8_2_170 admits no fusion-category categorification over* $\mathbb{C}$.

Proof. Its source tensor has a faithful C_2 grading with neutral labels $(0, 5, 6, 7)$. Put $g = 5$, $r = 6$, and $u = 3$. Its four invertibles form C_2^2, with u odd, and $uru = gr$. The subgroup $H = \{1, u\}$ maps isomorphically to the grading group. If a category with this tensor exists, multiply its associator by the inflation of the inverse of its pointed associator class on H. This is a normalized grading three-cocycle, so it preserves the pentagon and every fusion coefficient. After this twist $A = 1 \oplus u$ is a connected separable algebra. This step does not assume that A exists before twisting.

Consider the fusion category $\mathcal{B} = {}_A\mathcal{C}_A$. For a free bimodule $F(Y) = A \otimes Y \otimes A$, adjunction gives

$$\dim \mathrm{End}_{\mathcal{B}}(F(Y)) = \dim \mathrm{Hom}_{\mathcal{C}}(Y, A \otimes Y \otimes A) = |\{(h, k) \in H^2 : hYk \cong Y\}|.$$

The two pointed double orbits are $\{0, 3\}$ and $\{4, 5\}$, each of stabilizer size two. Each free bimodule therefore splits into two distinct simple bimodules, once each: its semisimple endomorphism algebra has dimension two. The underlying support of either summand is stable under H on both sides, so each contains both orbit objects once. Dividing its underlying FP dimension by $\mathrm{FPdim}(A) = 2$ shows that it is invertible. Thus these orbits give four invertibles in $\mathcal{B}$.

The four noninvertibles form one double orbit $\{1, 2, 6, 7\}$ with trivial stabilizer. Hence $F(r)$ is a single simple Z of dimension $2(2 + \sqrt{5}) = 4 + 2\sqrt{5}$. Every simple bimodule occurs in a free one, so this is the complete list. The four invertibles form a group G equal to C_4 or C_2^2. The unique noninvertible is self-dual and fixed by G; reciprocity and dimensions give

$$Z^2 = \sum_{h \in G} h + 8Z.$$

For $G = C_2^2$, this contradicts [25, Theorem 1.1]. For $G = C_4$, it contradicts the general cyclic-four central-support criterion of [16, Theorem 6.3], at parameter $k = 2$ and moment height six. The archived optimizer-free replay checks all six branches: two force a fusion coefficient $1/2$, and four have exact separating vectors with nonnegative $A^{\mathsf{T}}y$ and respective values $b^{\mathsf{T}}y = -800, -220, -320, -80$. This terminal criterion applies to arbitrary complex categorifications through its proved Galois pseudo-unitarity reduction. Both possible groups are excluded. No classification of the neutral $2D2$ category is used. □

8. The cyclic order-six Haagerup–Izumi algebra dual

Proposition 8.1. *The dual of the algebra* $1 + \rho$ *in a known unitary Haagerup–Izumi category for* C_6 *has Grothendieck ring R8_2_535.*

Proof. Set $d = 3 + \sqrt{10}$ and $D = 6(1 + d^2) = 120 + 36\sqrt{10}$. The category and its algebra $1 + \rho$ exist by [68, Example 4.5 and condition (2.24)]. Denote the unitary dual by $\mathcal{E}$. Izumi's description [8, Proposition 7.4 and Theorem 7.5] gives its simple objects and dimensions as

	1	b	r	rb	e_1	e_2	s_1	s_2
FPdim	1	1	d	d	$d+1$	$d+1$	$d-1$	$d-1.$

Here $b = \beta_3$, $e_j = \eta_j$, and s_1, s_2 are the two σ objects. Its canonical module has seven simples $(i_0, \dots, i_5, \tau)$ with relative dimensions $(1, \dots, 1, d-1)$.

Let P be the cyclic permutation matrix on the six i-labels. The upper-left six-by-six blocks of the eight action matrices are

$$I, \quad P^3, \quad I, \quad P^3, \quad P + P^{-1}, \quad P^2 + P^{-2}, \quad 0, \quad 0.$$

The matrices of $1, b$ have zero off-diagonal τ entries and τ diagonal 1. Each of the other six has every off-diagonal τ entry equal to 1, with respective τ diagonals $(5, 5, 6, 6, 4, 4)$. These formulas follow from the decompositions in [8, Proposition 7.4]; the last entries also follow from the relative FP dimension vector.

These matrices generate a commutative algebra of dimension five. Their three independent dependencies are

$$M_1 - M_b - M_r + M_{rb} = 0, \qquad M_1 - M_r + M_{s_1} = 0, \qquad M_1 - M_r + M_{s_2} = 0.$$

The complex fusion algebra is semisimple. A noncommutative simple block would have dimension at least four and would have to lie in the three-dimensional kernel of this representation. Hence the entire fusion algebra of $\mathcal{E}$ is commutative.

Write $d' = 3 - \sqrt{10}$ and $D' = 120 - 36\sqrt{10}$. Fourier diagonalization of the module yields five characters:

	1	b	r	rb	e_1	e_2	s_1	s_2	f
χ_{FP}	1	1	d	d	$d+1$	$d+1$	$d-1$	$d-1$	D
χ'_{FP}	1	1	d'	d'	$d'+1$	$d'+1$	$d'-1$	$d'-1$	D'
u	1	-1	1	-1	1	-1	0	0	6
v	1	1	1	1	-1	-1	0	0	6
w	1	-1	1	-1	-2	2	0	0	12.

There are three further one-dimensional characters. Orthogonality to the displayed rows forces each to have the form

$$(1, \epsilon, -1, -\epsilon, 0, 0, s, t), \qquad \epsilon \in \{1, -1\}, \quad s + t = 1 + \epsilon. \tag{1}$$

We determine their codegrees using the center of the known category, which is Morita invariant. The complete published inventory [68, Lemma 3.5 and Theorem 3.9] is

simple dimension	1	d^2	$1+3d$	$2+6d$	$3d$	$6d$
multiplicity	2	2	4	16	4	18.

The dimensions squared sum to D^2. Formal-codegree induction [24] associates to each character a simple constituent of $I(1)$ of dimension D/f. Thus the possible codegrees are

$$D, \quad D', \quad 12, \quad 6, \quad 4\sqrt{10}, \quad 2\sqrt{10}.$$

Formal codegrees are totally positive: in this commutative case each is the squared norm of a character and every Galois conjugate is another formal codegree. This eliminates the last two values. For a missing character, the Perron bound gives

$$f = 4 + |s|^2 + |t|^2 \le 4 + 2(d-1)^2 = 8 + 8d < D.$$

So no missing codegree is D. The five known rows form a Galois-stable set; a missing codegree D' would have a missing Galois conjugate of codegree D, also impossible. Finally $\sum_\chi f_\chi^{-1} = 1$ and $D^{-1} + (D')^{-1} = 1/6$ force the three missing codegrees to be $6, 6, 12$.

The object b permutes s_1, s_2. If it fixed both, a missing character with $\epsilon = -1$ would have $s = t = 0$ and codegree 4. All three missing rows would then have $\epsilon = 1$, but the vectors in (1) with $\epsilon = 1$ span a space of dimension at most two. Therefore b exchanges s_1 and s_2, and $t = \epsilon s$. There is exactly one missing row with $\epsilon = 1$, namely $(1, 1, -1, -1, 0, 0, 1, 1)$ of codegree 6. The two remaining rows have $\epsilon = -1$, one of each codegree $6, 12$. Complex conjugation preserves the missing set, ϵ, and

the codegree, so each of these two rows is real. Their norms and orthogonality determine, up to exchanging s_1, s_2, the full list

$$\begin{array}{c|cccccccc|c} x & 1 & 1 & -1 & -1 & 0 & 0 & 1 & 1 & 6 \\ y & 1 & -1 & -1 & 1 & 0 & 0 & 1 & -1 & 6 \\ z & 1 & -1 & -1 & 1 & 0 & 0 & -2 & 2 & 12. \end{array}$$

All characters are real, hence all simple objects are self-dual. Character inversion now determines every structure constant:

$$N_{ij}^{k} = \sum_{\chi} \frac{\chi(i)\chi(j)\chi(k)}{f_\chi}.$$

The resulting tensor is identified with $R8_2_535$ by the basis map $(0, 1, 3, 6, 4, 5, 2, 7)$. The verifier reconstructs all 512 coefficients exactly and checks the canonical module equations. No multiplicity bound or completeness claim about the census is used to determine the unknown dual ring. □

9. Algebraic rigidity of the 2D2 module graph

The next argument gives four general complex obstructions. It does not assert that every categorification of the rank-four $2D2$ ring is unitary: the specific module graph is part of the hypothesis.

Lemma 9.1. *Let $\mathcal{D}$ be a pseudo-unitary fusion category and $\mathcal{M}$ an indecomposable module category with a simple object X. Suppose its rooted principal graph is $2D2$, with modulus $\delta = \sqrt{3+\sqrt{5}}$. Then the associated algebraic shaded planar algebra is isomorphic to the known unitary $2D2$ planar algebra. In particular its dual even category has rank six.*

Proof. We first justify the algebraic planar-algebra construction. Schaumann's module-trace theorem [69, Proposition 5.8] supplies positive FP trace weights on $\mathcal{M}$. His Theorem 4.5 constructs natural double-adjoint isomorphisms for module functors between any two traced module categories, compatible with composition. Applied to $\mathcal{D}$ and $\mathcal{M}$, it equips the full Morita linking category with a pivotal structure. Corollary 4.6 and Proposition 5.10(iv) give the compatible spherical dual. Normalize the module trace so that the two loop values of X are δ.

For completeness, the graph embedding in this setting requires no star operation. Act on a column of the linking category, and decompose every object into simples. A morphism acts on the resulting path spaces as a direct sum of matrices, faithfully. Choose dual bases for the nondegenerate trace pairings. Composition is matrix multiplication, while evaluation and coevaluation are the weighted graph cups and caps. Thus all planar operations are preserved. This is the algebraic pivotal construction of [44, Theorems 3.45 and 3.70], with the cup and cap calculation of its Theorem 3.75 performed with dual bases. Its dagger-specific Theorem 3.77 is not needed. Denote the resulting planar algebra by $\mathcal{P}$ and its embedding into the graph planar algebra by $\mathcal{P} \subseteq \mathcal{G}$.

The graph is two-supertransitive. Its depth-three Jones–Wenzl corner is $f_3\mathcal{P}_{3,+}f_3 = \mathbb{C}p \oplus \mathbb{C}q$, where p, q are orthogonal primitive idempotents and $p + q = f_3$. The two depth-three PF weights coincide, so $T = p - q$ has trace zero and $T^2 = f_3$. The bilinear trace pairing is nondegenerate, and

$$\mathcal{P}_{3,+} = \mathrm{TL}_{3,+} \oplus \mathbb{C}T$$

is a trace-orthogonal decomposition. Pairing a cap of T with every element of $\mathcal{P}_{2,+}$ shows that it vanishes, since the corresponding cups lie in $\mathrm{TL}_{3,+}$. Rotation preserves the one-dimensional orthogonal complement, so T is an uncappable rotational eigenvector, whose eigenvalue is a cube root of unity. We verify the exclusion of the two primitive roots directly below; this also independently replays [70, Lemma 3.2] without a self-adjointness hypothesis.

We describe the exact complex calculation needed at this point. Label the graph vertices $0, \ldots, 7$, with edges

$$01,\ 12,\ 23,\ 24,\ 35,\ 45,\ 56,\ 67,$$

rooted at 0 and with even vertices $0, 2, 5, 7$. Put $s = \sqrt{5}$, $\varphi = (1+s)/2$, and assign weights

$$(w_0, \dots, w_7) = (1, 1, 2+s, \varphi, \varphi, 2+s, 1, 1).$$

These are the PF weights with odd entries divided by δ. For a path $p = (p_0, p_1, p_2, p_3)$, set $A_p = \sqrt{w_{p_1} w_{p_2}}$. The rectangular lopsided matrix is $T_{\mathrm{rect}} = A T_{\mathrm{sph}} A^{-1}$. Thus the coefficient at a loop $(v_0, \dots, v_5)$ is multiplied by $\sqrt{w_{v_1} w_{v_2} / (w_{v_5} w_{v_4})}$. In these coordinates a Temperley–Lieb projection has entries $w_{p_i} / \sum_{a \sim p_{i-1}} w_a$ on paths differing only at the backtracking vertex p_i, and zero otherwise. Caps therefore give unweighted sum equations. Two-click rotation replaces $(v_0, \dots, v_5)$ by $(v_4, v_5, v_0, v_1, v_2, v_3)$, with coefficient $w_{v_0} w_{v_5} / (w_{v_2} w_{v_1})$.

For a primitive root ω, work in $\mathbb{Q}(s, \omega)$, $\omega^2 + \omega + 1 = 0$. The cap and ω-rotation equations on the 152 loops have rank 144. After their elimination, let E_j be the 90 distinct normalized nonzero entries of $T^2 - f_3$, in the deterministic path-matrix order of the verifier. Direct polynomial arithmetic gives

$$\begin{aligned} 1 = {} & (-11 - 5s) E_{26} + \left(-25 - 11s + \frac{5 + 3s}{2} \omega \right) E_0 \\ & + \left(-12 - 6s + \frac{9 + 3s}{2} \omega \right) E_1 + (44 + 20s) E_4 \\ & + \big(4 + 2s - (7 + 3s) \omega \big) E_{13}. \end{aligned}$$

The portable checker reconstructs the linear elimination and every E_j from the graph before checking this identity. It excludes both primitive cubic roots simultaneously, without complex conjugation equations. Hence the eigenvalue is one.

For eigenvalue one the cap and rotation system again has rank 144. Its eight free loop coordinates $x_0, \dots, x_7$ may be chosen, in order, at

$$012321,\ 212323,\ 212354,\ 212453,\ 232323,\ 232354,\ 232453,\ 235453.$$

Put $k = (3 - s)/2$ and $c = (5s - 11)/2 > 0$. On this eight-dimensional linear space, the ideal of all path-matrix entries of $T^2 - f_3$ is generated by

$$\begin{gathered} x_0 + \frac{3 + s}{2} x_7, \qquad x_1 + \frac{3 - s}{2} x_7, \qquad x_2 + 2x_5, \qquad x_3 + 2x_6, \\ x_4 + \frac{s - 1}{2} x_7, \qquad x_5 x_6 - c, \qquad x_7^2 - k^2. \end{gathered} \tag{2}$$

This is an equality of polynomial ideals over $\mathbb{Q}(\sqrt{5})$, not just equality of their real or self-adjoint solution sets. The portable certificate generates all square equations from the graph and verifies both inclusions: reduction proves that (2) implies every square equation, and explicit polynomial identities of degree at most three express its seven generators in the original equations. The linear rank and all coefficients are checked exactly.

Thus every complex solution is $\pm T(\lambda)$ for $\lambda \in \mathbb{C}^\times$, with free coordinates

$$\left(-1, -k^2, -2\sqrt{c}\lambda, -2\sqrt{c}\lambda^{-1}, -\frac{s - 1}{2} k, \sqrt{c}\lambda, \sqrt{c}\lambda^{-1}, k \right).$$

The base solution $T(1)$ is self-adjoint after the diagonal change to spherical coordinates. Assign an oriented-edge character λ to the edge $3 \to 5$, its inverse to the reverse edge, and 1 to all others. Multiplying each loop coefficient by its edge-character product is a graph planar algebra automorphism: the factors on internal edges cancel in every state sum. The eight coordinates have winding numbers $(0, 0, 1, -1, 0, 1, -1, 0)$, so this gauge carries $T(1)$ to $T(\lambda)$. The two signs generate the same subalgebra.

After applying this gauge, $\mathcal{P}$ contains the planar subalgebra generated by $T(1)$. By [70, Lemma 3.4 and Theorems 7.8–7.9], this is the known unitary $2D2$ planar algebra. Its positive box dimensions are the rooted-loop dimensions of $2D2$, exactly those of $\mathcal{P}$. The containment is therefore equality in every positive box space; one-click rotation gives equality in the negative spaces of positive degree,

and both zero-box spaces are $\mathbb{C}$. This rules out additional higher-depth data without using positivity on $\mathcal{P}$. The known dual graph has level sizes $(1,1,1,2,4,1)$, hence six even vertices. □

Proposition 9.2. *The rings R8_2_152, R8_2_153, R8_2_166 and R8_2_495 admit no fusion-category categorification over* $\mathbb{C}$.

Proof. Each tensor has a faithful C_2 grading with rank-four neutral ring $g^2 = 1$, $gr = rg$, $r^2 = 1 + 2r + 2gr$. Its odd component contains X with $XX^* = X^*X = 1 + r$. The source witnesses are

ring	$(1, g, r, gr)$	X	graph vertices $0, \dots, 7$
R8_2_152	$(0, 6, 5, 7)$	1	$(0, 1, 5, 3, 4, 7, 2, 6)$
R8_2_153	$(0, 6, 5, 7)$	1	$(0, 1, 5, 3, 4, 7, 2, 6)$
R8_2_166	$(0, 5, 7, 6)$	3	$(0, 3, 7, 1, 2, 6, 4, 5)$
R8_2_495	$(0, 1, 6, 3)$	2	$(0, 2, 6, 4, 5, 3, 7, 1)$.

Both left and right incidence matrices for tensoring with X are the rooted graph of Lemma 9.1. The object r generates the full neutral category, so both even parts have rank four.

Suppose a complex categorification exists. Its neutral category has exactly two FP dimensions. By [25, Lemma 3.1], it is Galois conjugate to a pseudo-unitary category. Apply the same field automorphism to the entire category and its odd module; their based graphs are unchanged. Lemma 9.1 then forces the dual even part to have rank six, contradicting the rank-four graph computed from the tensor. Equivalently, the odd component is an invertible neutral bimodule and its dual category is the rank-four neutral category. No pivotal structure on the original rank-eight category is assumed. □

10. Pointed support in a twisted half-braiding

Proposition 10.1. *The rings R8_2_314 and R8_2_432 admit no fusion-category categorification over* $\mathbb{C}$.

Proof. Both tensors have a fusion subring $\mathcal{D}$ with basis $(1, g, r, gr)$ and rules

$$g^2 = 1, \qquad gr = rg, \qquad r^2 = g^\epsilon + r + gr,$$

where $\epsilon = 1$ for R8_2_314 and $\epsilon = 0$ for R8_2_432. The complement $\mathcal{M}$ is stable under both $\mathcal{D}$ actions. Its four simples are aX for $a \in \operatorname{Irr}\mathcal{D}$, with

$$Xa \cong F(a)X, \qquad X^2 = g^\epsilon + X + gX,$$
$$F(1) = 1, \quad F(g) = g, \quad F(r) = gr, \quad F(gr) = r.$$

The complete source witnesses are

ring	$(1, g, r, gr)$	X	(X, gX, rX, grX)
R8_2_314	$(0, 7, 1, 2)$	3	$(3, 4, 6, 5)$
R8_2_432	$(0, 3, 4, 5)$	6	$(6, 7, 1, 2)$.

Here F is initially only notation for the displayed label permutation. We explain why a hypothetical categorification forces the coherence needed below.

The functor $L : \mathcal{D} \to \mathcal{M}$, $a \mapsto aX$, is an equivalence of left module categories: it maps simples bijectively to simples and is fully faithful on their Hom spaces. Transport the right $\mathcal{D}$ action through L. Left module endofunctors of the regular category are right tensor functors, so this action supplies a strong monoidal functor $F : \mathcal{D} \to \mathcal{D}$ with coherent natural isomorphisms $Xa \cong F(a)X$. Its displayed permutation of simples makes it a tensor autoequivalence.

Applying this relation twice gives coherent isomorphisms $X^2a \cong F^2(a)X^2$. The decomposition $\mathcal{C} = \mathcal{D} \oplus \mathcal{M}$ is a decomposition of $\mathcal{D}$-bimodule categories, so its projection to $\mathcal{M}$ preserves these constraints. Write $z = 1 \oplus g$. The $\mathcal{M}$ part of X^2 is zX. Using $Xa \cong F(a)X$ and cancelling the fully faithful functor L gives

$$\eta_a : zF(a) \longrightarrow F^2(a)z.$$

If J^F, J^{F^2} denote the tensorators, its compatibility is

$$\eta_{ab}(\mathrm{id}_z \otimes J^F_{a,b}) = (J^{F^2}_{a,b} \otimes \mathrm{id}_z)(\mathrm{id}_{F^2(a)} \otimes \eta_b)(\eta_a \otimes \mathrm{id}_{F(b)}),$$

with associators understood. Equivalently this is the bimodule constraint of the $\mathcal{M}$ projection of multiplication $\mathcal{M} \boxtimes_{\mathcal{D}} \mathcal{M} \to \mathcal{C}$. Applying a monoidal quasi-inverse of F, and using $F^{-1}(z) \cong z$, gives an F-twisted half-braiding

$$\theta_a : za \longrightarrow F(a)z, \qquad \theta_{ab} = (J^F_{a,b} \otimes \mathrm{id}_z)(\mathrm{id}_{F(a)} \otimes \theta_b)(\theta_a \otimes \mathrm{id}_b). \tag{3}$$

No identification of F^2 with the identity is required.

Fix the two summands $z_0 = 1, z_1 = g$. Since $r \neq gr$ and $F(r) = gr$, the map θ_r must send $z_i r$ to $F(r)z_{i+1}$, with indices modulo two. Thus it exchanges the two support labels. Equation (3) for $a = b = r$ applies two exchanges, and hence $\theta_{r\otimes r}$ preserves these labels. Its associators and tensorator preserve the relevant direct-sum idempotents by naturality.

Choose a nonzero inclusion $v : r \to r \otimes r$, which exists in both rings. Naturality requires

$$\theta_{r\otimes r}(\mathrm{id}_z \otimes v) = (F(v) \otimes \mathrm{id}_z)\theta_r.$$

On $z_i r$ the left side lands in $F(r \otimes r)z_i$ and the right side in $F(r \otimes r)z_{i+1}$. Both maps are nonzero: $v, F(v)$ are split monomorphisms, tensoring preserves them, and the θ maps are isomorphisms. The two target direct summands have zero intersection, a contradiction. The argument uses only semisimplicity, rigidity and tensor coherence, with no pivotality assumption. □

11. A forced quantum-group Morita dual

Proposition 11.1. *The ring R8_2_342 has no categorification over* $\mathbb{C}$.

Proof. Suppose $\mathcal{C}$ categorifies this ring. The invertible $g = 5$ fixes simples $3, 7$. Its pointed associator is therefore trivializable, since either fixed simple gives a rank-one module category over $\langle g \rangle$. Thus $A = 1 \oplus g$ is a connected separable algebra. Put $\mathcal{M} = \mathcal{C}_A$ and let $\mathcal{E}$ be its Morita dual.

The simple module labels are

$$(0,0), (1,0), (3,+), (3,-), (4,0), (7,+), (7,-).$$

The exact certificate `verify_r342_inverse_morita.py` exhausts the two-dimensional signed representations on the two split labels. Certified annihilating polynomials and the three source generators $1, 4, 6$ leave eight generator tuples and exactly one full representation up to sign gauge. Consequently all seven-by-seven left action matrices L_i are forced. In particular $L_g = I_7$.

The distinguished induction Gram matrix is $2I_7$. Its nonnegative integral columns must be exactly two copies of each standard basis vector. Hence $\mathcal{E}$ has fourteen simples, two above each module label. Their right action matrices R_y commute with every L_i and satisfy

$$Q_a := \sum_i (L_i)_{aa} L_i = \sum_y R_y[:, a] R_y[:, a]^t, \qquad 0 \leq (R_y)_{ba} \leq \lfloor \sqrt{(Q_a)_{bb}} \rfloor.$$

These are linking-category associativity identities. The common commutant has dimension eleven; a prescribed standard first column leaves four free matrix entries. Exhausting the displayed bounds gives candidate counts $(2, 4, 2, 2, 2, 2, 2)$. The first domain contains the identity and a unique nontrivial involutive permutation P, representing the second invertible u. Right multiplication by P pairs every domain freely. Of the resulting two joint candidate tuples, exactly one satisfies all seven Gram identities.

The certificate reconstructs the full dual fusion tensor, rather than stopping at these necessary identities. For each product its coefficient vector n obeys

$$\sum_w n_w R_w = R_y R_z, \qquad n_1 = \delta_{z,y^*}, \qquad n_u = \delta_{zu,y^*}.$$

The first column bounds each coefficient by the corresponding entry of R_yR_z. Complete enumeration forces the ten fusion matrices with labels $0,1,4,5,6,7,10,11,12,13$. The fourteen words

$$1, u, 4, 5, 6, 7, 10, 11, 12, 13, 44, 46, 4\,11, 4\,13$$

have independent class vectors. Their associativity therefore forces all remaining matrices. The resulting tensor passes all original product domains and all 14^4 associativity equations.

The labels $(0,2,5,6,9,10,13)$ form a fusion subcategory $\mathcal{B}$, with complement $\mathcal{B}u$ and faithful C_2 grading. The restriction $\mathcal{M}|_{\mathcal{B}}$ is regular. Its based ring is $\mathrm{PSU}(3)_5$: the generator matrix in [73, Section 2.4] matches under the local map $(0,4,5,6,1,3,2)$, and its seven powers on the unit span the whole ring. Conjugation by u induces precisely its duality permutation $(0,1,3,2,4,6,5)$. The checker also verifies $\mathcal{B}_{\mathrm{ad}} = \mathcal{B}$ and that its based automorphism group is C_2.

The Kazhdan–Wenzl theorem, in the general monoidal form [74, Theorem 5.1], applies to $\mathcal{B}\boxtimes\mathrm{Vec}_{C_3}$, whose based ring is $\mathrm{SU}(3)_5$. Restrict its resulting equivalence to adjoint subcategories. The grading three-cocycle is trivial there, so $\mathcal{B}$ is a Galois conjugate of the standard quantum $\mathrm{PSU}(3)_5$ category. This step assumes neither a braiding nor pivotality.

The gauge-autoequivalence kernel of $\mathrm{SU}(3)_5$ is trivial by [51, Theorem 1.1 and Section 3]. Since the standard category factors as $\mathrm{PSU}(3)_5 \boxtimes \mathcal{P}$, with $\mathcal{P}$ pointed of rank three, any based-trivial tensor autoequivalence of $\mathrm{PSU}(3)_5$ extends by the identity on $\mathcal{P}$. It is therefore trivial. Galois transport proves the same statement for $\mathcal{B}$. Thus its nontrivial based duality permutation determines a unique tensor autoequivalence class. Moreover $U(\mathcal{B}) = 1$, so $\mathrm{Aut}_{\otimes}(\mathrm{Id}_{\mathcal{B}}) = 1$; the coherent C_2 action realizing this class is unique up to equivalence when it exists.

Finally, the first-column multiplicities give the actual internal-end algebra $\underline{\mathrm{End}}_{\mathcal{E}}(m_0) = 1 \oplus u$. Its associativity trivializes the pointed associator of u. The grading is consequently a crossed product $\mathcal{E} = \mathcal{B} \rtimes C_2$, and Morita reflection through this algebra identifies $\mathcal{C}$ with $\mathcal{B}^{C_2}$. The existing category in Proposition 12.1 has a central Tannakian boson, and its de-equivariantization is a Galois conjugate of $\mathrm{PSU}(3)_5$ by [73, Section 2.4]. Its action swaps exactly the two dual pairs and is the same nontrivial based action. The uniqueness above forces $K_0(\mathcal{C}) = R8_2_666$. Every simple in that ring is self-dual, whereas labels $1,2$ in $R8_2_342$ form a non-self-dual pair. This is a contradiction. □

12. Near-group centralizer constructions and a common spherical reduction

Proposition 12.1. *The ring $R8_2_666$ admits a unitary braided categorification: the centralizer of the full pointed subcategory in the center of a near-group $C_4 + 4$ category.*

Proof. The smallest hyperbolic case in [72, Section 3.2.3.1] is explicitly identified with the actual centers of the two complex-conjugate near-group $C_4 + 4$ categories. This finite case precedes the conjecture there for larger groups. A convenient description is given in [73, Section 2.4]:

$$G = C_4^2, \quad q_G(x,y) = i^{-xy}, \quad \theta_G(x,y) = (y,x),$$

$$H = C_{16} \times C_2, \quad q_H(x,y) = \zeta_{32}^{3x^2} i^{y^2}, \quad \theta_H(x,y) = (3x + 8y, x + y).$$

The fixed groups identify with $K = C_4$ through $k \mapsto (k,k)$ and $k \mapsto (4k, k \bmod 2)$. The center has 28 simple objects, labelled by K, $K \times \{\pi\}$ and representatives of the nonfixed involution orbits in G, H. Its pointed subcategory $\mathcal{P}$ has dimension four and twist $q_K(k) = i^{-k^2}$. Trivial monodromy with every pointed object selects exactly the eight labels

$$(0,0), (2,2), (0,0,\pi), (2,2,\pi), (3,1)_G, (2,1)_H, (6,1)_H, (4,0)_H.$$

Thus $\mathcal{P}'$ is an existing unitary braided fusion category, of dimension $96 + 64\sqrt{2}$. Its Müger center is the Tannakian order-two radical of $\mathcal{P}$.

The source-bound certificate `verify_c4_centralizer.py` reconstructs the full 28-label matrix from [73, Equation (1)]. It works in $\mathbb{Z}[\zeta_{32}]$, using $\zeta_{32}^{16} = -1$ and $\sqrt{2} = \zeta_{32}^4 + \zeta_{32}^{-4}$. It checks unitarity and all 1792 Verlinde coefficients whose first two labels belong to this centralizer, including vanishing

of every coefficient outside it. The resulting 512 coefficients identify its based ring with $R8_2_666$ under the unital map

$$(0, 1, 3, 4, 7, 2, 5, 6).$$

The proof note records the entire matrix formula, transversals, normalization and source hash, and the normal and optimized replays also verify associativity and self-duality. □

Proposition 12.2. *The rings $R8_2_820$ and $R8_2_869$ admit unitary modular categorifications. They are the nonpointed Müger factors of the Drinfeld centers of the first and second near-group $C_5 + 5$ categories in* [71, Proposition 6 and Table 2].

Proof. Use the actual C^*-fusion categories with parameters $(c, m) = (-1, 1)$ and $(e^{2\pi i/3}, 2)$ in that table. Their auxiliary metric groups are respectively $G' = C_3^2$, $Q'(x, y) = (x^2 + y^2)/3$, and $G' = C_9$, $Q'(x) = 2x^2/9$. The center formulas of [71, Proposition 7(b)] are identified with the actual centers for every odd group order at most 13 in the paragraph following its Conjecture 2. Thus the order-five case here is established independently of the conjecture for arbitrary groups.

Put

$$a = \frac{5 + 3\sqrt{5}}{2}, \qquad b = a + 1, \qquad c = a + 2.$$

Each near-group category has global dimension $5c$. The five invertible center objects a_g, $g \in C_5$, have pairing $\exp(2\pi i\, 2mgh/5)$, which is nondegenerate. Their Müger centralizer is consequently an existing unitary modular category of dimension $5c^2 = 45b$. Trivial monodromy with these invertibles selects exactly

$$a_0,\ b_0,\ c_{1,-1},\ c_{2,-2},\ d_{0,\gamma_1}, \ldots, d_{0,\gamma_4}, \qquad \{\gamma_i\} = (G' \setminus \{0\})/\{\pm 1\}.$$

Their dimensions are $(1, b, c, c, a, a, a, a)$; their squared dimensions sum to $45b$, confirming completeness.

The exact source-bound certificate `verify_neargroup_centralizer.py` reconstructs the restricted S matrix from those formulas and evaluates all 512 Verlinde coefficients over $\mathbb{Q}[q, z]/(q^2 - 5, z^3 - 3z + 1)$. It identifies the two complete tensors under the respective model-to-source maps

$$(0, 2, 6, 7, 1, 3, 4, 5), \qquad (0, 3, 6, 7, 1, 2, 5, 4).$$

Here the auxiliary representatives are $(1, 0), (0, 1), (1, 1), (1, 2)$ in C_3^2 and $1, 2, 3, 4$ in C_9. An independent certificate instead reconstructs every tensor entry from the integer fusion formulas following [71, Proposition 7(b)] and obtains the same identifications. Existence follows from the categories and their centralizers; the finite computations identify their based rings. □

The following result, obtained before these constructions, also controls all possible categorifications of the two rings.

Proposition 12.3. *Every complex fusion category with based ring $R8_2_820$ or $R8_2_869$ admits a spherical structure. Every spherical dimension character belongs to the FP Galois orbit. Consequently, categorifiability of either ring is equivalent to pseudo-unitary categorifiability, and any categorification has a pseudo-unitary Galois conjugate.*

Proof. Both tensors are self-dual. In the canonical sphericalization choose the positive-second-indicator lift of each simple. The resulting signed quotient tensor T is fully symmetric, has $|T_{ij}^k| \le N_{ij}^k$ and $T_{ij}^k \equiv N_{ij}^k \pmod 2$, and satisfies all associativity equations. The odd-channel parity theorem [63, Theorem 3.9] gives the stronger restriction

$$N_{jj}^i - T_{jj}^i \equiv 0 \pmod 4.$$

Every coefficient two in these tensors has a repeated label. Thus repeated coefficients one are fixed at $+1$, repeated coefficients two lie in $\{-2, 2\}$, and the remaining all-distinct coefficients one lie in $\{-1, 1\}$.

The source-bound checker reconstructs exactly 35 Boolean signs for $R8_2_820$ and 36 for $R8_2_869$. Expanding every ordered associativity identity and reducing sign squares to one leaves 215 distinct integer polynomial equations for either tensor. Exhaustive branching visits both signs at each unassigned variable and rejects a branch only when a fully assigned equation fails. It visits respectively 3071 and 1873 nodes and finds exactly one solution in each case: all signs are positive. Hence $T = N$. The chosen lifts form a full fusion subcategory of the sphericalization, and the forgetful functor restricts to a tensor equivalence with the original category. Restricting its spherical structure proves the first assertion.

The remaining reduction is common and theoretical. Write the basis as $(1, a_1, a_2, a_3, a_4, b, c_1, c_2)$, where b has source label 2 for $R8_2_820$ and label 3 for $R8_2_869$, the c-labels are $6, 7$, and the a-labels are the four other nonunit labels. Both integral based rings satisfy

$$b^2 = 1 + \sum_{i=1}^{4} a_i + b + c_1 + c_2, \qquad (b+1)a_i = b^2 - 1, \qquad (b-1)c_j = b^2 - 1.$$

Let $x = \dim(b)$ for any spherical structure. Nonzero simple dimensions exclude $x = 1$ and $x = -1$ by the last two identities. They therefore give $\dim(a_i) = x - 1$ and $\dim(c_j) = x + 1$. The first identity becomes

$$x^2 = 1 + 4(x-1) + x + 2(x+1) = 7x - 1, \qquad x = \frac{7 \pm 3\sqrt{5}}{2}.$$

These are exactly the two Galois-conjugate FP characters. Conjugate the category together with its spherical structure to the positive character. A spherical category with positive simple dimensions is pseudo-unitary. This statement concerns every categorification; it does not assert that each categorification is unitarizable. □

12.1. **A unified character-parity exclusion for the Fibonacci pair.**

Proposition 12.4. *Every complex categorification of $R8_2_190$ or $R8_2_489$ admits a spherical structure and has a pseudo-unitary Galois conjugate. For $R8_2_489$, the pointed C_2 subcategory has trivial associator, so $1 + g$ is a connected separable algebra. The following theorem uses this structural reduction to exclude both rings.*

Proof. Both rings have basis $g^i\tau^j x^k$, $i, j, k \in \{0, 1\}$, and commutative presentation

$$g^2 = 1, \qquad \tau^2 = 1 + \tau, \qquad x^2 = g^\epsilon + \tau(x + gx), \qquad x^* = g^\epsilon x,$$

where $\epsilon = 1$ for $R8_2_190$ and $\epsilon = 0$ for $R8_2_489$. In the canonical sphericalization, choose the lift of g with dimension 1. It squares to the unit. A lift of τ is self-dual, and its signed square has the form $1 \pm \tau$; change the lift to make the sign positive. Choose a lift of x, and define the other lifts by the simple products $g\tau, gx, \tau x, g\tau x$. The relevant left and right products have the same underlying simple and equal nonzero spherical dimensions, so the chosen generators commute in the signed quotient. The same argument gives $x^* = g^\epsilon x$. The unit channel of xx^* therefore fixes the constant coefficient in

$$x^2 = g^\epsilon + a\tau x + b\tau gx, \qquad a, b \in \{1, -1\}.$$

These relations determine the entire signed multiplication tensor. When $\epsilon = 1$, duality fixes x^2 and exchanges the last two terms, so $a = b$.

For $\epsilon = 0$, suppose instead $a = -b$. Writing $t = \dim(\tau)$ and $z = \dim(x)$ gives $t^2 = t + 1$, $z^2 = 1$, and global dimension $\delta = 4(t + 2)$. A source character with values $g \mapsto 1$, $\tau \mapsto t$, $x \mapsto d$, where $d^2 = 1 + 2td$, has formal codegree $f_E = 2(t + 2)(1 + d^2)$. Central induction in the sphericalization requires δ/f_E to be an algebraic integer [75, Theorem 2.13]: the sphericalization dimension and the source formal codegree both acquire a factor two. However,

$$d^4 - 2d^3 - 6d^2 + 2d + 1 = 0, \qquad y = \frac{\delta}{f_E} = \frac{2}{1 + d^2}$$

has primitive minimal polynomial

$$5y^4 - 20y^3 + 25y^2 - 10y + 1.$$

Indeed the polynomial of d is irreducible modulo three, and $d = ((d^2)^2 - 6d^2 + 1)/(2(d^2 - 1))$ gives $\mathbb{Q}(y) = \mathbb{Q}(d)$; elimination gives the displayed polynomial. Its leading coefficient excludes integrality. Thus $a = b$ in both cases. Changing the lift of x and its three translates if necessary makes both signs positive. Now $T = N$, so the chosen lifts form a tensor-closed spherical section of the forgetful functor.

This section has $\dim(g) = 1$, $t^2 = t + 1$, and $z^2 = 1 + 2tz$. The irreducible quartic above shows that these four dimension characters form a single Galois orbit, containing the positive character

$$t = \varphi = \frac{1 + \sqrt{5}}{2}, \qquad z = \varphi + \sqrt{\varphi^2 + 1}.$$

Conjugating to that character gives pseudo-unitarity.

For $R8_2_489$, work with this positive spherical structure. A homogeneous pairing on a self-dual object V induces one on V^n of sign $\nu_2(V)^n$. Any opposite-indicator self-dual summand consequently has even multiplicity. Since $\tau^2 = 1 + \tau$, we have $\nu_2(\tau) = 1$. The exact congruences

$$x^3 \equiv x + \tau + g\tau \pmod 2, \qquad (g\tau)^3 \equiv g \pmod 2$$

then imply $\nu_2(x) = \nu_2(g\tau) = \nu_2(g) = 1$. For the positive spherical structure on $\mathrm{Vec}^{\omega}_{C_2}$, the nontrivial associator class has indicator -1 on g. Hence ω is trivial, and $1 + g$ is its regular separable algebra. This algebra also exists before Galois conjugation. No half-braiding on g is inferred.

The source-bound replay `verify_fib_pair_theoretical.py` checks both complete multiplication tensors, the duality and cube identities, irreducibility modulo three, and the elimination polynomial. The proof does not require an enumeration of signed tensors. □

Theorem 12.5. *Neither* $R8_2_190$ *nor* $R8_2_489$ *admits a complex fusion-category categorification.*

Proof. By Proposition 12.4, it suffices to exclude pseudo-unitary categorifications. Retain the common presentation above, put $y = \tau x$, and write

$$\varphi = \frac{1 + \sqrt{5}}{2}, \qquad d = \varphi + \sqrt{\varphi + 2}, \qquad D = 2(\varphi + 2)(1 + d^2).$$

The four dimension coordinates $1, \varphi, d, \varphi d$ are rationally independent: the norm of $\varphi + 2$ from $\mathbb{Q}(\varphi)$ to $\mathbb{Q}$ is the nonsquare 5, so d has degree two over $\mathbb{Q}(\varphi)$.

For a character χ, let Z_χ denote its central simple summand of $I(1)$ and let $\beta_\chi = D/f_\chi$. The character correspondence preceding [75, Theorem 2.13], together with its Example 2.9, gives $\dim Z_\chi = \beta_\chi$, $[FZ_\chi : 1] = 1$, and

$$S_{Z_\chi, W} = \beta_\chi \chi(FW)$$

for the unnormalized modular matrix of the center. Symmetry therefore yields

$$\beta_\chi \chi(FZ_\psi) = \beta_\psi \psi(FZ_\chi). \tag{4}$$

Take $\omega = 1$ for $\epsilon = 0$ and $\omega = i$ for $\epsilon = 1$. The two characters

$$\chi_\pm(g) = -1, \qquad \chi_\pm(\tau) = \varphi, \qquad \chi_\pm(x) = \pm\omega$$

have formal codegree $4(\varphi + 2)$ and central dimension $\beta_\pm = 1 + \varphi d$. Formal codegrees here use $\chi(b)\chi(b^*)$, which also gives this value when $x^* = gx$ and $\omega = i$. Their central restrictions consequently satisfy

$$FZ_\pm \in \{1 + y,\ 1 + gy\},$$

by the independence of the four dimension coordinates and unit multiplicity one.

Now consider the character

$$\psi(g) = 1, \qquad \psi(\tau) = 1 - \varphi, \qquad \psi(x) = z = (\varphi - 1)d - \varphi.$$

The relations give $z^2 = 1 + 2(1-\varphi)z$, and direct exact reduction gives

$$f_\psi = 2(3-\varphi)(1+z^2), \qquad \beta_\psi = (1+\varphi) + (2+3\varphi)d.$$

Write $B = Z_\psi$. Because $\psi(g) = 1$, the values $\psi(FZ_+)$ and $\psi(FZ_-)$ coincide for every permitted choice of the two restrictions. Applying (4) twice and using $\beta_+ = \beta_-$ gives $\chi_+(FB) = \chi_-(FB)$.

The dimension of B requires total multiplicity two for x, gx in FB and total multiplicity three for y, gy. If Δ_x, Δ_y are the respective integer multiplicity differences, subtracting these two character evaluations gives

$$0 = 2\omega(\Delta_x + \varphi\Delta_y).$$

Irrationality of φ forces both differences to vanish. The two equal multiplicities of y and gy cannot sum to three, which is the required contradiction.

The source-bound replay `verify_fib_character_parity.py` checks all 512 tensor entries for each presentation, the three characters, the formal codegrees, and the exact dimension identities. The obstruction itself requires no induction-column enumeration. □

12.2. **Cyclic induction phase multiplicities exclude $R8_1_9$.**

Lemma 12.6. *Let a be an invertible object of order n in a pseudo-unitary fusion category of dimension D. Every phase occurring in $I(a)$ is an nth root of $\nu_n(a)$, and the total dimension in each of these n phases, counted with its induction multiplicity, is D/n.*

Proof. The rotation spaces $\mathrm{Hom}(1, a^m)$ vanish for $0 < m < n$, so $\nu_m(a) = 0$. For $m = n$ that space is one-dimensional, and $\eta = \nu_n(a)$ is a root of unity. The positive weighted trace $\mathrm{Tr}(\theta^n I(a)) = D\eta$ has maximal absolute value $D = \dim I(a)$, so every constituent has $\theta^n = \eta$. Choose $\lambda^n = \eta$ and write W_j for the weight of phase $\lambda\zeta_n^j$. The zeroth and first $n-1$ trace identities are $\sum_j W_j = D$ and $\sum_j W_j \lambda^m \zeta_n^{jm} = 0$ for $1 \le m < n$. Discrete Fourier inversion gives $W_j = D/n$. □

Theorem 12.7. *The ring $R8_1_9$ has no complex fusion categorification.*

Proof. By [25, Lemma 3.1], take a pseudo-unitary Galois conjugate and put $d = 2+\sqrt{5}$, $D = 8+16d$. The odd-cube pivotal identities force the second indicators, in source order, to be $(1,0,0,0,0,1,1,1)$. The pointed generator a has source label1, with a^{-1} of label2 and a^2 of label7. Its associator class $k \in \mathbb{Z}/4$ satisfies $\nu_2(a^2) = (-1)^k$, so $k = 0$ or2. The canonical pointed indicators give $\nu_4(a) = \nu_4(a^{-1}) = (-1)^{k/2}$.

The pointed induction Gram block is $8I_4$. Hence a central row v occurring over a has no other pointed coordinate, and its dimension has the form $v_a + Bd$, with $B \ge 0$. Lemma 12.6 gives weight $D/4 = 2 + 4d$ in each actual phase. Comparing constant coefficients therefore gives $\sum_{Z \text{ in one phase}} v_a(Z)^2 = 2$. Thus $v_1, v_2 \le 1$ for every central row. This argument is uniform in the pointed associator class and does not average away the two imaginary phases.

The complete formal codegrees are $D, D', 8, 8, 8, 8, 8, 8$; the six other characters lie in $\mathbb{Q}(i)$. Actual character-labelled Galois symmetry transports a forgetful row by

$$T(v) = v + (0,0,0,1,1,1,1,0)\left(v_0 + v_1 + v_2 + v_7 - \frac{v_3+v_4+v_5+v_6}{2}\right),$$

up to the sign making it nonnegative. Partner rows have equal normalized relative moments, because their twists are related by σ^2 and σ^2 fixes $\mathbb{Q}(\sqrt{5})$. The two forced $I(1)$ rows and the six remaining character-labelled rows leave exactly eight possible low frames.

For arbitrary roots of unity, the first two relative traces lie in a fixed 17-vertex hull. It consists of the nine pairs from orders $1, 2, 4, 5, 10, 20$, together with the needed negative-half endpoints: an extra odd factor coprime to 10 scales both moments by $\mu(m)/\varphi(m) \in [-1/2, 1]$. No root-order bound or prime-support assumption is used. The full elementary Gauss-sum proof is supplied in the evidence packet.

For $k = 0$, constituents over labels1 and2 have $\theta^4 = 1$; for $k = 2$ they have $\theta^4 = -1$ and their first two relative traces vanish. Those over label7 have $\theta^2 = 1$. Use these phase restrictions, the exact dimension-integrality condition $D/\dim(Z)$ integral, Galois pairing, and the residual Gram bound for the entire row orbit. With the Fourier bound, class zero leaves 64 row orbits and 405 moment columns per frame. The original, weaker bounds suffice for class two and leave 92 orbits and 389 columns.

The Gram and two induction trace identities have 56 integer coordinates. The two covectors in `r9_center_trace_certificate.json` have nonnegative pairing with every column, but target pairings -64 in class zero and -6432 in class two, for all eight low frames. Their standalone standard-library verifier reconstructs every bounded domain and every positive semidefinite condition with exact rational arithmetic. Neither target is a nonnegative combination of its columns, excluding both possible pointed classes. The initial Galois reduction proves the general complex assertion. □

12.3. **The full modular identity and a fourth-moment projection.**

Lemma 12.8. *Let $\mathcal{C}$ be a pseudo-unitary fusion category with commutative Grothendieck ring, and let b_χ be the forgetful vector of the actual summand $A_\chi \subset I(1)$ corresponding to the character χ. For a simple $Z \in Z(\mathcal{C})$ write v_Z for its forgetful vector. Then*

$$\sum_Z \theta_Z (v_Z)_i (v_Z)_j = \sum_\chi (b_\chi)_i \overline{\chi(X_j)}. \tag{5}$$

Proof. Let $C_{\chi i} = \chi(X_i)$, let f_χ be the formal codegrees, and put $X = \operatorname{diag}(f_\chi^{-1})C$. The character identification in [75, Example 2.9 and Theorem 2.13] gives $S_{A_\chi,Z} = \chi(FZ)/f_\chi$ for every central simple Z, with $S_{1Z} = \dim(Z)/\dim(\mathcal{C})$. All A_χ have twist one, and the center has trivial anomaly. Thus the low block of $STS = T^{-1}ST^{-1}$ is

$$X \left(\sum_Z \theta_Z v_Z v_Z^{\mathsf{t}} \right) X^{\mathsf{t}} = X B^{\mathsf{t}},$$

where B has rows b_χ. Character orthogonality gives $C\overline{C}^{\mathsf{t}} = \operatorname{diag}(f_\chi)$, so $X^{-\mathsf{t}} = \overline{C}$. Cancellation proves the identity. □

Theorem 12.9. *The ring R8_1_18 has no complex fusion categorification.*

Proof. By [25, Lemma 3.1], pass to a pseudo-unitary Galois conjugate. Put $d = 2 + \sqrt{5}$, $K = \mathbb{Q}(\sqrt{5})$ and $D = 8 + 16d$. In model order $(1, a, a^2, a^3, \rho, a\rho, a^2\rho, a^3\rho)$, mapped to source $(0, 1, 7, 2, 3, 6, 5, 4)$, the ring is commutative and satisfies

$$a^4 = 1, \qquad \rho^2 = a + (1 + a + a^2 + a^3)\rho.$$

Its six non-FP characters have formal codegree eight and are

$$\chi_{k,\pm}(a) = \zeta_8^{2k}, \qquad \chi_{k,\pm}(\rho) = \pm\zeta_8^k, \qquad k = 1, 2, 3.$$

The other two codegrees are D, D'. The two forced low rows are 1 and $1 + X_3 + X_4 + X_5 + X_6$; the other six have form 1+ two nonpointed constituents. Actual character reciprocity and duality leave eight low assignments. Taking normalized relative trace t_K in (5) gives, for nonunit indices,

$$\sum_Z (v_Z)_i (v_Z)_j t_K(\theta_Z) = \begin{cases} 4 - d & i, j \in \{3, 4, 5, 6\}, \\ 0 & \text{otherwise.} \end{cases}$$

Indeed every nonpointed value of the six $\mathbb{Q}(\zeta_8)$ characters has relative trace zero; the pointed entries cancel in each of the eight assignments.

The pointed induction Gram is $8I_4$. Lemma 12.6 therefore gives $(v_Z)_1, (v_Z)_2 \le 1$ exactly as in the preceding theorem, uniformly over all four pointed associator classes. Besides the quadratic transport T displayed there, full Galois symmetry acts on rows by the source permutation

$$P = (0, 2, 1, 6, 5, 4, 3, 7).$$

It is induced by an automorphism fixing K and sending ζ_8 to ζ_8^3: on the model basis it sends $a \mapsto a^{-1}$ and $a^j\rho \mapsto a^{3j+1}\rho$. Include also source duality $(0,2,1,4,3,6,5,7)$. Every full row orbit has a common relative moment histogram: Galois transport changes a twist by σ^2, which fixes quadratic K, and duality preserves twists.

The Cauchy theorem for spherical fusion categories [62, Theorem 3.9] applies to the nonintegral dimension D. Its norm is $320 = 2^6 \cdot 5$, so the exponent has only primes two and five. For powers $1, 2, 4$, every normalized relative twist triple is one of the triples from fortieth roots or the zero triple. This follows directly from the quadratic Gauss sums: fifth-adic exponent at least two, or 2-adic exponent at least four, gives zero in all three coordinates. There are exactly thirteen triples; this is not an assertion that the exponent divides forty.

There is no need to determine fourth indicators. Each $\nu_4(X_i)$ is a sum of fourth roots, so $t_K(\nu_4(X_i)) \in \mathbb{Z}$. The functional $\ell(a + bd) = 2a - b$ annihilates $D\mathbb{Q}$. Applying it to the fourth induction identity gives

$$\ell\left(\sum_Z (v_Z)_i \dim(Z) t_K(\theta_Z^4)\right) = 0.$$

Together with the first two induction traces, the full modular identity above, and the Gram identity, this gives 119 rational linear coordinates after low terms are subtracted.

The exact domain uses the diagonal Gram bounds, integrality of $D/\dim(Z)$, the Fourier multiplicity bound, full row-orbit closure, and positive semidefiniteness after subtracting that entire orbit. It has 25 row orbits for every low assignment. The pointed classes $k = 0$, $k = 1, 3$ and $k = 2$ impose their respective cyclic twist constraints and leave 167, 145 and 153 columns. The three integer covectors in `r18_sts_moment_certificate.json` pair nonnegatively with every column and pair to -256 with every corresponding target, for all eight low assignments. The standard-library verifier reconstructs the source, characters, thirteen twist triples and all bounded domains using exact rational arithmetic. Thus none of the four pointed classes is possible. The initial Galois reduction proves the general complex assertion. □

13. The open case $R8_2_24$

The ring $R8_2_24$ has no unitary categorification by Proposition 18.1, but its general complex categorifiability remains open. The following necessary conditions reduce the question to a specific tensor-autoequivalence problem.

Lemma 13.1. *Let $\mathcal{B}$ be pseudo-unitary with simple objects $1, g, r, gr$ and rules*

$$g^2 = 1, \qquad gr = rg, \qquad r^2 = 1 + 2r + 2gr.$$

Then $\nu_2(g) = -1$, $\nu_2(r) = \nu_2(gr) = 1$, and the cubic rotations on $\mathrm{Hom}(1, r^3)$ and $\mathrm{Hom}(1, (gr)^3)$ are identities.

Proof. Put $d = 2 + \sqrt{5}$ and $D = 4 + 8d = 20 + 8\sqrt{5}$. The formal codegrees are $D, D', 4, 4$, and $1/D - 1/D' = -1/\sqrt{5}$. The proof of Lemma 14.2, with the primitive character modulo five in place of that modulo seventeen, gives the same equality conclusion at conductor five: every central simple with nonrational dimension ratio z/D has twist of order dividing five.

We use the signed version of the saturated indicator identity. Let $\ell(z)$ be a rational coordinate of the nontrivial quadratic part of z/D, scaled so that its values on $(1, g, r, gr)$ are $(2, 2, -1, -1)$. For a cyclotomic automorphism σ with residue k modulo five, write π and ε_A for its modular permutation and signs. The Chinese remainder theorem lifts each chosen unit residue to an exponent coprime to the full cyclotomic conductor. The dimension transformation gives

$$z_{\pi A} = \varepsilon_A \frac{D}{\sigma D} \sigma(z_A), \qquad \ell(z_{\pi A}) = \varepsilon_A \chi_5(k) \ell(z_A).$$

The first induction identity, weighted by ℓ, is

$$\sum_A \ell(z_A) z_A \theta_A = D\ell(1).$$

The signs in the two preceding transformations cancel, giving

$$\sigma\big(\ell(z_A)z_A\big) = \chi_5(k)\frac{\sigma D}{D}\ell(z_{\pi A})z_{\pi A}.$$

Apply σ to the weighted identity and reindex by π. If h is the inverse of k modulo five, its nonzero terms obey $\sigma(\theta_A) = \theta_{\pi A}^h$, since their twists divide five and $\theta_{\pi A} = \sigma^2(\theta_A)$. The terms belonging to $I(1)$ have twist one and satisfy the same equation. Cancel σD and use the induction formula for the hth indicator. Inversion permutes the units modulo five and $\chi_5(h) = \chi_5(k)$, proving

$$\sum_i \ell(\dim X_i)\nu_k(X_i) = \chi_5(k)\ell(1) \qquad (k,5) = 1.$$

At $k = 2$ this reads $2 + 2\nu_2(g) - \nu_2(r) - \nu_2(gr) = -2$. All three unknowns are signs, forcing the stated values. At $k = 3$ it reads $\nu_3(r) + \nu_3(gr) = 4$. Each rotation acts on a two-dimensional space and has order three. Equality in the trace bound forces both rotations to be identities. □

Proposition 13.2 (Necessary inner-square condition)**.** *A complex categorification of R8_2_24 has a pseudo-unitary Galois conjugate with a faithful C_2 grading. Its neutral subcategory $\mathcal{B}$ has the fusion rules and indicators of Lemma 13.1. There is a tensor autoequivalence T of $\mathcal{B}$ fixing every simple label and satisfying*

$$T^2 \simeq_\otimes \mathrm{Ad}(g).$$

The pointed subcategory generated by g has nontrivial associator class in $H^3(C_2, \mathbb{C}^\times)$.

Proof. The source has exactly two FP dimensions, 1 and $2 + \sqrt{5}$. Apply [25, Lemma 3.1] to the whole category to obtain a pseudo-unitary Galois conjugate. The source tensor has neutral labels $\{0, 5, 6, 7\}$, with $r = X_5$, $g = X_7$, and the displayed rank-four fusion rules. The odd invertible $a = X_1$ has $a^2 = g$. Commutativity of the source ring makes $T = \mathrm{Ad}(a)|_{\mathcal{B}}$ fix every neutral simple label. The coherent composition of inner tensor functors gives $T^2 \simeq_\otimes \mathrm{Ad}(a^2) = \mathrm{Ad}(g)$. The neutral category inherits pseudo-unitarity, so Lemma 13.1 applies. In the canonical spherical structure on a pointed C_2 category, $\nu_2(g) = -1$ is precisely the nontrivial associator class [60]. □

This is a necessary condition, not an exclusion. In particular, the nontrivial pointed class on $\langle g\rangle$ is compatible with restriction of an odd associator class from $C_4 = \langle a\rangle$. Pseudo-unitarity alone does not identify $\mathcal{B}$ with the known unitary $2D2$ category.

13.1. **The categorical transport question.** A possible local approach uses the composition-paired spaces

$$H^+ = \mathrm{Hom}(r^2, r), \quad H^- = \mathrm{Hom}(r, r^2), \qquad J^+ = \mathrm{Hom}(r^2, gr), \quad J^- = \mathrm{Hom}(gr, r^2).$$

Each has dimension two. For any tensor functor, the two induced maps on opposite Hom spaces preserve composition. Consequently their matrices obey $K^- = (K^+)^{-\mathsf{t}}$. This inverse-transpose identity is intrinsic; identifying a numerical partial-bend formula with these maps requires a common choice of oriented charts.

In a proposed paired right-bend frame, write $E = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $\bar{i} = 1 - i$. If the decorated outgoing cubic rotation is $\Omega = \mathrm{diag}(\omega_0, \omega_1)$, the oriented left bends are

$$L^- = E\Omega, \qquad L^+ = \Omega^{-1}E.$$

Thus the incoming bend on the composition-dual vector with index i has coefficient $\omega_{\bar{i}}^{-1}$. The two decorated phases cannot be cancelled by identifying that coefficient with ω_i^{-1}. No equality of the two phases is available from these matrix identities alone.

A route avoiding decorated left bends retains all eight even slots in one tetrahedral block:

$$A=\begin{pmatrix} t & 0 & 0 & a_U\\ 0 & t & t & 0\\ 0 & t & t & 0\\ a_L & 0 & 0 & t\end{pmatrix}, \qquad D_0=\begin{pmatrix} x & 0 & 0 & Z\\ 0 & X & y & 0\\ 0 & Y & W & 0\\ z & 0 & 0 & w\end{pmatrix}.$$

Here W, w are independent. For clarity, let $F_0 = F_r^{r,r,r}$ and $F_1 = F_{gr}^{r,r,r}$, and let A, B be the $p = r, gr$ blocks of F_0 over $q = r$ and C, D_0 the corresponding blocks of F_1, in the convention of [30, pp. 528–530]. A superscript $\vee$ denotes the opposite-tree coefficient of the inverse associator, not complex conjugation. For $v = (i, j, k, l)$ put

$$\begin{aligned} p(v) &= (k, \bar{j}, i, \bar{l}),\\ c(v) &= (j, \bar{k}, \bar{i}, l), \qquad c^2(v) = (\bar{k}, i, \bar{j}, l), \qquad b_l = -i(-1)^l. \end{aligned}$$

The desired common-frame identities are

$$A_v^\vee = A_{p(v)}, \qquad (D_0)_v^\vee = (D_0)_{p(v)},$$

and, crucially, the four oriented transports

$$\begin{aligned} B_v &= b_l (D_0)_{c(v)}, \quad B_v^\vee = b_l^{-1} (D_0)_{c(v)}^\vee,\\ C_v &= b_l (D_0)_{c^2(v)}, \quad C_v^\vee = b_l^{-1} (D_0)_{c^2(v)}^\vee. \end{aligned} \tag{6}$$

Equation (6) is an *unproved transport requirement*, not a lemma applied in the classification. In particular, the opposite B, C transports cannot be inferred by applying p to the forward equations: the slot maps p and c do not commute.

The unresolved task is to derive all four transports in a single oriented pivotal frame, including the evaluation of g, the chosen g–r crossing and the mixed-channel charts. Reversing a g cap can contribute a sign because $\nu_2(g) = -1$. The unitary numerical formulas of [30] therefore do not establish these identities for an arbitrary pseudo-unitary category. The general tetrahedral action of [31, Lemma 3.13(ii), Proposition 3.22 and Corollary 4.9] retains partial-dual maps on its tensor factors; those maps need not preserve chosen bases. It does not by itself supply the required numerical transport table.

The finite algebra explains the relevance of this question. If the paired coefficient identities and (6) hold, three column contractions of F with its inverse give, with $S = xX + Ww$ and $d = 2 + \sqrt{5}$,

$$1 = 2t^2 + S + 1/d, \qquad 1 = S + 2zZ, \qquad 1 = S + 2yY.$$

The unit term is obtained from normalized cup and cap contractions. If the annular-corner parameter has the compatible value $t = (2 - \sqrt{5} + \tau)/4$, $\tau = \pm 1$, subtraction gives

$$yY = zZ = t^2 + \frac{1}{2d} = \begin{cases} (\sqrt{5}-1)/8, & \tau = 1,\\ (3\sqrt{5}-5)/8, & \tau = -1. \end{cases}$$

Both values are nonzero. In a compatible diagonal gauge the weights of y, z are $u^{-2}v^2, u^{-2}v^{-2}$; their invariance would force $u^4 = 1$, preventing the required projective square on H^+. This is a conditional coefficient calculation. Its exact replay in `verify_eight_slot_algebra.py` verifies the slot algebra and the two nonzero field values, but proves none of the four categorical identities in (6). It contributes no exclusion of $R8_2_24$.

14. A conductor-seventeen indicator balance

Theorem 14.1. *The five rings*

$$R8_2_33, \quad R8_2_36, \quad R8_2_76, \quad R8_2_412, \quad R8_2_446$$

have no complex fusion categorification.

The same short indicator identity excludes all five rings, including the noncommutative cases. No enumeration of possible center rows or associator equations is needed.

Lemma 14.2. *Let S, T be the normalized unitary modular representation of a pseudo-unitary Drinfeld center, and let P project onto the T-fixed space. Suppose that a T-fixed vector w has diagonal-torus character the primitive quadratic character modulo* 17*. If a known subspace L of the T-fixed space contains w and*

$$\operatorname{proj}_L Sw = -\frac{w}{\sqrt{17}},$$

then $PSw = -w/\sqrt{17}$ and the cyclic representation generated by w factors through level 17*.*

Proof. The compression bound in [64, Lemma 4.1] is already attained on L, so the full projection cannot have any additional coordinate. For completeness, the equality case also gives the asserted level bound. By the congruence theorem [61], decompose the finite representation into local irreducible tensor factors. At 17, an irreducible representation with an upper-unipotent-fixed vector embeds in the primitive-vector permutation representation over $\mathbb{Z}/17^a\mathbb{Z}$. In the quadratic torus eigenspace, the maximal singular subspace of the compression is spanned by the functions

$$F(p,q) = \chi_{17}(q)[q \text{ is a unit}], \qquad G(p,q) = \chi_{17}(-p)[q = 0 \pmod{17}].$$

This is the equality subspace in the proof of the cited local bound. Both functions are pullbacks from level 17. An irreducible constituent containing an equality vector therefore factors through that level. At every other prime the character is trivial; equality in the bound one makes the vector fixed by both upper and lower unipotents, and hence by the entire local group. In a tensor product of contractions, the maximal singular subspace is the tensor product of the local maximal singular subspaces, as follows by diagonalizing the positive operators A^*A. Thus all constituents supporting w factor through level 17. The full equality argument is also recorded in the evidence note `compression/CONDUCTOR_EQUALITY_LEMMA.md`. □

Lemma 14.3 (Saturated Galois indicator balance)**.** *Let $\mathcal{C}$ be pseudo-unitary, with simple dimensions in a real quadratic field K of conductor f and irrational global dimension D. Suppose every center simple A outside $I(1)$ with $\dim(A)/D \notin \mathbb{Q}$ satisfies $\theta_A^f = 1$. Let $\ell : K \longrightarrow \mathbb{Q}$ be a nonzero rational linear map with $\ell(D) = 0$. If k is coprime to f and $\zeta_f \mapsto \zeta_f^k$ fixes K, then*

$$\sum_{i \neq 0} \ell(\dim X_i)\nu_k(X_i) = 0. \tag{7}$$

Proof. Choose a common cyclotomic twist field $\mathbb{Q}(\zeta_N)$, with $f \mid N$. The Chinese remainder theorem gives an exponent congruent to k modulo f and coprime to N: at each prime dividing N but not f, impose the additional congruence 1. The corresponding Galois automorphism τ fixes K and raises fth roots to their kth powers.

For $z_A = \dim A$, induction reciprocity and the first trace identity [75, Theorem 2.5] give

$$\sum_A \ell(z_A) z_A \theta_A = D\ell(1).$$

All coefficients belong to K. The terms with $z_A/D \in \mathbb{Q}$ vanish since $\ell(z_A) = 0$. In every other term outside $I(1)$ the twist is an fth root by hypothesis; inside $I(1)$ it is one. Applying τ therefore replaces every remaining θ_A by θ_A^k, while fixing the coefficients and right side. The general induction indicator formula [60, Theorem 4.1] now gives

$$D \sum_i \ell(\dim X_i)\nu_k(X_i) = D\ell(1).$$

Cancel D and the unit term. The argument proves an exact complex identity, without taking real parts or relative traces, and does not assume that the source ring is commutative. □

Proof of Theorem 14.1. Every source ring has four invertibles, forming a group G of order four, and four noninvertibles $\mathcal{O}$, all of dimension

$$d = 4 + \sqrt{17}, \qquad d^2 = 1 + 8d, \qquad D = 4 + 4d^2 = 8 + 32d = 136 + 32\sqrt{17}.$$

By [25, Lemma 3.1], any complex categorification is Galois conjugate to a pseudo-unitary one. This two-FP-dimension theorem does not assume commutativity. Work with the conjugate and its canonical positive spherical structure.

The source tensors give

$$\Omega = \sum_X XX^* = 8\,1 + 8\sum_{\rho\in\mathcal{O}} \rho, \qquad \det(xI - L_\Omega) = (x-8)^6(x^2 - 272x + 1088).$$

Here L_Ω is the Casimir multiplication matrix; in the noncommutative cases we do not identify it with the full induction Gram matrix. If an irreducible ring representation E has dimension e_E, the Casimir eigenvalue is $e_E f_E$ with multiplicity e_E^2 [75, Remark 2.11]. Thus D and $D' = 136 - 32\sqrt{17}$ are the codegrees of one-dimensional representations, and every other ordinary codegree $f_E = 8/e_E$ is rational.

By [75, Theorem 2.13], the corresponding center constituent A_E of $I(1)$ has multiplicity e_E and dimension D/f_E. Let J correspond to D'. Then

$$[I(1) : J] = 1, \qquad \dim J = D/D' = d^2.$$

All these constituents have twist one. Also $F(I(1)) = \Omega$ implies that none contains a nonunit invertible in its restriction.

Normalize $S_{00} = 1/D$. The multiplicity vector m of $I(1)$ satisfies $Sm = m$ [75, Proposition 2.8]. Therefore its support is Galois-stable with positive Galois signs. A Galois automorphism fixing $K = \mathbb{Q}(\sqrt{17})$ sends the central unit to the unique constituent of $I(1)$ of dimension one; one flipping K sends it to J, the unique constituent of dimension D/D'. Thus $w = e_1 - e_J$ has quadratic torus character χ_{17}.

The known twist-one block sends w to

$$\left(\frac{1}{D} - \frac{1}{D'}\right) w = -\frac{w}{\sqrt{17}}.$$

Indeed its $1, J$ block has diagonal $1/D$ and off-diagonal $1/D'$, and the other entries in these two columns agree because their codegrees are rational. This remains true when $e_E = 2$. Lemma 14.2 now gives $PSw = -w/\sqrt{17}$ and level 17 on the cyclic representation. For a center simple A of dimension z, the coordinate of Sw is

$$\frac{z}{D} - \sigma\left(\frac{z}{D}\right).$$

If z/D is nonrational and A is outside $I(1)$, its twist is therefore a primitive seventeenth root: it has order dividing 17, and twist one is excluded by the exact formula for PSw. Apply Lemma 14.3 with

$$\ell(a + b\sqrt{17}) = 4a - 17b, \qquad k = 2.$$

Indeed $\ell(D) = 0$, and 2 is a quadratic residue modulo 17, so $\zeta_{17} \mapsto \zeta_{17}^2$ fixes K. The source weights are $\ell(1) = 4$ on the invertibles and $\ell(d) = -1$ on the noninvertibles. We obtain the common balance

$$\boxed{\sum_{\rho\in\mathcal{O}} \nu_2(\rho) = 4 \sum_{g\in G\setminus\{1\}} \nu_2(g).} \tag{8}$$

No restrictions on twists of constituents of $I(g)$, no relative moment hull, and no enumeration of center rows are needed.

A group of order four has an odd number of nonunit involutions: one for C_4, three for C_2^2. Its nontrivial pointed indicator sum is therefore a nonzero odd integer, so the right side of (8) has

absolute value at least four. The left side has absolute value at most the number of self-dual noninvertibles. The exact source values are

ring	G	self-dual noninvertible labels	number
$R8_2_33$	C_4	$5, 6$	2
$R8_2_36$	C_4	$\varnothing$	0
$R8_2_76$	C_4	$\varnothing$	0
$R8_2_412$	C_2^2	$6, 7$	2
$R8_2_446$	C_2^2	$\varnothing$	0

Each row contradicts the balance, proving all five exclusions over $\mathbb{C}$. □

The source-bound verifier `verify_q17_indicator_balance.py` checks the original tensors, hashes, associativity, FP dimensions, pointed groups, duality lists, Casimir polynomial, and the displayed arithmetic. It uses only exact integer operations and runs identically under standard and optimized Python. The proof and independent audits are stored in `evidence/unified_20261016/rank8/`, including `Q17_INDICATOR_BALANCE_PROOF.md`.

14.1. A spherical reduction for the two $\sqrt{10}$ cases.

Proposition 14.4. *Every complex categorification of $R8_2_630$ or $R8_2_671$ admits a spherical structure and has a pseudo-unitary Galois conjugate.*

Proof. Both tensors are self-dual. Choose the positive-second-indicator lifts in the canonical sphericalization. These lifts are self-dual, because their duals have the same nonzero spherical dimensions, whereas the other lifts have the opposite dimensions. Their signed quotient tensor T is fully symmetric and associative, with $|T_{ij}^k| \le N_{ij}^k$ and $T_{ij}^k \equiv N_{ij}^k \pmod 2$. The odd-channel parity theorem [63, Theorem 3.9] further gives $N_{jj}^i - T_{jj}^i \equiv 0 \pmod 4$.

Thus repeated coefficients one are positive, repeated coefficients two have signs ± 2, and all-distinct coefficients one have signs ± 1. There are 27 Boolean signs for each tensor. Only $R8_2_671$ has an all-distinct coefficient two, with possible values $-2, 0, 2$. The source-bound exact checker regenerates every ordered associativity equation. For $R8_2_630$, exhaustive branching on the 157 distinct equations visits 195 nodes and leaves only $T = N$. For the three values of the extra coefficient of $R8_2_671$, it visits $57, 5, 103$ nodes; the only solution is again $T = N$, in the last branch. Every rejection evaluates an integer associativity equation exactly. Consequently the chosen actual lifts are tensor closed, and restriction of the forgetful functor identifies their spherical fusion category with the original category.

The same checker verifies the complete tables of eight distinct characters over $\mathbb{Q}(\sqrt{3}, \sqrt{10})$. The formal codegrees are

$$120 \pm 36\sqrt{10}, \quad 4, 6, 6, 12, 12, 12.$$

All six characters outside the FP conjugate pair vanish on a simple basis object. Since spherical dimensions of simple objects are nonzero, the spherical dimension character belongs to that pair. A Galois automorphism therefore takes it to the positive FP character and produces a pseudo-unitary category. □

The complete character tables, signed domains, source-bound replay, and independent audit are retained in `evidence/unified_20261016/rank8/HI6_PAIR_GALOIS_PU.md`. This proposition is a structural reduction; by itself it makes no existence decision and does not identify either center with a known Haagerup–Izumi center.

15. A third-indicator triangle obstruction at conductor forty

Theorem 15.1. *The fusion ring $R8_2_630$ has no complex categorification.*

Proof. Proposition 14.4 reduces a hypothetical complex categorification to a pseudo-unitary Galois conjugate. Work with its canonical positive spherical structure. Put $s = \sqrt{10}$ and $K = \mathbb{Q}(s)$. In census order, the FP dimensions and global dimension are

$$(d_0, \dots, d_7) = (1, 1, 2+s, 3+s, 3+s, 2+s, 4+s, 4+s), \qquad D = 120 + 36s.$$

The complete character table gives formal codegrees

$$D, \quad D' = 120 - 36s, \quad 4, 6, 6, 12, 12, 12.$$

The source ring is commutative, so each constituent A_E of $I(1)$ has multiplicity one and dimension D/f_E [75, Theorem 2.13]. Let J denote the constituent for D'; then $\dim J = D/D' = 19 + 6s$.

The multiplicity vector of $I(1)$ satisfies $Sm = m$, so its support is Galois-stable with positive signs. The central unit has Galois orbit exactly $\{1, J\}$: these are the unique constituents of $I(1)$ of dimensions 1 and D/D'. Thus $w = e_1 - e_J$ has the primitive quadratic torus character χ_{40}. The known twist-one block satisfies

$$\operatorname{proj}_{I(1)} Sw = \left(\frac{1}{D} - \frac{1}{D'}\right) w = -\frac{s}{20} w = -\frac{w}{\sqrt{40}}.$$

Indeed the $1, J$ submatrix has diagonal $1/D$ and off-diagonal $1/D'$, while all other entries in the two columns agree since their codegrees are rational.

The local compression bound [64, Lemma 4.1] is therefore attained. Its equality refinement, proved in the argument of Lemma 14.2, works at each prime power: for conductor p^c, the equality vectors are pullbacks from level p^c; trivial local characters force the trivial representation. Applying this to the nontrivial local conductors 8 and 5 gives level 40 for the cyclic representation generated by w. The general equality proof is recorded in `compression/CONDUCTOR_EQUALITY_LEMMA.md`. Since the coordinate of Sw at a center simple of dimension z is

$$\frac{z}{D} - \sigma\left(\frac{z}{D}\right),$$

where σ flips K, every such object with z/D nonrational has $\theta^{40} = 1$.

Apply the same saturated Galois indicator lemma used for the five conductor-seventeen cases, Lemma 14.3, now with

$$\ell(a + bs) = 3a - 10b, \qquad f = 40, \qquad k = 3.$$

We have $\ell(D) = 0$, and $\chi_{40}(3) = 1$ means that $\zeta_{40} \mapsto \zeta_{40}^3$ fixes K. The source weights are

$$(\ell(d_i))_{i=0}^{7} = (3, 3, -4, -1, -1, -4, 2, 2),$$

so the lemma gives

$$\sum_{i \neq 0} \ell(d_i) \nu_3(X_i) = 0. \tag{9}$$

This is an exact complex identity. The first trace is transformed into the third by a field automorphism fixing every coefficient; no reality of the third indicators is assumed.

Finally the source tensor gives

$$\left(\dim \operatorname{Hom}(1, X_i^3)\right)_{i=0}^{7} = (1, 0, 0, 1, 1, 1, 0, 0).$$

The only nonzero nonunit third indicators are consequently those at labels $3, 4, 5$. Each is the trace of an order-three rotation on a one-dimensional space, hence a cube root of unity of modulus one. Equation (9) becomes

$$\nu_3(X_3) + \nu_3(X_4) + 4\nu_3(X_5) = 0.$$

The first two terms have sum of absolute value at most 2, whereas the last has absolute value 4. This strict triangle contradiction excludes the hypothetical category and proves the theorem. □

The source-bound standard-library verifier `verify_r630_weighted_indicator.py` replays the structural sphericality gate, complete character table, dimensions, codegrees, conductor arithmetic, weights, and cubic invariant dimensions. The theoretical contradiction uses no enumeration of center rows. Its normal and optimized receipts and independent audit accompany `R630_WEIGHTED_GALOIS_INDICATOR_PROOF.md` and the common lemma `SATURATED_GALOIS_INDICATOR_LEMMA.md` in `evidence/unified_20261016/rank8/`.

16. A conductor-forty center trace obstruction

Theorem 16.1. *The ring R8_2_671 has no complex fusion categorification.*

Proof. Proposition 14.4 reduces a hypothetical categorification to a pseudo-unitary one. Put $q = \sqrt{10}$. Its simple dimensions and global dimension are

$$d = (1, 1, 3+q, 4+q, 4+q, 3+q, 2+q, 2+q), \qquad D = 120 + 36q.$$

The complete ordinary formal codegrees are $D, D', 4, 6, 6, 12, 12, 12$, where $D' = 120 - 36q$. The six remaining characters are defined over $\mathbb{Q}(\sqrt{3})$. All eight source objects are self-dual, and the odd-isotypic pivotal identities force all their second indicators to be $+1$. The source induction row is

$$[FI(1)] = (8, 2, 4, 6, 8, 6, 6, 6).$$

Write a central forgetful row as v, with $\dim FZ = A + Bq$, where

$$A = v_0 + v_1 + 3v_2 + 4v_3 + 4v_4 + 3v_5 + 2v_6 + 2v_7, \qquad B = \sum_{i=2}^{7} v_i.$$

Choose a cyclotomic automorphism σ flipping q and fixing $\sqrt{3}$. The character-labelled correspondence [75, Example 2.9 and Theorem 2.13] and modular Galois symmetry give the exact transport

$$T(v) = v + (0, 0, 1, 1, 1, 1, 1, 1)\left(A - \frac{10B}{3}\right). \tag{10}$$

An actual row has partner $T(v)$ or $-T(v)$, with the sign making it nonnegative; the entries must be integral. In particular $3 \mid B$. The unit row is paired with $(1, 0, 1, 1, 1, 1, 1, 1)$.

Let J be the Galois image of the central unit and put $w = e_1 - e_J$. The multiplicity vector of $I(1)$ is fixed by S, so its Galois signs are positive. Its known twist-one block gives

$$\operatorname{proj}_{I(1)} Sw = \left(\frac{1}{D} - \frac{1}{D'}\right) w = -\frac{w}{\sqrt{40}}.$$

The torus character of w is the primitive quadratic character χ_{40}. The equality refinement of the compression bound [64, Lemma 4.1] implies $PSw = -w/\sqrt{40}$ and level 40 for its cyclic representation. The quadratic Gauss-sum refinement gives negative χ_{40} on primitive 40th twist exponents. These refinements, with their full local proofs, are supplied in the common compression evidence packet. Since the Z-coordinate of Sw is $z/D - \sigma(z/D)$, every central simple outside $I(1)$ with nonrational z/D has twist order dividing 40 and cannot have twist one. The last assertion uses the exact full projection PSw, whose right side has no such coordinate.

Let t denote normalized relative trace to $\mathbb{Q}(q)$. For a root of unity of arbitrary order, the pair $(t(\theta), t(\theta^2))$ lies in the convex hull of

$$(1,1),\ (-1,1),\ (\tfrac{1}{2},-\tfrac{1}{2}),\ (-\tfrac{1}{2},-\tfrac{1}{2}),\ (0,-1),\ (\pm q/8, 0),\ (0, \pm q/8).$$

This is a root-order independent statement: rational traces are Ramanujan sums; an irrational first trace requires order $40m$, with m squarefree and coprime to 10, and has absolute value $q/(8\varphi(m))$ and second trace zero. An irrational second trace instead requires order $80m$ and has first trace zero. The displayed segments contain all these values. For a high row with nonrational dimension ratio, the allowed pair list reduces to

$$(-1,1),\ (0,-1),\ (-\tfrac{1}{4},-\tfrac{1}{4}),\ (0,0),\ (\tfrac{1}{4},-\tfrac{1}{4}),\ (0,\tfrac{1}{4}),\ (-q/8, 0).$$

These are the orders $2, 4, 5, 8, 10, 20, 40$, respectively. If $v_1 > 0$, positivity and $\mathrm{Tr}(\theta^2 I(X_1)) = D = \dim I(X_1)$ force $\theta^2 = 1$. Thus only the two signs remain, and only -1 remains when the ratio is nonrational. Galois partners have equal relative moments. On the restricted rows one may also choose σ with exponent 59 modulo 120; its square is one modulo 40, so their actual twists agree.

Here is the complete finite arithmetic needed to finish the proof. The source tensor determines

$$C_{ij} = \sum_{k,l} N_{ki}^l N_{kj}^l = \sum_Z v_i(Z) v_j(Z).$$

Each constituent Z_χ of $I(1)$ has unit multiplicity and known dimension D/f_χ. Its actual character label imposes

$$\frac{\chi_i(v_j)}{f_i} = \frac{\chi_j(v_i)}{f_j}.$$

After the two forced rows, the six exact bounded domains have sizes $86, 74, 74, 22, 22, 22$. Their prescribed sum and the entrywise and positive semidefinite residual Gram conditions leave exactly six frames. For every remaining row, integral closure of the modular formal-codegree relation forces D/z integral; equivalently,

$$A^2 - 10B^2 \mid 120A - 360B, \qquad A^2 - 10B^2 \mid 36A - 120B.$$

Together with (10) and the residual Gram bound for the whole partner orbit, the six frames have respectively $73, 85, 73, 28, 36, 28$ allowable row orbits.

Combine their Gram columns with the first and second induction trace identities

$$\mathrm{Tr}(\theta I(X_i)) = 0, \qquad \mathrm{Tr}(\theta^2 I(X_i)) = D \quad (i > 0).$$

After subtracting the known twist-one terms and multiplying the trace coordinates by eight, these are 56 integer equations. The moment domains above produce respectively $360, 398, 360, 138, 168, 138$ columns. An actual center would express the target as a nonnegative real combination of them.

Two fixed integer covectors contradict all six systems. Their pairings with every allowable column are nonnegative; their target pairings are -225800939520 for the first three frames and -960 for the last three. The covectors are given explicitly in `r671_center_trace_certificate.json`. The standalone verifier reconstructs every domain, tests every rational positive semidefinite condition, and checks every integer pairing directly from the source tensor. It uses no optimizer or floating-point arithmetic. Thus every possible center is excluded, and the initial Galois reduction proves the general complex claim. □

17. Galois-invariant pointed associators

The next obstruction is intrinsic to a pointed fusion subcategory; it therefore does not require a pivotal structure on an ambient category.

Lemma 17.1. *Let D be a fusion ring with precisely two FP dimensions, 1 and $d > 1$, and let g be a specified order-two invertible. If every pseudo-unitary categorification of D has the same associator class on $\langle g \rangle$, then every complex categorification has that class. If the class is nontrivial, no fusion category containing such a subcategory can have a simple M with $gM \cong M$.*

Proof. By the published [25, Lemma 3.1], every categorification of D is Galois conjugate to a pseudo-unitary one. The two elements of $H^3(C_2, \mathbb{C}^\times)$ have normalized representatives $\omega(g,g,g) = +1$ and -1, each fixed by every field automorphism. Thus their classes are preserved separately by Galois conjugation. If $gM \cong M$, the orbit $\{M\}$ is a rank-one module over $\mathrm{Vec}_{C_2}^\omega$. After choosing $gM \to M$, its scalar module pentagon says $d\psi = \omega$ for a two-cochain ψ. Hence $[\omega] = 0$, a contradiction. This is also the pointed module-category criterion of [22]. □

Write the ordinary and pseudo-$PSU(2)_6$ rings in the basis $(1, g, x, y)$, with $g^2 = 1$, $gx = xg = y$ and $gy = yg = x$. Both have FP dimensions $(1, 1, 1+\sqrt{2}, 1+\sqrt{2})$. Their distinguishing products and dualities are

	$x^2 = y^2$	$xy = yx$	duality
ordinary	$1 + x + y$	$g + x + y$	$x^* = x,\ y^* = y$
pseudo	$g + x + y$	$1 + x + y$	$x^* = y.$

Proposition 17.2. *The rings R8_2_312, R8_2_315 and R8_2_317 have no fusion-category categorification over* $\mathbb{C}$.

Proof. In a pseudo-unitary pseudo-$PSU(2)_6$ category, Larson's [57, Proposition 5.5 and proof of Theorem 5.8, Case 1(c)(1)] forces $\nu_2(g) = -1$. In the $c = 1$ case at issue, the positive-sign branch would require at least eight roots of unity when at most seven are available. The canonical spherical structure has dimension one on pointed simples, so $\nu_2(g) = \omega(g, g, g)$ in $\mathrm{Vec}^{\omega}_{C_2}$ [60]. Thus the pseudo ring forces the nontrivial pointed class. Lemma 17.1 makes this a constraint on every complex categorification.

For completeness, the ordinary ring forces the trivial class. Equip a hypothetical pseudo-unitary categorification with its canonical spherical structure. The odd-multiplicity parity theorem [63, Theorem 3.9] gives $\nu_2(x) = \nu_2(y) = 1$. The remaining alternative $\nu_2(g) = -1$ is excluded by the following exact indicator certificate. Let A be the restriction matrix from the center and $C = AA^{\mathsf{T}}$ its induction Gram matrix. The complete low-column lists are precisely

$$L_1 = [(1,0,0,0), (1,0,0,2), (1,0,1,0), (1,0,1,0)],$$
$$L_2 = [(1,0,0,0), (1,0,1,1), (1,0,0,1), (1,0,1,0)],$$
$$L_3 = [(1,0,0,0), (1,0,2,0), (1,0,0,1), (1,0,0,1)].$$

They follow from the complete formal-codegree list; these columns have trivial twist, and every other column has unit coordinate zero and belongs to the residual Gram matrix. For each L_i, the necessary center data yield an integer system $B_i n = b_i$, $n \geq 0$. The stored integer vectors v_i certify

$$v_i^{\mathsf{T}} B_i \geq 0, \qquad (v_1^{\mathsf{T}} b_1, v_2^{\mathsf{T}} b_2, v_3^{\mathsf{T}} b_3) = (-48, -12, -48).$$

Each system has 38 equations and respectively 102, 152, 102 variables, including nonnegative slack variables. All three are therefore infeasible by direct integer arithmetic.

We specify the hypotheses of this finite certificate. Besides reciprocity, it imposes the center indicator identities

$$\sum_{Z \in \mathrm{Irr}\, Z(\mathcal{D})} [F(Z) : X] \dim(Z) \theta_Z^j = \dim(\mathcal{D}) \nu_j(X), \qquad 1 \leq j \leq 4,$$

and the trace bounds for an order-j pivotal rotation on $\mathrm{Hom}(1, X^j)$ [60]. The Cauchy theorem [62], applied to $\dim(\mathcal{D}) = 8 + 4\sqrt{2}$ of norm 32, makes every twist order a power of two. The ten joint relative-trace vectors through height four to $\mathbb{Q}(\sqrt{2})$ cover all such orders: orders larger than the divisor box 32 have the zero vector, which is included. This is a finite trace classification, not a bound on the exponent. The remaining columns satisfy the d-number and dimension-quotient integrality conditions [24, 22]. Finally Galois symmetry of the normalized center S matrix [61] pairs simple dimensions by

$$z \longmapsto \left| \frac{(8 + 4\sqrt{2}) z'}{8 - 4\sqrt{2}} \right|,$$

with equal counts in each pair. These conditions give the displayed systems. The source-bound checker reproduces every column, trace vector, matrix and separating product; it uses no optimization solver. Hence $\nu_2(g) = 1$ in every pseudo-unitary ordinary categorification, and Lemma 17.1 makes its pointed class trivial in every complex categorification.

The exact census witnesses, in the displayed rank-four basis, are

ring	ordinary subring	pseudo subring	g
$R8_2_312$	none needed	$(0,7,1,2)$	7
$R8_2_315$	$(0,7,3,4)$	$(0,7,1,2)$	7
$R8_2_317$	$(0,7,5,6)$	$(0,7,1,2)$	7.

For $R8_2_312$, each of the outside simples $3,4,5,6$ is fixed by g, contradicting the nontrivial pseudo-ring class by Lemma 17.1. For $R8_2_315$ and $R8_2_317$, the two subcategories restrict the same ambient associator to the same pointed subcategory, but force opposite classes. This is impossible. The argument conjugates the rank-four subcategories separately; it assumes no simultaneous pseudo-unitarization of the ambient category. □

18. A scoped obstruction for the open case

The following obstruction applies to unitary categorifications and to a specified Galois class of neutral categories. Its source subring and grading support are checked separately. It does not settle general complex categorifiability.

Proposition 18.1. *The ring $R8_2_24$ has no unitary categorification. More generally, it admits no categorification whose neutral $2D2$ category is Galois conjugate to the known unitary one.*

Proof. Let $\mathcal{Q}$ be the known unitary $2D2$ category. The construction in Proposition 7.1 gives a tensor autoequivalence $T = \mathrm{Ad}(b)|_{\mathcal{Q}}$ swapping r, gr, with $T^2 = \mathrm{Ad}(g)$. Grossman's [56, Lemma 4.1] shows that $\mathrm{Ad}(g)$ exchanges two inequivalent algebra structures on $1+r$, so it is nontrivial. Thus T has order four. The factor automorphism fixing M and sending λ to $-\lambda$ induces an outer, label-fixing involution S by [56, Lemma 4.2]. It commutes with b on both M and λ, hence S commutes with T and is independent of T^2. Grossman's order-eight computation [56, Theorem 4.19 and its proof] therefore gives

$$\mathrm{Aut}_{\otimes}(\mathcal{Q}) = \langle T, S\rangle \cong C_4 \times C_2.$$

Its label-fixing subgroup is $\langle T^2, S\rangle \cong C_2^2$; every label-swapping element has square $T^2 = \mathrm{Ad}(g)$.

The neutral category of a unitary categorification is equivalent to $\mathcal{Q}$ by [30, Theorem 3.31]. In $R8_2_24$, an odd invertible u has $u^2 = g$ and fixes every neutral label by conjugation. Then $\mathrm{Ad}(u)$ would be a label-fixing autoequivalence whose square is the nontrivial $\mathrm{Ad}(g)$, impossible. Applying a Galois automorphism to the entire category proves the stated stronger scoped claim. The two-dimension Galois lemma supplies pseudo-unitarity of an arbitrary neutral category, not equivalence to $\mathcal{Q}$. General complex categorifiability therefore remains open; the necessary conditions and the outstanding local transport question appear in Section 13. □

19. Compression over all constituents of the induced unit

The modular-representation argument in [64, Lemma 4.1 and the proof of Theorem 5.1] can retain every simple constituent of $I(\mathbf{1})$, rather than only the Galois orbit of the dimension character. This gives two further exclusions in the pivotal and spherical settings.

Proposition 19.1. *Let $\mathcal{C}$ be a spherical fusion category, with dimension field $K = \mathbb{Q}(\dim X : X \in \mathrm{Irr}(\mathcal{C}))$ and $G = \mathrm{Gal}(K/\mathbb{Q})$. Write f_E for the formal codegrees of its fusion algebra. Then $f_E \in K$ and, for every character χ of G with conductor q,*

$$\sum_E \left| \sum_{\sigma \in G} \chi(\sigma)\sigma(f_E^{-1}) \right|^2 \le \frac{|G|}{q}.$$

Proof. Put $D = \dim \mathcal{C}$ and $s = S/D$ for the normalized matrix of $\mathcal{Z}(\mathcal{C})$. Let A_E be the constituent of $I(\mathbf{1})$ associated to E. It has dimension D/f_E and twist one, so $f_E \in K$. If A_σ corresponds to $\sigma(\dim)$, then $s_{A_\sigma, Y} = \sigma(\dim Y/D)$. Set $w_\chi = \sum_\sigma \chi(\sigma) e_{A_\sigma}$ and let P project onto the twist-one

subspace. The cited local compression estimate gives $\|Psw_\chi\|^2 \le |G|/q$. Its A_E coordinate is the inner sum in the assertion. Summing over these distinct simple coordinates proves the inequality; their multiplicities in $I(\mathbf{1})$ do not repeat coordinates. □

Corollary 19.2. *Let $B = \sum_i N_i N_i^{\mathsf{T}}$ be the regular Casimir of a fusion ring. Suppose the irreducible factors of $\det(xI - B)$ belonging to a fixed real quadratic field of discriminant q are $(x^2 - t_j x + n_j)^{m_j}$. A spherical categorification requires*

$$q \sum_j m_j \frac{t_j^2 - 4n_j}{n_j^2} \le 1.$$

Proof. The Casimir eigenvalue attached to E is $\lambda_E = (\dim E) f_E$, with regular multiplicity $(\dim E)^2$. The quadratic field lies in K. If its index in K is e', pull back its nontrivial character to G. Each conjugate pair of eigenvalues contributes $2e'^2 m_j (t_j^2 - 4n_j)/n_j^2$ to the preceding bound, whose right side is $2e'/q$. Cancelling and using $e' \ge 1$ proves the claim. □

Proposition 19.3. *The rings R8_2_318 and R8_2_735 admit no pivotal or spherical categorification over $\mathbb{C}$.*

Proof. For both source tensors, direct multiplication gives

$$\det(xI - B) = (x-16)^3 (x-8)(x^2 - 192x + 1024)(x^2 - 16x + 32).$$

Both quadratic factors belong to $\mathbb{Q}(\sqrt{2})$, of discriminant 8. Hence the necessary bound would read

$$8\left(\frac{192^2 - 4\cdot 1024}{1024^2} + \frac{16^2 - 4\cdot 32}{32^2}\right) = \frac{5}{4} \le 1,$$

a contradiction.

For completeness, the eight characters of each commutative ring have formal codegrees

$$16, 16, 16, 8, 8 + 4\sqrt{2}, 8 - 4\sqrt{2}, 96 + 64\sqrt{2}, 96 - 64\sqrt{2}.$$

Every character in the first six positions vanishes on at least one basis element. The remaining two are the Frobenius–Perron character and its quadratic conjugate, and are duality invariant. These statements follow by substitution in the source tensor; the complete exact character tables are supplied in the associated `rank8_full_I1_certificate.json`. Simple pivotal dimensions cannot vanish. Thus a pivotal structure would have one of the last two dimension characters and would be spherical, which has just been excluded. □

These compression arguments alone give scoped exclusions. The independent module obstruction in Section 5 now excludes arbitrary complex categorifications of both rings.

An independent general proof for the four quadratic cases. The same estimate gives a second proof of Proposition 4.1, without a pseudo-unitary reduction. Each of its four rings has Casimir polynomial

$$(x-8)^4 (x^2 - 160x + 640)(x^2 - 32x + 128).$$

The sphericalization's dimension field therefore contains both $\mathbb{Q}(\sqrt{10})$ and $\mathbb{Q}(\sqrt{2})$. Its index e over $\mathbb{Q}(\sqrt{10})$ is at least two. The first quadratic factor has quantity $Q = 9/4$ in Corollary 19.2. In the sphericalization's positive-sign sector, formal codegrees are doubled, so compression requires $eQ/4 \le 1$. But $eQ/4 \ge 9/8 > 1$, a contradiction.

20. The conductor-32 obstruction

Theorem 20.1. *The ring R8_2_857 has no categorification over $\mathbb{C}$.*

Proof. Reorder the source basis as $(0,1,7,2,5,4,3,6)$ and denote it by $X_0 = \mathbf{1}, X_1, \dots, X_7$. All objects are self-dual. Multiplication by X_1 is the symmetric tridiagonal matrix J_2 whose off-diagonal entries are one and whose diagonal is $(0,1,1,1,1,1,1,2)$. If M_j denotes multiplication by X_j, then

$$M_0 = I, \qquad M_1 = J_2, \qquad M_{j+1} = (J_2 - I)M_j - M_{j-1} \quad (1 \le j \le 6).$$

These identities are direct consequences of the source tensor.

Put $\zeta = e^{2\pi i/32}$ and $L = \mathbb{Q}(\zeta + \zeta^{-1})$. The positive dimension character is $d_j = \sum_{r=-j}^{j} \zeta^r = \sin((2j+1)\pi/32)/\sin(\pi/32)$. Its eight Galois conjugates are distinct and give all ring characters. Their formal codegrees are the conjugates of

$$D = \sum_{j=0}^{7} d_j^2 = 4\csc^2(\pi/32), \qquad \frac{1}{D} = \frac{2 - \zeta - \zeta^{-1}}{16}.$$

For any primitive even Dirichlet character χ of conductor 32, the reciprocal Fourier sum over $\mathrm{Gal}(L/\mathbb{Q})$ is

$$\widehat{g}(\chi) = -\frac{\tau(\chi)}{16}, \qquad |\widehat{g}(\chi)|^2 = \frac{32}{16^2} = \frac{1}{8}.$$

Here $\tau(\chi)$ is the primitive Gauss sum: summing $\zeta^a + \zeta^{-a}$ over representatives modulo sign gives precisely its full sum. The eight constituents of $I(\mathbf{1})$ thus violate Proposition 19.1: if the spherical dimension field has index e over L, their squared Fourier norm is e^2, whereas the bound is $e/4$. Hence the ring has no spherical categorification.

Now suppose an arbitrary categorification exists, and take its canonical sphericalization with sign object ε of dimension -1. In the $\varepsilon = 1$ sector the eight formal codegrees are $2\sigma(D)$. They belong to the spherical dimension field K, so $L \subseteq K$. Put $e = [K : L]$. The same eight coordinates in the full induced-unit compression have squared norm $e^2/4$, with upper bound $e/4$. Thus $e = 1$ and $[K : \mathbb{Q}] = 8$.

Both lifts of every simple are self-dual, since duality preserves its nonzero spherical dimension while exchanging the lifts negates it. Choose the positive-second-indicator lifts and pass to the signed quotient $\varepsilon = -1$. Its constants satisfy

$$|T_{ij}^k| \le N_{ij}^k, \qquad T_{ij}^k \equiv N_{ij}^k \pmod 2, \qquad T_{ii}^j \equiv N_{ii}^j \pmod 4.$$

For the last congruence, a negative-indicator lift can occur in $X_i^* \otimes X_i$ only with even multiplicity by [63, Theorem 3.9]. The signed difference is therefore the original multiplicity minus a multiple of four. Self-duality gives full symmetry of T. Consequently $T_{1j}^j = T_{jj}^1 = 1$ for $1 \le j \le 6$.

The off-diagonal entries of the signed multiplication matrix of X_1 are ± 1. Successively switching the lifts of $X_2, \dots, X_7$ makes them all positive, while leaving the unit, X_1, and all diagonal entries unchanged. Its matrix is now J_t, with diagonal $(0,1,1,1,1,1,1,t)$ and $t \in \{-2, 0, 2\}$. The first seven rows give the same polynomial recurrence for all signed multiplication matrices. Thus $\dim X_1$ generates K.

If $t = -2$, the bottom-right entry of $M_2 = J_t^2 - J_t - I$ is 6, exceeding the original multiplicity $N_{27}^7 = 2$. If $t = 0$, then

$$\det(xI - J_0) = (x-1)(x+1)(x^2 - 2x - 1)(x^4 - 4x^3 + 2x^2 + 4x - 1).$$

The degree of $\dim X_1$ is at most four, contrary to $[K : \mathbb{Q}] = 8$. Finally, if $t = 2$, the recurrence makes every signed multiplication matrix equal to the original nonnegative one. The negative lift multiplicities all vanish, so the eight chosen lifts form a spherical fusion subcategory equivalent to the original categorification. This contradicts the spherical exclusion already proved. □

The source-bound certificate verifies all 512 dimension-character products in $\mathbb{Q}(\zeta_{32})$, all eight exact Fourier norms, and the three displayed matrix branches. The argument is independent of any construction of the corresponding folded quantum-group datum.

20.1. **A central-lift obstruction for the $2D2$ rules.**

Proposition 20.2 (No central invertible lift for the $2D2$ rules)**.** *Let $\mathcal{Q}$ be a pseudo-unitary fusion category with simple objects $1, f, r, s$ and fusion rules*

$$f^2 = 1, \qquad fr = rf = s, \qquad r^2 = s^2 = 1 + 2r + 2s, \qquad rs = sr = f + 2r + 2s.$$

There is no invertible object $z \in Z(\mathcal{Q})$ whose underlying object is f.

Proof. Put $d = 2 + \sqrt{5}$, so $d^2 = 1 + 4d$. The displayed rules imply $\mathcal{Q}_{\mathrm{ad}} = \mathcal{Q}$. Consequently the only central lift of the tensor unit is the tensor unit: such lifts are tensor automorphisms of the identity, equivalently characters of the universal grading group. Thus a hypothetical z satisfies $z^2 \cong 1$. In its pointed braided subcategory the positive ribbon twist $q = \theta_z$ belongs to $\{1, -1, i, -i\}$.

If $q = 1$, then $1 \oplus z$ is the regular algebra of a central Tannakian $\mathrm{Rep}(C_2)$. Its module category has two simple free orbits, represented by $1, r$, and their rule is $\bar{r}^2 = 1 + 4\bar{r}$. This contradicts the rank-two fusion-category classification [23].

Suppose $q = \pm i$. Write I for induction to the center. The formal-codegree induction formula, as recalled in [30, Appendix A.2], gives the distinct constituents of $I(1)$ dimensions

$$1, \quad d^2, \quad 1 + 2d, \quad 1 + 2d.$$

Every constituent has twist 1. Each has tensor-unit multiplicity one after forgetting, and none contains f, because $FI(1)$ has no f-summand. The projection formula gives $I(f) \cong z \otimes I(1)$, disjoint from $I(1)$. Since $z^2 = 1$, the monodromy of z with any simple central object is ± 1. Hence every constituent of $I(f)$ has twist $\pm i$.

In $I(r) \oplus I(s)$ the constituents belonging to $I(1)$ have total dimension

$$K = 4d^2 + 4(1 + 2d) = 8 + 24d.$$

Indeed their total multiplicities are, respectively, $0, 4, 2, 2$. The constituents belonging to $I(f)$ have the same total dimension. Since

$$\dim\big(I(r) \oplus I(s)\big) = 2d(4 + 8d) = 16 + 72d,$$

the other constituents have total dimension $24d$. The first-indicator identity gives zero twist trace on $I(r) \oplus I(s)$. Its $I(1)$ part has real trace $8 + 24d$, its $I(f)$ part has purely imaginary trace, and the absolute real trace of the remaining part is at most $24d$. This is impossible.

Finally suppose $q = -1$. Let $\mathcal{I}$ be an Ising category with fermion ψ and noninvertible σ. The object $z \boxtimes \psi$ is a central boson in $\mathcal{Q} \boxtimes \mathcal{I}$. Condensing its regular algebra gives a pseudo-unitary fusion category $\mathcal{D}$ of rank six. All twelve simples before condensation have free orbits. Representatives give objects $1, f, u, r, s, v$ of dimensions

$$1, 1, \sqrt{2}, d, d, \sqrt{2}d,$$

with the above rules on $1, f, r, s$ and additional rules

$$fu = u, \quad fv = v, \quad u^2 = 1 + f, \quad ur = us = v,$$

$$uv = r + s, \quad rv = sv = u + 4v, \quad v^2 = 1 + f + 4r + 4s.$$

We prove directly that these rules have no pseudo-unitary categorification.

Put $D = \dim(\mathcal{D}) = 8 + 16d$ and $T = Dd = 16 + 72d$. The six formal codegrees show that $I(1)$ has two constituents of each dimension $1, d^2, 2 + 4d$, all of twist 1, each with tensor-unit multiplicity one. Positivity and linear independence over $\mathbb{Q}(\sqrt{5})$ show that their restrictions have no u or v summands. The two constituents of dimension d^2 restrict to

$$1 + a_j r + (4 - a_j)s \qquad (0 \le a_j \le 4),$$

and the two of dimension $2 + 4d$ restrict to

$$1 + f + b_j r + (4 - b_j)s \qquad (0 \le b_j \le 4).$$

The r-coefficient of $FI(1)$ is 8, so if $a = a_1 + a_2$ then $b_1 + b_2 = 8 - a$. The total dimension K of the twist-one summands of $I(r)$ supplied by $I(1)$ is therefore

$$K = ad^2 + (8-a)(2+4d) = 16 - a + 32d \geq 8 + 32d.$$

Let $R = T - K$ be the dimension of the other summands. The first two indicator identities [60], with $\nu_2(r) \leq 1$, imply

$$\sum_j w_j \cos t_j = -K, \qquad \sum_j w_j \cos(2t_j) = D\nu_2(r) - K \leq D - K, \qquad \sum_j w_j = R,$$

where $w_j > 0$ and e^{it_j} are their dimensions and twists. If $R = 0$ the first equation is already impossible; otherwise Cauchy–Schwarz gives

$$2K^2 \leq R(R + D - K), \qquad 0 \leq T(T+D) - K(3T+D).$$

But the last expression is at most

$$T(T+D) - (8+32d)(3T+D) = -64 - 512d - 1088d^2 < 0.$$

This final contradiction excludes $q = -1$. □

Remark 20.3. The published Galois pseudo-unitarity reduction [25, Lemma 3.1] for categories having two Frobenius–Perron dimensions upgrades this statement to arbitrary complex categorifications of the displayed ring: a central lift survives Galois conjugation. This uses the corrected published formulation "Galois conjugate to a pseudo-unitary category", not a claim that every categorification is itself pseudo-unitary.

Remark 20.4 (Scope of the inner-conjugation construction). Let $\mathcal{P} = \mathcal{Q}_{\mathrm{pt}}$ and $\mathcal{D} = \mathcal{Z}_{\mathcal{P}}(\mathcal{Q})$. By [48, Remark 2.4(i)], this relative center is Morita equivalent to $\mathcal{P} \boxtimes \mathcal{Q}^{\mathrm{rev}}$. Its invertibles are precisely those of $\mathcal{Z}(\mathcal{P})$, whose group is $C_2 \times C_2$ for either associator on $\mathcal{P} \simeq \mathrm{Vec}_{C_2}^{\omega}$. Thus this natural inner-conjugation construction cannot realize the C_4-based rings $R8_1_9$ or $R8_1_18$; no indicator assumption is needed.

The displayed rank-four rules have trivial universal grading, so the theorem implies $\mathrm{Inv}\,\mathcal{Z}(\mathcal{Q}) = 1$. Morita invariance then gives $|\mathrm{Inv}\,\mathcal{Z}(\mathcal{D})| = 4$. If $\mathcal{D}$ had the rules $R8_1_49$, its universal grading would also be trivial. The forgetful map on invertibles would therefore be injective, since its kernel is the group of tensor automorphisms of the identity, and hence bijective. Every invertible would have a central lift, contradicting the noncommuting invertible action in that ring. This rules out the proposed construction only. General exclusions are proved independently in Theorems 12.7 and 12.9 and Proposition 7.3.

21. Reproduction and mathematical limits

Run `python3 tools/verify_current_publication.py` from the repository root, and repeat with `-O`. The verifier checks the published documents and their website copies, the source identities, the current decision ledger and the accepted finite certificate applications, and the three unresolved cases across the rank-six and rank-eight studies. Its `-historical` option also replays the full chain of source-bound certificate applications in isolated inputs. The exact replay instructions and the scope of every recorded run appear in `docs/VERIFY.md`.

The rank-eight full-certificate audit covers all 973 input tensors, including the 75 signed proof trees and four rank-seven subring proof transfers used by the corresponding base certificates. The theoretical applications and positive tensor comparisons are checked separately. All finite checks use integer or algebraic arithmetic. They validate their specified equations and source bindings; they do not independently prove categorical transport identities, cited theorems or census completeness. A timeout or an unfinished branch never constitutes an exclusion.

The cyclic induction and center STS arguments in Theorems 12.7 and 12.9, together with Proposition 7.3, complete the multiplicity-one classification. At multiplicity two the conclusions are

847 general exclusions, 29 unitary constructions and the single open ring $R8_2_24$. Proposition 13.2 gives necessary conditions for that case; Section 13.1 identifies an unproved categorical transport step. The conditional coefficient calculation in that section contributes no classification decision.

The finite applications are conditional on the stated mathematical arguments, cited theorems and source census completeness. This work does not certify a multiplicity-three census or classify categories up to tensor equivalence. Nothing here implies that no new criterion is needed beyond multiplicity two. Noncommutative Extended Haagerup models at rank eight have multiplicities four and seven [44]; they are genuine models outside the present census bound.

AI disclosure and verification status. GPT-6 Astra and GPT-6.1 Sol assisted with literature exploration, mathematical research, code development and manuscript preparation. The results, proofs and accompanying data are AI-generated and have not been independently verified by a human expert or formalized in Lean. They should be used with caution. Computational replay provides evidence for its explicit finite checks and does not replace mathematical review.

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